

STUDENT LESSON CONTENT

Grade 6 (Year 7) | Phase 3 / Cycle 2 | AI Solution Builder

Bridging Program - Building Trustworthy AI

Transition-Year readiness | Term 1, Weeks 1-6 (Core + Lab)

Field Detail

Program Grade 6 Bridging (Transition Year)

Readiness bar Prior-grade Advanced (Grade 5; Critical Evaluation from Grade 4)

Readiness bar note 'Prior-grade Advanced' means students are baselined against the Grade 5 Advanced-

tier outcomes for AI Conceptual Understanding, AI Solution Design & Development, and Ethics & Governance; the exception is Critical Evaluation & Informed Interaction, which is probed at the Grade 4 Advanced bar because Critical Evaluation paused at Grade 5. See the Start-of-Year Diagnostic Kit for the full baseline procedure.

Delivery model 2 x 45-minute periods per week - a Core lesson + a paired Lab lesson each week

Structure Wk 1-4 strand catch-up (Core + Lab) | Wk 5-6 integrated hands-on mastery task (Core + Lab)

Diagnostic Start-of-year baseline, administered outside teaching hours

Lessons in this pack 12 lessons (6 Core + 6 Lab) + start-of-year diagnostic kit

Week 1 - Core | Data Quality and AI Outputs

Imagine teaching a younger cousin to recognise dates using a photo album. If half the photos are blurry and one

photo is actually a stone, incorrectly labelled as a date, what will your cousin get wrong? An AI learns the same

way you would teach - from the examples it is shown. Today you become a data detective who can predict an

AI's mistakes just by studying its data.

I can...

- I can explain, using a clear example, how the quality of data changes what an AI produces.
- I can predict whether an output will get better or worse when the data is made cleaner or noisier, and say why.
- I can tell the difference between the amount of data and the quality of data.

Words to know

Word What it means

data quality How accurate, complete, consistent, and representative a dataset is.

label The name attached to an example that tells the model what it is.

noise Irrelevant or inconsistent detail in data that confuses the pattern.

representative The data covers the real range of cases the model will meet.

causal chain A step-by-step link from a cause to its effect.

The model learns from its examples

An AI model is not filled with rules that a person typed. Instead it studies a dataset and adjusts itself to match the patterns it finds.

This means the dataset is really the model's teacher. If the teacher shows wrong examples, the model learns the wrong lesson - and it does so confidently.

Quality is not the same as quantity

It is tempting to think more data always helps. But a huge dataset full of mistakes just teaches the wrong pattern more strongly.

A small, clean, well-chosen dataset can beat a giant messy one. Before trusting any output, a careful builder checks

the quality of the data behind it - and never puts personal or identifying information into a tool.

Let's try it - hands-on

You will study a dataset card, predict the exact mistake the AI will make, then watch the real outputs to check your detective work.

1. Read your dataset card and note how each item is labelled.
2. Decide the quality problem: error, gap, or noise.
3. Predict one exact output mistake and write it down before the reveal.
4. Compare your prediction with the revealed clean and degraded outputs.
5. Write one sentence linking the data cause to the output effect.

Think about it

- When could a bigger dataset actually make an AI worse?
- How can an output be wrong and still sound completely sure of itself?

My reflection

- Write one thing you can now predict about an AI just by looking at its data.
- Name one place in daily life where poor-quality data could lead to an unfair result.

Week 1 - Lab | Data Quality Investigation (Lab)

In the Core lesson you predicted one AI mistake from its data. Today you run a proper investigation: four

versions of the same dataset, four sets of outputs, and one big question - which data quality problem does the

most damage? By the end you will rank them and defend your order with evidence, just like a real AI auditor.

I can...

- I can run a fair comparison of clean and noisy data and record what changes in the output.
- I can rank several datasets from best to worst expected output and justify my ranking.
- I can use evidence from the comparison to explain the data-to-output link.

Words to know

Word What it means

fair test A comparison where you change only one thing so you can tell what caused a result.

baseline The clean result you compare all other versions against.

evidence A specific observation that supports a claim.

ranking Ordering items from best to worst using a clear reason.

audit A careful check of data or a system before you trust it.

Make the comparison fair

To find out how data quality changes an output, you must be fair: keep the task and the model the same, and change only the dataset.

Then any difference you see in the output can be traced back to the data - that is what makes your evidence trustworthy.

Rank with evidence

After you see how each dataset changed the output, you rank them from best to worst expected quality.

The hard part is the middle: when two datasets are both weak, you decide which flaw hurt the output more, and you point to the exact output change that proves it.

Let's try it - hands-on

Your team compares four output sets against a clean baseline, records what changed, and ranks the datasets from best to worst with evidence.

1. Study the clean baseline output so you know what correct looks like.
2. For each degraded version, write exactly how the output changed.
3. Match each change to its data flaw.
4. Score each dataset on accuracy, completeness, and consistency.
5. Place all four on the ranking line and justify the order.

Think about it

- Which data flaw did the most damage, and what evidence proves it?
- When two datasets are both weak, how did your team decide which was worse?

My reflection

- Write the one output change that most convinced you of your top ranking.
- Describe how you would audit a dataset before trusting an AI built on it.

Week 2 - Core | Refining Prompts for Better Outputs

You ask an AI to 'write about AI Ain' and get a dull, general paragraph. Was the AI wrong - or was the prompt?

Today you become a prompt engineer who sets a clear goal, judges the output against it, and makes one smart

change at a time. Best of all, you will be able to say exactly why your change worked.

I can...

- I can turn a weak prompt into a stronger one that better meets the goal.
- I can name the exact change I made and explain why it improved the output.
- I can judge an output against the goal instead of just accepting it.

Words to know

Word What it means

prompt The instruction and context you give an AI to guide its output.

goal What you want the output to achieve, stated clearly.

constraint A limit you set, such as length, tone, or format.

ambiguous Unclear or open to more than one meaning.

refinement loop Goal, judge, name the weakness, change, re-judge - repeated.

Judge before you change

You cannot improve an output until you judge it honestly against the goal. Ask: what is missing, and what is off-target?

Common weaknesses in a prompt are a missing audience, missing format, missing constraint, or an ambiguous task.

Naming the weakness tells you exactly what to fix.

Change one thing, and know why

Strong refiners change one thing at a time and predict why it should help before they test it.

That way, when the output

improves, they know which change did it.

Specificity comes from the task, the audience, and the format - never from personal or private information. A precise

prompt is not a personal one.

Let's try it - hands-on

You will take a weak prompt with a clear goal, name one weakness, write a targeted v2, predict its effect, and check

against the revealed output.

1. Read the weak prompt and its goal.
2. Judge the v1 output and name one weakness.
3. Write a v2 that targets only that weakness.
4. Predict how the output should change and why.
5. Compare with the revealed v2 output and log it.

Think about it

- Why is changing one thing at a time smarter than changing many?
- How can you make a prompt specific without adding any private information?

My reflection

- Write the one change that most improved your output and why it worked.
- Describe a time a clearer instruction would have helped you get a better result in real life.

Week 2 - Lab | Prompt Refinement Workshop (Lab)

In the Core lesson you made one smart change to a prompt. Today you run a full workshop: version 1, 2, 3,

maybe 4 - each a single change you can defend. Then comes the real challenge of a designer: chasing two goals

at once, clarity and accuracy, and discovering that helping one can cost the other. Your log will prove exactly

what worked.

I can...

- I can iterate a prompt across several versions and log each change and its effect.
- I can refine for two goals at once and explain the trade-off between them.
- I can use my log as evidence for which change helped the most.

Words to know

Word What it means

iterate To improve something step by step across several versions.

trade-off When improving one goal makes another goal worse.

balance The best compromise between competing goals for a purpose.

accuracy How correct and on-topic the output is.

clarity How easy the output is to understand.

Iterate and keep the receipts

A designer does not stop at the first improvement. You iterate - version by version - changing one thing each time and

logging its effect.

The log is your evidence. When someone asks which change helped most, you point to the row that proves it,

because you changed only one thing that version.

Two goals, one trade-off

Now aim for two goals at once: clarity and accuracy. Sometimes a change that makes the output clearer quietly drops

an accurate detail.

That is a trade-off. Your job is not to pretend it away but to choose the version that best balances both goals for your

purpose - and to say why.

Let's try it - hands-on

Your team iterates a weak prompt across several versions, logging each change and effect, then balances clarity and

accuracy and chooses the best version with evidence.

1. Set the goal and judge the v1 output.
2. Make v2 with one change; log the reason and revealed effect.
3. Iterate to v3 and v4, one change each, logging cause and effect.
4. Add accuracy as a second goal alongside clarity.
5. Find and record a trade-off where one goal cost the other.
6. Choose the best-balanced version and justify it from your log.

Think about it

- Which single change gave the biggest quality jump, and how does your log prove it?
- Where did clarity and accuracy pull against each other, and how did you decide?

My reflection

- Write the best-balanced version your team chose and one sentence on why.
- Explain what balancing two goals taught you about making design decisions.

Week 3 - Core | How Data Shapes a Solution's Outcome

Two apps can share the exact same code and still disagree about the same photo. The secret is not in the code -

it is in what each app was shown. Today you become a data detective, tracing how the examples an app learns

from decide the answer it gives.

I can...

- I can explain, in my own words, how the data fed into a solution shapes the outcome it produces.
- I can predict what would happen to the outcome if the data were changed, and give a reason.
- I can point to a specific part of the data that most affects the result.

Words to know

Word What it means

solution The whole system that solves a problem - the data plus the model built from it.

data The examples a solution learns from.

outcome The answer or result the solution gives.

pattern The regularity a model finds across many examples.

balance Whether the data has roughly fair amounts of each category.

A solution is data plus a model

When people say an app 'decides', they often picture a programmer writing every rule by hand.

Machine-learning

solutions do not work that way. They build a model by studying data - lots of examples - and capturing the patterns inside.

Because the pattern comes from the examples, the examples quietly control the answer. Feed the model different

data and it learns a different pattern, so it gives a different outcome, even with the very same code.

Amount and balance both matter

Data influences an outcome in two ways. First, the amount: if the model saw 100 sunny photos and only 3 rainy ones, it treats sunny as normal and stumbles on rain.

Second, the balance: adding more of what the model already has changes little, but adding examples of a

rare or missing category can move the outcome a lot. A data detective looks for the gap.

Let's try it - hands-on

Your teacher will run a fruit-ripeness sorter on the board. Your job is not to run it - it is to trace why it answers the way

it does, using the data list on your card.

1. Read the data list on your card and say what the sorter mostly saw.
2. Watch one fruit go through and note the outcome.
3. Draw an arrow from the data batch you think caused that outcome.
4. Write: 'The outcome is ____ because the data mostly showed ____.'
5. Check your reason against a second fruit - does the story still hold?

Think about it

- Which data batch had the biggest effect on the outcome, and how do you know?
- If you could add only one batch of new data, which would you add to make the sorter fairer?

My reflection

- Write one sentence explaining how data shaped today's outcome.
- Note one thing you would check before trusting a solution's answer.

Week 3 - Lab | Trace-the-Data Investigation (Lab)

In the Core lesson you predicted how a data change would move the outcome. A prediction is only a guess until

you test it. Today you run a fair test: change one thing, watch the outcome, and let the evidence tell you which

data matters most.

I can...

- I can run a controlled data change and record the outcome before and after.
- I can compare data changes and identify which one moved the outcome the most.
- I can explain, using my recorded evidence, why that change had the biggest effect.

Words to know

Word What it means

controlled change Changing one thing at a time so you know what caused the result.

before/after The outcome measured just before and just after a single change.

shift How far the outcome moved because of a change.

influential Having a large effect on the outcome.

verify To re-test and confirm an effect is real, not a one-off.

Change one thing at a time

To find out how much a data batch matters, you must isolate it. Measure the outcome before, change only that one

batch, and measure again. The difference is the influence of that batch.

If you change two batches at once and the outcome moves, you cannot tell which batch did it. A

fair test changes one

variable at a time - the same rule scientists use in any experiment.

Evidence beats opinion

Your log is your evidence. Without a written before-and-after, you are relying on memory, and memory blurs across

many swaps.

Once you have run several swaps, the most influential batch is the one with the biggest shift.

Verify it by re-running on

a fresh fruit - if the effect repeats, you have proof, not a coincidence.

Let's try it - hands-on

Work in threes with roles: swapper, recorder, checker. You will run three single data swaps, log each outcome, and

find the most influential change.

1. Write your Core prediction at the top of the log.
2. Record the BEFORE outcome for your test fruit.
3. Swap in one data batch and record the AFTER outcome.
4. Note the shift: none, a little, or a lot.
5. Repeat for two more single swaps.
6. Rank the swaps and star the most influential.
7. Verify the winner on a fresh fruit and write your conclusion.

Think about it

- Did the evidence match your prediction? Where did it surprise you?
- Why might a small batch beat a large one for influence?

My reflection

- Write your evidence-based conclusion about the most influential data change.
- Note one thing a fair test taught you that a guess could not.

Week 4 - Core | Misinformation Risks and Verification

An AI can write an answer that sounds completely sure and is completely wrong. It is built to sound right, not to be right. Today you learn to spot the exact risk in an output and pair it with a check that makes it trustworthy -

before you ever share it.

I can...

- I can explain a specific way an AI output could be wrong and spread misinformation.
- I can propose a concrete verification step and say why it makes the claim more trustworthy.
- I can move from 'it might be wrong' to naming the exact risk and the exact check.

Words to know

Word What it means

misinformation False information that spreads, whether or not anyone meant to deceive.

claim A specific statement that could be true or false, like a number or a date.

verify To confirm a claim against a trusted, independent source.

trusted source A source kept accurate by people who check facts, like an official website.

layered check Verifying with more than one independent source for extra reliability.

Why fluent is not the same as true

An AI language system predicts the next likely word, over and over, to build fluent text. It is optimised to sound

plausible, not to check facts, so it can state a wrong number as confidently as a right one.

That is what makes AI output a misinformation risk. The output gives no signal for which parts are reliable, so a made-

up statistic can look exactly like a real one and spread when people repeat it.

Name the risk, then check it

Good evaluation is specific. Instead of 'this might be wrong', say exactly what is at risk: the number, the name, the date, or the source.

Then pair that risk with a concrete check against a trusted source that is independent of the AI.

If you can, layer two

sources - one source can be wrong too.

Let's try it - hands-on

You will study three curated AI outputs. For each, name the exact risk and propose a concrete check. Your teacher

will verify one live against a trusted source.

1. Read each output and underline the exact claim you doubt.
2. Name the risk type: number, date, name, or source.
3. Choose a trusted source that could settle it.
4. Write the check and why it helps.
5. Watch one live verification and record the verdict.

Think about it

- Which output was the hardest to spot a risk in, and why?
- Why is verifying before sharing part of using AI responsibly?

My reflection

- Write the strongest risk-and-verify pair you made today.

- Note one habit you will use before sharing any AI answer.

Week 4 - Lab | Verify and De-risk Workshop (Lab)

One suspect output is easy to check. Ten is where you need a system. Today you run a verification routine on a

whole batch, reach an evidence-backed verdict on each, and then judge which checking strategy you would

actually trust.

I can...

- I can run a verification routine on several suspect outputs and record a clear verdict for each.
- I can compare two verification strategies and judge which is more reliable, with a reason.
- I can back my verdicts with evidence from a trusted source.

Words to know

Word What it means

routine A fixed set of steps applied the same way to every output.

verdict Your conclusion on a claim: verified, false, or unclear.

evidence The trusted-source information that backs your verdict.

independent sources Sources that do not copy each other, so they are unlikely to share the same error.

reliable A method you can trust to give the right result repeatedly.

A routine keeps you fair

When you check many outputs, it is tempting to judge each one by how confident it sounds. A

verification routine

stops that: the same fixed steps apply to every output, so a slick output gets no easy pass.

Our routine has four steps - isolate the claim, pick an independent trusted source, compare, and record a verdict. The

verdict can be verified, false, or unclear, and 'unclear' is an honest answer when the evidence is not there.

Not all checks are equally reliable

A single-source check is quick, but if that one source is wrong or out of date, your verdict is wrong too. A two-source

cross-check is slower but far harder to fool, because two independent sources rarely share the same mistake.

Judging which strategy is more reliable means weighing speed against confidence and asking a sharp question: could

these sources fail together? The more independent they are, the more you can trust their agreement.

Let's try it - hands-on

Work in groups with roles: claim-isolator, source-checker, verdict-recorder, reporter. Run the routine on the whole

batch, then compare two strategies on a tricky output.

1. Copy the four routine steps onto your sheet.
2. For each output, isolate the claim and pick a trusted source.
3. Compare and record a verdict with its evidence.
4. Choose one tricky output and check it single-source, then two-source.

5. Judge which strategy is more reliable and why.
6. For one false output, write the harm and the check that de-risks it.

Think about it

- Which verdict was hardest to reach, and what made it hard?
- When is the slower two-source check worth the extra time?

My reflection

- State which verification strategy you would use and the trade-off you accept.
- Note one responsible-use habit this workshop reinforced.

Week 5 - Core | Trustworthy AI Helper - Design & Test the Data (Part 1 Core)

Over four lessons (two weeks, Core and Lab) you will build a Trustworthy AI Helper and prove it can be trusted.

Today you design it and test how the data quality changes its output.

I can...

- I can design a helper with a clear task, a structured prompt, and a data set.
- I can test how data quality changes the output and explain why.

Words to know

Word What it means

Helper A small AI solution that completes a task using a prompt and data.

Structured prompt A precise instruction: goal, details, and the form of the answer.

Data quality How accurate, balanced, and complete the data is.

A helper has three parts

A helper is a small solution: a task, a structured prompt, and data.

Its output comes from the data and the prompt, so if the data is poor, the output is poor - even with a good prompt.

Test the data quality

Feed your helper clean data, then noisy data, and compare the outputs.

This shows how much the data quality matters. Poorer data usually gives a poorer, less fair output.

Let's try it - hands-on

Design your helper, then test clean vs noisy data.

1. Write the task.
2. Write a structured prompt.
3. Prepare the data.
4. Run on clean data; note the output.
5. Run on noisy data; note the output.
6. Explain the difference.

Think about it

- Why is a good prompt not enough on its own?
- How did the noisy data change your output?
- What will you build in the Lab?

My reflection

- My helper's task is ...
- Data quality mattered because ...
- In the Lab I will build ...

Week 5 - Lab | Trustworthy AI Helper - Build & Refine the Prompt (Part 1 Lab)

Now build it! Today you turn your design into a working helper and make it better by refining the prompt, one

change at a time.

I can...

- I can build a working prototype of my helper.
- I can refine the prompt across versions and explain what improved.

Words to know

Word What it means

Prototype A first working version you can test and improve.

Iterate To improve something step by step, testing each change.

Version One attempt at your prompt that you can compare with the next.

Build, run, refine

Run your prompt on the improved data and read the output.

Judge it against your task. Then change one part of the prompt and run it again.

Changing one thing at a time tells you what actually helped.

Let's try it - hands-on

Build your prototype and refine the prompt across versions.

1. Run prompt v1 on your data.
2. Judge it against your task.
3. Change one part; run again.
4. Log the change and its effect.
5. Repeat until it meets the task.
6. Record your best version.

Think about it

- Which change improved your output the most?
- Why refine one change at a time?
- Is your best prompt precise or just long?

My reflection

- My best prompt version is ...
- The change that helped most was ... because ...
- Next I will check ...

Week 6 - Core | Trustworthy AI Helper - Trace the Data & Plan Checks (Part 2 Core)

Your helper works - now make sure it can be trusted. Today you trace how the data shapes its outcome and plan

how you will verify it and check its risks.

I can...

- I can explain how the data shapes my helper's outcome and predict a change.
- I can plan a verification and a risk check for my helper.

Words to know

Word What it means

Trace To follow how the data leads to the outcome.

Verify To check a claim against a trusted, independent source.

Risk A way the output could be wrong, unsafe, biased, or a privacy problem.

The data drives the outcome

Change the data and the outcome changes. So you can trace which data drives which part of the output.

Knowing this helps you trust - or fix - your helper.

Plan your checks

Verification checks a claim against a trusted source - not by asking the AI again.

A risk check asks: could this be wrong, unsafe, biased, or a privacy problem, and how would I reduce it?

Let's try it - hands-on

Trace the data's influence, then plan your verification and risk check.

1. Name the data that drives one part of your output.
2. Predict how a data change would alter the outcome.
3. Choose a claim to verify against a trusted source.
4. Name one risk and a way to reduce it.
5. Write your check plan.

Think about it

- How does your data shape your outcome?
- What is the most trustworthy way to verify a claim?
- What is your helper's biggest risk?

My reflection

- The data that most shapes my outcome is ...
- I will verify ... against ...
- My top risk is ... and I will reduce it by ...

Week 6 - Lab | Trustworthy AI Helper - Verify, De-risk & Showcase (Part 2 Lab)

Time to prove it. Today you verify your helper, reduce a risk, and show that it can be trusted - then present it and

reflect on everything you have learned.

I can...

- I can verify my helper's output and reduce a risk, refining if needed.
- I can present my helper and reflect on my thinking.

Words to know

Word What it means

Mitigation A change that reduces a risk.

Trustworthy Something you can rely on because you have checked it.

Evidence The proof you give for a claim or a judgement.

Verify, then de-risk

Verify a claim in your output against a trusted source. If it fails, refine and re-verify.

Run your risk check and apply a mitigation - then re-check that it worked.

Let's try it - hands-on

Verify, de-risk, then showcase and reflect.

1. Verify your claim against a trusted source.
2. If it fails, refine and re-verify.
3. Run your risk check; apply one mitigation.
4. Confirm your helper can be trusted, with evidence.

5. Prepare a one-minute showcase.

6. Present and reflect.

Think about it

- What did your verification find?
- What risk did you reduce, and how?
- Which of the four skills was hardest?

My reflection

- My helper is trustworthy because ...
- The check that mattered most was ...
- The skill I found hardest was ... and I worked through it by ...

Figure list (for the graphic designer)

Every visual to be produced, in order. Full briefs appear at the point of use in the lessons above.

Figure ID Lesson One-line description

G6-W1-01 Week 1 - Core Trace the chain: a data problem becomes an output problem.

G6-W1-02 Week 1 - Core Bigger is not always better - quality can outweigh quantity.

G6-W1L-01 Week 1 - Lab Change only the data - keep everything else the same.

G6-W1L-02 Week 1 - Lab Order the datasets and defend the hardest placements.

G6-W2-01 Week 2 - Core Refining is a loop, not a lucky guess.

G6-W2-02 Week 2 - Core One targeted change turns a vague output into a focused one.

G6-W2L-01 Week 2 - Lab Iterate one change at a time and log each effect.

G6-W2L-02 Week 2 - Lab The best version is the best balance, not the last one.

G6-W3-01 Week 3 - Core Trace the arrows: the data shapes the model, and the model shapes the outcome.

G6-W3-02 Week 3 - Core A gap in the data shows up as a weakness in the outcome.

G6-W3-04 Week 3 - Lab The gap between before and after is the influence of that one change.

G6-W3-05 Week 3 - Lab The biggest shift wins - and then you verify it.

G6-W4-01 Week 4 - Core Sounding true and being true are not the same thing.

G6-W4-02 Week 4 - Core Name which type of claim is at risk before you check it.

G6-W4-04 Week 4 - Lab Same steps, every time - that is what makes it fair.

G6-W4-05 Week 4 - Lab Weigh speed against reliability before you choose.

G6-W5-01 Week 5 - Core A helper needs a clear task, a structured prompt, and good data.

G6-W5-02 Week 5 - Lab Refine one change at a time so you know what helped.

G6-W6-01 Week 6 - Core Trace the data, then verify and risk-check before you trust.

G6-W6-02 Week 6 - Lab Rank risks by likelihood and impact, and mitigate the biggest first.

Word bank

Rule-based system - follows fixed IF-THEN rules a person wrote.

Learning system (model) - learns patterns from many examples (training data) instead of fixed rules.

Training data - the examples a learning system learns from.

Prompt - the words we give an AI system to tell it what to do.

Data quality / noisy data - how complete and correct data is; noisy data is messy or has errors.

Bias - when a model is unfair because its training data was unbalanced or noisy.

Trustworthy AI - an AI whose answers are reliable, fair and safe to use.

Personal information - facts about a real person that we keep private.

Two kinds of AI

A rule-based system follows fixed rules a person wrote (IF ... THEN ...). A learning system (model) is

different: it learns patterns from lots of examples called training data. If the training data is unbalanced or

messy, the model can be biased - even though no one wrote a wrong rule.

Fair or unfair? A quick story

We train a helper with many cat photos but very few dogs. Shown a new dog, it says 'cat'. No one wrote a

wrong rule - the training data was unbalanced, so the model learned a biased pattern.

Balanced, clean

training data gives fairer, more accurate answers.

My check

- Is the training data balanced?
- Is the data clean or noisy?
- Did I test different kinds of examples?
- Who might be left out or treated unfairly?
- What one change to the data would make it more trustworthy?

Keep it private

We never type a real name, photo, home or age into a tool. We use made-up (synthetic) data only. Once

information goes into a system it is hard to take back.

What changed in version 2.0

This book was updated to match your Teacher Pack: a word bank, a data-quality story, a simple rule-based

vs learning explanation, a trustworthy-AI check, and a privacy reminder were added. The success criteria

here line up with the checks in the Teacher Pack.

STUDENT WORKSHEETS

Grade 6 (Year 7) | Phase 3 / Cycle 2 | AI Solution Builder

Bridging Program - Building Trustworthy AI

Transition-Year readiness | Term 1, Weeks 1-6 (Core + Lab)

Field Detail

Program Grade 6 Bridging (Transition Year)

Readiness bar Prior-grade Advanced (Grade 5; Critical Evaluation from Grade 4)

Readiness bar note 'Prior-grade Advanced' means students are baselined against the Grade 5 Advanced-

tier outcomes for AI Conceptual Understanding, AI Solution Design & Development, and Ethics & Governance; the exception is Critical Evaluation & Informed Interaction, which is probed at the Grade 4 Advanced bar because Critical Evaluation paused (was not taught) at Grade 5. See the Start-of-Year Diagnostic Kit for the full baseline procedure.

Delivery model 2 x 45-minute periods per week - a Core lesson + a paired Lab lesson each week

Structure Wk 1-4 strand catch-up (Core + Lab) | Wk 5-6 integrated hands-on mastery task (Core + Lab)

Diagnostic Start-of-year baseline, administered outside teaching hours

Lessons in this pack 12 lessons (6 Core + 6 Lab) + start-of-year diagnostic kit

Week 1 - Core - Worksheet | Data Quality and AI Outputs

Name: _____ Class: _____ Date: _____

Trace the chain

For your dataset card, fill each box: the data problem, the learned pattern problem, and the predicted output

problem. Use an arrow between each.

The data problem is ____.

So the model learns ____.

So the output will ____.

Quality or quantity?

Read four short scenarios and decide whether the dataset's problem is about quality or quantity.

Justify each choice

in one sentence.

This is a ____ problem because ____.

Prediction vs reveal log

Record your prediction before the reveal, then what the outputs actually showed, then whether you were right.

Dataset version My predicted output error Actual revealed output Match? (Y/N) and why

Exit ticket

In one or two sentences, explain how making a dataset noisier would change an AI's output, and why.

Challenge (stretch)

Design your own dataset card with a hidden quality problem, swap with a partner, and see if they can predict the

exact output mistake it would cause.

Week 1 - Lab - Worksheet | Data Quality Investigation (Lab)

Name: _____ Class: _____ Date: _____

Change from baseline

For each degraded version, describe exactly how the output changed from the clean baseline, and name the data

flaw responsible.

Compared to the baseline, this output ____.

The flaw responsible is ____.

Defend the ranking

Write your full best-to-worst ranking and justify the two placements you found hardest to separate, using output

evidence.

My order is ____.

The hardest pair was ____ because the output ____.

Investigation log

Record each dataset version, its output change, its data flaw, and its dimension scores.

Version Output change vs

baseline Data flaw Accuracy/Complete/Consistent score

Exit ticket

Which single data flaw would you fix first to improve the output the most, and why?

Challenge (stretch)

Invent a fifth dataset that combines two flaws. Predict where it lands on the ranking line and justify your prediction

with the evidence you gathered.

Week 2 - Core - Worksheet | Refining Prompts for Better Outputs

Name: _____ Class: _____ Date: _____

Name and target

For the weak prompt given, name one weakness and write a v2 that targets only that weakness.

The weakness is ____.

My v2 prompt is ____.

Say why it worked

After the reveal, explain the causal reason your change improved the output.

My change was ____.

It improved the output because ____.

Refinement log

Record your v1, the weakness, your v2 change, the predicted effect, and the observed effect.

v1 prompt Weakness named v2 change Predicted vs observed effect

Exit ticket

Explain in one or two sentences which prompt change improved your output and the reason it worked.

Challenge (stretch)

Write a v3 that improves the output on a second goal (for example, making it both clearer and more accurate) without losing the first improvement.

Week 2 - Lab - Worksheet | Prompt Refinement Workshop (Lab)

Name: _____ Class: _____ Date: _____

Iteration log entries

For each version, write the one change you made, your reason, and the effect on the output.

Version ____ change: ____.

Reason: ____.

Effect: ____.

The trade-off

Describe one point where improving clarity hurt accuracy (or the reverse), then choose your best-balanced version and justify it.

Improving ____ hurt ____ when I ____.

My best-balanced version is ____ because ____.

Workshop iteration log

Record each version's change, reason, clarity effect, and accuracy effect.

Version Single change Reason Effect on clarity / accuracy

Exit ticket

Which version was your best balance of clarity and accuracy, and what did choosing it cost you?

Challenge (stretch)

Add a third goal, tone, and describe in a few lines how balancing three goals is harder than balancing two, using your log.

Week 3 - Core - Worksheet | How Data Shapes a Solution's Outcome

Name: _____ Class: _____ Date: _____

Trace the data

Use the worked solution card. Draw an arrow from the data batch to the outcome, then explain the link in one sentence.

The outcome is ____

because the data mostly showed ____

What if the data changed?

Predict how the outcome changes if we add 50 green unripe date photos. Point to the exact data that causes your predicted change.

I predict the outcome will ____

because the added data ____

Data change log

For each data change, record what changed and how the outcome moved.

Data change Predicted outcome Which data caused it Confidence (high/med/low)

Exit ticket

Explain in two sentences how the data shaped the sorter's outcome, and predict one data change that would flip it.

Challenge (stretch)

Design the smallest possible data change that would flip the sorter from 'ripe' to 'unripe' on a brown date, and explain why so small a change can have such a big effect.

Week 3 - Lab - Worksheet | Trace-the-Data Investigation (Lab)

Name: _____ Class: _____ Date: _____

Run three single swaps

For each swap, record the before outcome, the after outcome, and the size of the shift. Change only one batch at a time.

Before: ____

After: ____

Shift: ____

Name and verify the winner

State which swap was most influential, justify it from your log, and describe how you verified it on a fresh fruit.

The most influential swap was ____

My log shows ____

I verified it by ____

Investigation log

Log every single swap and its effect so you can rank them fairly.

Swap (one batch) Before outcome After outcome Shift size Rank

Exit ticket

State the most influential data change you found and give the recorded evidence that proves it.

Challenge (stretch)

Design a data swap you predict will produce zero outcome change, then test it. Explain why some data has no influence on this solution.

Week 4 - Core - Worksheet | Misinformation Risks and Verification

Name: _____ Class: _____ Date: _____

Name the risk

For each curated output, underline the doubtful claim and name the risk type and why it is risky.

The risky claim is ____

The risk type is ____

It is risky because ____

Propose the check

For one output, write a concrete verification strategy naming a trusted source and explaining why it reduces the risk.

I would check it by ____

using the trusted source ____

This helps because ____

Risk-and-verify log

Record the risk and matching check for each output you evaluated.

Output Risk (specific claim) Verification step Trusted source Why it helps

Exit ticket

Name one misinformation risk in an AI output and propose one concrete verification strategy, with a reason it helps.

Challenge (stretch)

Design a layered verification strategy using two independent trusted sources for one claim, and explain why two sources are more reliable than one.

Week 4 - Lab - Worksheet | Verify and De-risk Workshop (Lab)

Name: _____ Class: _____ Date: _____

Run the routine on the batch

Apply the four-step routine to each suspect output. Record a verdict and the trusted-source evidence for each.

The claim is ____

My verdict is ____

Evidence: ____

Compare two strategies

Verify one tricky output single-source, then two-source. Judge which is more reliable and explain using source independence.

Single-source result: ____

Two-source result: ____

The more reliable strategy is ____ because ____

Verification workshop log

Log every suspect output, your verdict, the evidence, and the strategy used.

Output Claim checked Verdict Evidence (trusted source) Strategy used

Exit ticket

Give a verdict on one output with its evidence, and judge which verification strategy is more reliable and why.

Challenge (stretch)

Design a three-layer verification for a high-stakes claim about a UAE record, and explain when the extra effort is worth it and when it is not.

Week 5 - Core - Worksheet | Trustworthy AI Helper - Design & Test the Data (Part 1 Core)

Name: _____ Class: _____ Date: _____

Task 1: Design your helper

Write the task, a structured prompt, and the data.

Task: ...

Structured prompt (goal | details | form): ...

Data: ...

Task 2: Data-quality test

Record the output on clean data and on noisy data, and explain the difference.

Clean-data output: ...

Noisy-data output: ...

The difference is because ...

Trustworthy AI Helper Log - Part 1 (design + data test)

Record your design and your data-quality test.

Helper part My design Data-quality finding

Exit ticket

Write one sentence: why does your helper need good data as well as a good prompt?

Challenge (stretch)

Predict how much noise your data can take before the output becomes untrustworthy, and explain your reasoning.

Week 5 - Lab - Worksheet | Trustworthy AI Helper - Build & Refine the Prompt (Part 1 Lab)

Name: _____ Class: _____ Date: _____

Prompt refinement

Log each prompt version, the change you made, and its effect.

v1: ... | effect: ...

v2: ... | change: ... | effect: ...

Best version: ...

Prompt-refinement log

Record each version, the change, and the effect on the output.

Version Change made Effect on the output

Exit ticket

Name your best prompt version and the one change that improved it most, with a reason.

Challenge (stretch)

Refine your prompt for accuracy AND clarity at once. Explain the trade-off you had to balance.

Week 6 - Core - Worksheet | Trustworthy AI Helper - Trace the Data & Plan Checks (Part 2 Core)

Name: _____ Class: _____ Date: _____

Trace the data

Explain how the data influences your outcome and predict a change.

The data ... drives the outcome because ...

If the data changed to ..., the outcome would ...

Plan your checks

Plan a verification and a risk check.

Claim to verify: ... | trusted source: ...

Risk: ... | how I will reduce it: ...

Trustworthy AI Helper Log - Part 2 (trace + check plan)

Record your data trace and your verification and risk plan.

Data -> outcome link Verification plan Risk + mitigation

Exit ticket

Explain how your data shapes your outcome, and name one claim you will verify and one risk you will reduce.

Challenge (stretch)

Find the single data change that would most alter your helper's outcome, and explain why it matters most.

Week 6 - Lab - Worksheet | Trustworthy AI Helper - Verify, De-risk & Showcase
(Part 2 Lab)

Name: _____ Class: _____ Date: _____

Verify & de-risk

Record your verification and your risk mitigation.

Claim verified: ... | source: ... | result: ...

Risk reduced: ... | how: ... | re-check: ...

Trust argument

Argue, with evidence, why your helper can be trusted.

My helper can be trusted because ...

Trustworthy AI Helper Log - Part 2 Lab (verify, de-risk, reflect)

Record your checks and reflect across the four skills.

Skill used What I did / found How trustworthy? (evidence)

Exit ticket

State whether your helper can be trusted and give two reasons (one verification, one risk mitigation).

Challenge (stretch)

Your helper is safe for one use but risky for another. Explain the difference and propose a rule for when it should and should not be used.

Word bank

Rule-based system - follows fixed IF-THEN rules a person wrote.

Learning system (model) - learns patterns from many examples (training data) instead of fixed rules.

Training data - the examples a learning system learns from.

Prompt - the words we give an AI system to tell it what to do.

Data quality / noisy data - how complete and correct data is; noisy data is messy or has errors.

Bias - when a model is unfair because its training data was unbalanced or noisy.

Trustworthy AI - an AI whose answers are reliable, fair and safe to use.

Personal information - facts about a real person that we keep private.

Two kinds of AI

A rule-based system follows fixed rules a person wrote (IF ... THEN ...). A learning system (model) is

different: it learns patterns from lots of examples called training data. If the training data is unbalanced or

messy, the model can be biased - even though no one wrote a wrong rule.

Fair or unfair? A quick story

We train a helper with many cat photos but very few dogs. Shown a new dog, it says 'cat'. No one wrote a

wrong rule - the training data was unbalanced, so the model learned a biased pattern.

Balanced, clean

training data gives fairer, more accurate answers.

My check

- Is the training data balanced?
- Is the data clean or noisy?
- Did I test different kinds of examples?
- Who might be left out or treated unfairly?
- What one change to the data would make it more trustworthy?

Keep it private

We never type a real name, photo, home or age into a tool. We use made-up (synthetic) data only. Once

information goes into a system it is hard to take back.

TEACHER PACK

Grade 6 (Year 7) | Phase 3 / Cycle 2 | AI Solution Builder

Bridging Program - Building Trustworthy AI

Transition-Year readiness | Term 1, Weeks 1-6 (Core + Lab)

Field Detail

Program Grade 6 Bridging (Transition Year)

Readiness bar Prior-grade Advanced (Grade 5; Critical Evaluation from Grade 4)

Readiness bar note 'Prior-grade Advanced' means students are baselined against the Grade 5 Advanced-

tier outcomes for AI Conceptual Understanding, AI Solution Design & Development, and Ethics & Governance; the exception is Critical Evaluation & Informed Interaction, which is probed at the Grade 4 Advanced bar because Critical Evaluation paused (was not taught) at Grade 5. See the Start-of-Year Diagnostic Kit for the full baseline procedure.

Delivery model 2 x 45-minute periods per week - a Core lesson + a paired Lab lesson each week

Structure Wk 1-4 strand catch-up (Core + Lab) | Wk 5-6 integrated hands-on mastery task (Core + Lab)

Diagnostic Start-of-year baseline, administered outside teaching hours

Lessons in this pack 12 lessons (6 Core + 6 Lab) + start-of-year diagnostic kit

1. About this Teacher Pack

This pack contains everything needed to teach the Grade 6 bridging program: a start-of-year diagnostic kit and

six weeks of lessons that rebuild the required prior-grade Advanced skills before the main course begins. Grade

6 opens Phase 3 (AI Solution Builder) and steps up from Grade 5: students reason about data quality, refine

prompts, trace how data shapes a solution's outcome, and verify and de-risk outputs. Each week is delivered as

a Core lesson (teaches the outcome) and a paired Lab lesson (deeper hands-on work on the same outcome).

Weeks 1-4 rebuild one strand each; Weeks 5-6 integrate them in the mastery task, 'Trustworthy AI Helper'.

How to use this pack

- Run the start-of-year diagnostic first (outside teaching hours) and record each student's level per strand.
- Teach each week as Core (period 1) then Lab (period 2) on the same outcome; use the 10-minute

Prerequisite Primer to rebuild any flagged rung.

- Keep all AI use teacher-mediated and account-free; prefer a board demo or simulated-AI cards; never use personal data.
- Teach Weeks 5-6 as the four-lesson integrated mastery task; use it to confirm readiness.
- Pair this Teacher Pack with the Student Lesson Content and Student Worksheets (same order).
- Any strand still 'not-yet' after bridging is closed just-in-time via the Primer in the relevant Week 7+ lesson.
- Strand-naming note: the ethics strand is labelled 'Responsible AI Use' (Grade 4), 'Ethics & Responsible Use' (Grade 5), and 'Ethics & Governance' (Grade 6); the data strand is labelled 'Data Awareness' (Grade 4), 'Data Literacy' (Grade 5), and folded into AI Conceptual Understanding from Grade 6 onward - these track the same strand across grades, not separate strands.

Safeguarding & responsible use

Grade 6 builds on AI interaction and adds solution-building. Keep all AI use teacher-mediated and account-free for students; prefer a board demonstration, a simulated-AI card routine, or curated output sets, or a teacher-managed school-approved child-safe assistant. Never enter personal, identifying, or sensitive data into any prompt. Build the evaluation, verification, and risk-check habit into every activity. Keep all activity supervised, use only school-approved devices, post nothing online, and follow ADEK safeguarding and responsible-use guidance. Contexts and names are UAE-appropriate and inclusive.

Materials & tools (UAE-accessible)

Tool / material What it is & why Access route Unplugged fallback

Data-quality & solution

cards (unplugged)

Clean/noisy data sets and worked-solution

cards for data-quality and data-influence

reasoning (W1, W3, W5).

School-printed; no

account/device.

Data on the board.

Simulated-AI card set

(unplugged)

A teacher-run card 'AI' returning prepared outputs, so students practise prompting/refining and evaluation safely (W2, W5-W6).

School-printed; account-free; teacher-run.

Teacher demonstration only.

Verification workshop pack (printable)

Suspect-output cards, a trusted reference sheet, and a verification flowchart (W4, W6).

School-printed. Read outputs aloud with a reference sheet.

Trustworthy AI Helper kit (printable)

Helper planner, clean/noisy data, prompt-refinement log, verification/risk planners, and the two-part Log (W5-W6).

School-printed. Plan and record on paper.

School-approved AI assistant (optional,

A child-safe, school-approved assistant used only by the teacher on the board for a live demo,

Teacher-managed class account only; no student

Use the simulated-AI card set or curated

teacher-managed) if permitted. sign-in; no personal data. output sets.

Start-of-Year Diagnostic Kit (administered outside teaching hours)

A short, hands-on baseline taken at the very start of the year, OUTSIDE timetabled teaching hours (about 30 minutes,

in small groups or stations). It establishes each student's level across the four strands so the Week 1-4 sessions

target real gaps. Critical Evaluation is probed at the Grade 4 Advanced bar (it paused at Grade 5). It is not graded and

no mark is reported to students.

Administration guide

- Run before bridging teaching begins, outside teaching hours.
- Set up four short stations, one per strand; students rotate in small groups.

- Any AI step is teacher-mediated or simulated, account-free, with no personal data.
- Accept spoken, shown, or written evidence; allow extra time and reading support.
- Record each student as mastery / partial / not-yet per strand on the class readiness tracker.
- Use results to set Week 1-4 focus groups and the just-in-time Prerequisite Primer plan for Weeks 7+.

Baseline probes (one per strand, at the prior-grade Advanced bar)

Strand Prior-grade Advanced skill Hands-on probe What mastery looks like

AI Conceptual

Understanding

Explains how data quality affects

AI outputs. [AIF·CU·G5·A]

Show clean vs noisy data outputs;
the student explains how quality
changed the output.

Explains the data-quality-to-
output link with an example.

AI Solution Design &

Development

Refines prompts to improve
output quality and explains why.

[AIF·SD·G5·A]

Give a weak prompt and its output;
the student refines it and explains
the improvement.

Improves the prompt and
gives a causal reason.

Critical Evaluation &

Informed Interaction

Explains how data influences the
outcome of a solution.

[AIF·CE·G4·A]

Show a worked solution; the
student explains how its data
shapes the outcome and predicts a
change.

Explains the data-to-outcome
link and predicts a change.

(Probed at Grade 4

Advanced.)

Ethics & Governance Explains misinformation risks and
proposes verification strategies.

[AIF·GE·G5·A]

Show a suspect output; the student
names a misinformation risk and a

verification strategy.

Names a plausible risk and a workable verification.

Record each student as mastery / partial / not-yet per strand on the class readiness tracker; no grade is reported to students.

Program at a glance

Lesson Strand Framework ref(s)

Week

1 -

Core

Data Quality and AI Outputs AI Conceptual Understanding AIF·CU·G5·A

Week

1 - Lab

Data Quality Investigation (Lab) AI Conceptual Understanding AIF·CU·G5·A

Week

2 -

Core

Refining Prompts for Better Outputs AI Solution Design & Development AIF·SD·G5·A

Week

2 - Lab

Prompt Refinement Workshop (Lab) AI Solution Design & Development AIF·SD·G5·A

Week

3 -

Core

How Data Shapes a Solution's Outcome Critical Evaluation & Informed Interaction

AIF·CE·G4·A

Week

3 - Lab

Trace-the-Data Investigation (Lab) Critical Evaluation & Informed Interaction

AIF·CE·G4·A

Week

4 -

Core

Misinformation Risks and Verification Ethics & Governance AIF·GE·G5·A

Week

4 - Lab

Verify and De-risk Workshop (Lab) Ethics & Governance AIF·GE·G5·A

Week

5 -

Core

Trustworthy AI Helper - Design & Test the Data
(Part 1 Core)

Integration - all four strands AIF·CU·G5·A AIF·SD·G5·A
AIF·CE·G4·A AIF·GE·G5·A

Week

5 - Lab

Trustworthy AI Helper - Build & Refine the Prompt

(Part 1 Lab)

Integration - all four strands AIF·CU·G5·A AIF·SD·G5·A

AIF·CE·G4·A AIF·GE·G5·A

Week

6 -

Core

Trustworthy AI Helper - Trace the Data & Plan

Checks (Part 2 Core)

Integration - all four strands AIF·CU·G5·A AIF·SD·G5·A

AIF·CE·G4·A AIF·GE·G5·A

Week

6 - Lab

Trustworthy AI Helper - Verify, De-risk & Showcase

(Part 2 Lab)

Integration - all four strands AIF·CU·G5·A AIF·SD·G5·A

AIF·CE·G4·A AIF·GE·G5·A

Week 1 - Core | Data Quality and AI Outputs

Strand: AI Conceptual Understanding | Framework ref(s): AIF·CU·G5·A | 1 x 45-minute period
(Core)

Learning outcome (framework): Explains how data quality affects AI outputs.

Objectives & success criteria

- Explain, with an example, how higher- or lower-quality data changes an AI output.
- Predict the effect on the output when the data quality is degraded or improved.

Student-facing success criteria:

- I can explain, using a clear example, how the quality of data changes what an AI produces.
- I can predict whether an output will get better or worse when the data is made cleaner or noisier, and say why.
- I can tell the difference between the amount of data and the quality of data.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In Grade 5 you learned that AI systems learn patterns from examples rather than following rules a person types

in. You compared good and poor training examples and noticed that messy examples confused a model. This lesson

rebuilds and deepens that skill: instead of only spotting messy data, you now reason from a specific data problem to the specific output problem it causes.

Activate: Show two class photos described in words: one caption is accurate and detailed, the other is blurry, mislabelled, and

half-missing. Ask which description would help a friend who has never seen the photo, and why. This surfaces the everyday idea

that the quality of the information in shapes the quality of the answer out.

Teach: Explain that an AI model learns from a dataset the way a student learns from worked examples. If the examples are wrong, biased, or inconsistent, the model learns the wrong pattern - not because it is broken, but because it faithfully learned from poor data. Introduce three quality problems: errors (labels are wrong), gaps (some cases are missing), and noise (irrelevant or inconsistent detail). Name the link out loud: garbage in, garbage out.

Apply: In pairs, students examine one curated dataset card showing a small labelled set (for example: dates-fruit images labelled 'date', 'dat', and 'DATE' inconsistently, and one image of a stone incorrectly labelled 'date'). They identify the specific data-quality problem and predict one specific output error it would cause.

Check: Cold-call two pairs: What is the quality problem, and what exact mistake will the output make? Listen for a named cause linked to a named effect, not just 'it will be bad'.

Critical & analytical thinking focus

Analyse - reason from data quality to output quality (Analyse | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

A machine learning model does not store rules that a programmer wrote; it adjusts itself to fit patterns in a training dataset.

Because of this, the dataset is effectively the model's teacher, and the model can only be as trustworthy as the examples it

was shown. Data quality has several dimensions that matter here: accuracy (are the labels correct), completeness (are important cases present), consistency (is the same thing labelled the same way every time), and representativeness (does the data reflect the real range of situations the model will meet). When any of these drop, the model still learns confidently -

it just learns the wrong thing, which is why a low-quality dataset can produce an output that is wrong and sounds sure of

itself. This is different from the amount of data: a huge dataset full of mislabelled or one-sided examples is still low quality,

so more is not automatically better. A small, clean, well-chosen dataset can beat a large, messy one. Understanding this

lets you predict failures before they happen - if you know the input has a gap, you can predict the output will be blind to that

gap. For Grade 6, the key mental model is a chain: data problem leads to learned pattern problem leads to output problem, and you can trace each link.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Starter

7 min

Display the two photo descriptions (accurate vs blurry/mislabelled). Pose the

Compare the two descriptions and decide

- Which description would help a stranger picture the photo, and hook question and take three quick responses. Introduce today's driving question: how exactly does the quality of data change what an AI produces? which would help someone reconstruct the photo.

Share reasons with a shoulder partner before responding.

why?

- What is the difference between having more information and having better information?

DirectTeach

12 min

Teach the model-as-learner idea and the three quality problems (errors, gaps, noise) plus the four quality dimensions. Draw the causal chain on the board: data problem to pattern problem to output problem. Model one full example aloud using the mislabelled 'date' dataset card.

Take structured notes using a chain organiser (three boxes joined by arrows).

Copy one worked example of the chain.

- Why can a model be confident and wrong at the same time?
- Which quality dimension is broken when the same fruit is labelled three different ways?

Guided

12 min

Hand each pair a curated dataset card. Guide them to identify the quality

problem, name the dimension, and predict the specific output error. Circulate and push for specificity: not 'wrong' but 'it will read the stone as a date'.

In pairs, complete the chain organiser for the card:

quality problem, dimension, predicted output error.

Prepare to justify the prediction.

- What exact mistake will this output make, and which input example causes it?
- If we fixed only the labels, would the problem fully go away?

Independent

9 min

Reveal the actual pre-generated outputs for clean vs degraded versions of the same dataset. Ask students to check their prediction against the reveal and write a one-sentence explanation of the difference.

Compare your prediction to the revealed outputs. Write an explanation that names the data cause and the output effect for both versions.

- Did the output fail where you predicted? What surprised you?
- Could you have known this from the data alone, before seeing the output?

Plenary

5 min

Collect two strong explanations and one common misconception ('more data always wins'). Restate the causal chain and the responsible-use point: we check data quality before we trust an output, and we never feed personal or identifying data into any tool.

Contribute or record a

peer's best explanation. Add
the responsible-use
reminder to notes.

- When is a bigger dataset still a worse dataset?
- Why should we always check the data before trusting the answer?

Anticipated misconceptions

Misconception How to address it

More data always makes an AI better. Volume and quality are different. A million mislabelled or one-sided

examples teach a million-times-wrong pattern. Show a small clean set beating a large messy set to make quality the deciding factor.

If the AI sounds confident, the data must have been good.

Confidence comes from the maths of fitting patterns, not from truth.

A model trained on wrong labels states wrong answers just as firmly.

Separate 'sounds sure' from 'is correct'.

Bad outputs mean the AI is broken. Usually the code works fine; it faithfully learned from poor data.

Reframe the fix as improving the dataset, not repairing a machine.

Hands-on activity guide

Overview: A short guided contrast: students trace one dataset card through the causal chain and check their prediction

against a revealed output pair. The deeper build happens in the Lab.

Grouping / timing: Pairs, mixed reading level, with a whole-class reveal.

Tool: Curated dataset cards and a pre-generated clean-vs-degraded output pair (teacher-mediated; no student accounts).

- Teacher displays outputs on the board or hands out printed output sets. Students never enter any prompt or data themselves.

Facilitation steps:

1. Read your dataset card and list what each example is labelled.
2. Name the quality problem (error, gap, or noise) and the dimension it breaks.
3. Predict one specific output error the model will make.
4. Watch the reveal of the clean and degraded outputs.
5. Write one sentence linking the data cause to the output effect.

Unplugged fallback: Unplugged: compare pre-generated outputs from a clean dataset and a noisy dataset on paper. Students

underline where the noisy-data output goes wrong and label the matching data problem.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Grade 5 Advanced: explains how data quality affects AI outputs

- Check: Data-quality contrast task: predict, reveal, explain the output difference

- Success criteria: Explains the data-quality to output link with an example and predicts the effect of a quality change.

- Evidence: Contrast task record / think-aloud.

- Feedback / adjustment: Contrasting cases for students stuck at 'more data = better'; secure students distinguish quality from quantity.

Differentiation

Representation: Provide the causal chain as a three-box visual, a spoken walkthrough, and a written worked example so the same idea arrives three ways.

Action & expression: Students may respond by speaking, annotating the card, or filling the chain organiser. Predictions can be drawn or written.

Engagement: Use familiar UAE contexts (dates, majlis photos, class lists) so the data feels real and the stakes are clear.

People of Determination: For students with fine-motor needs, provide pre-printed chain boxes and label stickers to place rather

than write. For attention needs, chunk the task into one card at a time with a checklist.

EAL: Give a bilingual term card for accuracy, completeness, consistency, noise. Provide sentence stems: 'The data problem is ____, so the output will ____.'

Stretch / next tier: Ask secure students to design a dataset card with two overlapping quality problems and predict how the errors combine.

Answer key / model responses

Task / item Model response Marking note

Identify the quality problem on the 'date' card.

The labels are inconsistent ('date', 'dat', 'DATE') and one label is an error (a stone labelled 'date').

Accept either consistency or accuracy named, but push for the specific mislabelled example.

Predict the output error. The model will sometimes read a small stone or pebble as a date, because it learned that a rounded brown object with that label counts as a date.

Look for a specific predicted mistake tied to a specific bad example.

Explain quality vs quantity.

Adding thousands more mislabelled photos would make the wrong pattern stronger, not weaker, so the dataset is worse despite being bigger.

Strong answers separate

volume from correctness.

Assessment rubric (observable look-fors)

Criterion Mastery Partial Not yet

Explains the data-to-output

link with an example

Traces a clear causal chain from

a named data problem to a

specific output error using an

example.

States that poor data causes

poor output but the example or

the link is vague.

Says the output is bad

without connecting it to a

data cause.

Predicts effect of a quality

change

Predicts a specific, correct

change in the output when data

is degraded or improved, and

Predicts a general direction of

change with weak justification.

Cannot predict, or predicts

unrelated to the data

change.

justifies it.

Resources & links

- Dataset quality cards (curated) - Printable cards showing small labelled sets with built-in

errors, gaps, and noise for the UAE

date example. (teacher-prepared)

- Clean vs degraded output set - Pre-generated output pairs the teacher reveals; keeps all tool

use teacher-mediated and

account-free. (teacher-prepared)

Week 1 - Lab | Data Quality Investigation (Lab)

Strand: AI Conceptual Understanding | Framework ref(s): AIF·CU·G5·A | 1 x 45-minute period

(Lab)

Learning outcome (framework): Explains how data quality affects AI outputs.

Objectives & success criteria

- Run a hands-on comparison of clean vs noisy data and record how the output changes.

- Rank data sets by expected output quality and justify the order.

Student-facing success criteria:

- I can run a fair comparison of clean and noisy data and record what changes in the output.
- I can rank several datasets from best to worst expected output and justify my ranking.
- I can use evidence from the comparison to explain the data-to-output link.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Core lesson you traced the causal chain from a data problem to an output problem and checked one

prediction against a reveal. In this Lab you run a fuller investigation: you compare several datasets, collect evidence, and

rank them. Same outcome, deeper hands-on work.

Activate: Re-show the Core causal chain (data problem to output problem) and ask one pair to restate it in their own words. Then

set the Lab challenge: today we do not just predict one case, we test several and rank them by evidence.

Teach: Introduce the idea of a fair test: to compare data quality honestly we change only the data and keep the task the same.

Explain a simple quality-ranking method - score each dataset on accuracy, completeness, and consistency, then order them.

Apply: Model scoring one dataset on the three dimensions and placing it on a ranking line, thinking aloud.

Check: Ask: what must we keep the same for the comparison to be fair? Listen for 'same task, only the data changes'.

Critical & analytical thinking focus

Evaluate - rank datasets by expected output quality using evidence (Evaluate | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

This Lab turns the Core idea into an investigation with controls. To compare data quality fairly, you hold the task and the

model fixed and vary only the dataset, so any change in the output can be traced to the data - this is the core logic of a

controlled comparison used across science and engineering. You will work with several versions of the same dataset that

differ in known ways: one clean, one with label errors, one with gaps, one with noise. By observing how the pre-generated

output shifts across versions, you build evidence rather than opinion. Ranking then forces a judgement: you weigh the

quality dimensions (accuracy, completeness, consistency, representativeness) against each other, because real datasets

are rarely bad in only one way. Sometimes two datasets are poor for different reasons and you must decide which failure

hurts the output more, which is a genuine evaluation. This mirrors what real AI builders do when they audit training data

before trusting a system. Throughout, all tool interaction stays teacher-mediated and account-free, and no personal or

identifying data is ever used - the datasets are about dates, weather, or shapes, never real people.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Starter

6 min

Re-activate the causal chain and introduce the fair-test rule. Show the four dataset versions students will investigate and name how each is degraded.

Restate the causal chain aloud and note the fair-test rule: keep the task the same, change only the data.

- What must stay the same for this comparison to be fair?
- Which single thing are we deliberately changing?

Investigate

16 min

Distribute the four output sets (clean, error, gaps, noise) generated from the same task. Guide teams to record the specific change in each output and match it to the data flaw. Circulate for evidence, not guesses.

In teams, examine each output set, record the exact change from the clean baseline, and label the responsible data flaw.

- How is this output different from the clean baseline, exactly?
- Which data flaw best explains that difference?

Rank

12 min

Introduce the ranking line and scoring on three dimensions. Support teams to place all four datasets from best to worst expected output and write a justification for the order, including the hardest two to separate.

Score each dataset, order

them on the ranking line,
and justify the placement of
the two that were hardest to
tell apart.

- Which flaw damaged the output most, and how do you know?
- When two datasets are both weak, how do you decide which is worse?

Plenary

6 min

Compare team rankings, surface disagreements, and resolve them with evidence. Reinforce that a defensible ranking cites the output changes, not hunches. Restate the responsible-use and no-personal-data reminders.

Defend or revise your ranking using evidence from the output sets. Record the class's agreed reasoning.

- Whose ranking is best supported by the evidence?
- What evidence would change your mind?

Anticipated misconceptions

Misconception How to address it

A ranking is just a matter of opinion. A defensible ranking cites specific output changes as evidence. Show two rankings and ask which is better supported to make evidence the deciding factor.

If we change the data and the task at the same time, the test still works.

Then you cannot tell which change caused the effect. Model the fair-test rule: vary only the data.

Every data flaw hurts the output equally. Different flaws cause different and sometimes larger damage. Use

the evidence to show one flaw shifting the output more than another.

Hands-on activity guide

Overview: A substantial four-way investigation: teams compare outputs from clean, error, gap, and noise versions of one

dataset, gather evidence, and rank the datasets by expected output quality with justification.

Grouping / timing: Teams of three with defined roles (recorder, evidence-checker, ranker), rotated once.

Tool: Four pre-generated output sets from the same task with datasets that differ only in a known flaw (teacher-mediated, account-free). - Teacher provides printed or displayed output sets and the dataset descriptions. Students analyse; they do not run prompts or enter data.

Facilitation steps:

1. Read the clean-baseline output and note what a correct result looks like.
2. For each degraded version, record exactly how the output changed from the baseline.
3. Match each change to its data flaw (error, gap, or noise).
4. Score each dataset on accuracy, completeness, and consistency.
5. Place all four on a ranking line from best to worst expected output.
6. Write a justification for the order, focusing on the two hardest to separate.

Unplugged fallback: Unplugged: compare pre-generated outputs from clean vs noisy data on paper cards. Teams sort the cards

along a best-to-worst line and annotate each with the data flaw and its output effect.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Grade 5 Advanced: explains how data quality affects AI outputs

• Check: Four-way contrast plus ranking task with written justification

• Success criteria: Ranks datasets by expected output quality and justifies the order using evidence from the output changes.

• Evidence: Investigation log and ranking line with justification.

• Feedback / adjustment: For rankings without evidence, prompt 'point to the output change that proves it';

extend secure teams to weigh two flaws against each other.

Differentiation

Representation: Present each dataset version as a description, a sample, and its output so students can enter through whichever mode suits them.

Action & expression: Teams can record findings as annotations, a table, or a card sort. Ranking can be a physical card line or a written list.

Engagement: Team roles give everyone a job; the ranking debate raises the stakes and invites argument from evidence.

People of Determination: Provide movable cards for students who find writing effortful so they can rank by arranging. Offer a quiet role (evidence-checker) for students who prefer less verbal load.

EAL: Supply a scoring grid with icons for the three dimensions and stems: 'Dataset ____ ranks ____ because the output ____.'

Stretch / next tier: Ask top teams to create a fifth dataset with two combined flaws and predict where it lands on the ranking, then justify.

Answer key / model responses

Task / item Model response Marking note

Describe how the error-version output differs from the baseline.

The error version confidently mislabels some items because its training labels were wrong, so it repeats the wrong label as if it were correct.

Look for a specific output difference tied to the label error.

Rank the four datasets. Clean best, then noise, then gaps, then error worst, because wrong labels teach an actively false pattern while noise only adds confusion.

Accept other defensible orders if the justification cites output evidence.

Explain the fair-test rule. We kept the task and model the same and changed only the dataset, so any output change must come from the data.

Reward explicit mention of holding variables constant.

Assessment rubric (observable look-fors)

Criterion Mastery Partial Not yet

Collects evidence from the comparison

Records specific, accurate output changes for each version and links each to its data flaw.

Records some changes but links are partial or a version is missed.

Records vague impressions with no link to a data flaw.

Ranks datasets with justification

Produces a defensible full ranking justified with output evidence, including the hardest pair.

Ranks most datasets with partial justification.

Ranks by opinion with no evidence.

Resources & links

- Four-version output packs - Pre-generated outputs from clean, error, gap, and noise datasets on the same task. (teacher-prepared)

- Ranking line and scoring grid - A printed best-to-worst line and a three-dimension scoring grid for teams. (teacher-prepared)

Week 2 - Core | Refining Prompts for Better Outputs

Strand: AI Solution Design & Development | Framework ref(s): AIF·SD·G5·A | 1 x 45-minute period (Core)

Learning outcome (framework): Refines prompts to improve output quality and explains why.

Objectives & success criteria

- Improve a weak prompt so the output better meets the goal.
- Explain which change improved the output and why it worked.

Student-facing success criteria:

- I can turn a weak prompt into a stronger one that better meets the goal.
- I can name the exact change I made and explain why it improved the output.
- I can judge an output against the goal instead of just accepting it.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In Grade 5 you learned that clearer instructions get better results, and you added detail to simple prompts. This

lesson deepens that into a design skill: you set a goal, judge the output against it, name the weakness, and make a targeted

change - then explain the causal reason it worked, rather than changing things at random.

Activate: Show a weak prompt ('write about AI in') and its vague pre-generated output. Ask: did the output meet a clear goal?

What goal was even given? This surfaces that a prompt without a clear goal cannot produce a focused answer.

Teach: Teach a refinement loop: state the goal, judge the output against it, name the specific weakness (missing audience, missing format, missing constraint, ambiguous task), make one targeted change, and re-judge. Stress that a good refinement is

reasoned, not random - you can say why the change should help before you see the result.

Apply: In pairs, students take the weak prompt and a stated goal, name one weakness, and write a v2 prompt that targets it.

Check: Ask two pairs to read v1 and v2 and name the single change and the reason it should improve the output.

Critical & analytical thinking focus

Create & evaluate - iterative prompt refinement (Create/Evaluate | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

A prompt is the instruction and context you give an AI system, and the quality of the output depends heavily on how

precisely that instruction defines the goal. Language models generate the most likely continuation of your words, so vague

or ambiguous prompts leave the model to guess your intent - and it often guesses wrong or too generally. Refining a prompt

means reducing that ambiguity by adding the information the model needs: the audience, the format, the length, the

constraints, and the exact task. The skill that lifts this from trial-and-error to design is reasoning about cause: before you

test a change, you predict why it should help, so you are testing a hypothesis rather than guessing. Evaluation is the other

half - you cannot improve an output unless you first judge it honestly against the goal, spotting what is missing or off-target.

Good refiners change one thing at a time so they can tell which change made the difference, just like a fair test. Importantly,

prompting well never means adding personal or identifying data; a strong prompt gets specificity from the task and

constraints, not from private information. For Grade 6, the model to internalise is a loop: goal, judge, name the weakness,

change, re-judge - repeated until the output meets the goal.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Starter

7 min

Display the weak prompt and its vague output. Ask students to judge it against an unstated goal, then reveal the intended goal and show the mismatch.

Judge the output: does it meet the goal? Identify what is missing once the goal is revealed.

- What goal was the prompt actually aiming at?
- What is missing from the prompt that made the output so general?

DirectTeach

12 min

Teach the refinement loop and the four common weaknesses (missing audience, format, constraint, or ambiguous task).

Model refining the weak prompt aloud,

Note the loop and the four weakness types. Watch the modelled prediction-then-check.

- Why should naming the audience change the output?
- How is a reasoned change different from a random change?
- predicting why the change should help

before revealing v2's output.

Guided

12 min

Give pairs a weak prompt and a clear goal.

Support them to name one weakness and write a v2 that targets it, stating the predicted effect. Reveal v2 outputs for each.

In pairs, write v2 targeting one named weakness and predict its effect. Compare with the revealed output.

- Which single change did you make and why?
- Did the output improve where you predicted it would?

Independent

9 min

Each student logs their v1 to v2 refinement with a reasoned justification.

Circulate to push justifications from 'it is better' to 'this change fixed this weakness because...'.
Complete a refinement log:

v1, weakness, change,
predicted reason, observed
effect.

- Can you name the change and its effect in one sentence?
- Would this change help a different prompt with the same weakness?

Plenary

5 min

Share two strong refinements and the reasoning behind them. Reinforce judging against the goal and the rule that specificity never means adding personal data. Note tool use stays teacher-mediated.

Contribute or record a strong refinement and its reason. Add the no-personal-data reminder.

- What made this refinement reasoned rather than lucky?
- How do we add detail to a prompt without adding private information?

Anticipated misconceptions

Misconception How to address it

A longer prompt is always a better prompt. Extra words that do not target a weakness just add clutter. Show a

shorter, sharper prompt beating a long rambling one to make targeting the point.

If the output improved, my change must have been the reason.

Changing several things at once hides the real cause. Model changing one thing at a time so the improvement can be attributed correctly.

Refining a prompt means adding lots of personal detail to make it specific.

Specificity comes from task, audience, and format - never private data. Reframe strong prompting as precise, not personal.

Hands-on activity guide

Overview: A short guided refinement: pairs turn one weak prompt into a stronger v2 targeting a named weakness, predict

the effect, and check against a revealed output. The full workshop is in the Lab.

Grouping / timing: Pairs, then individual logging.

Tool: Weak-prompt cards with stated goals and pre-generated v1/v2 outputs (teacher-mediated, account-free). - Teacher

reveals v2 outputs on the board after pairs predict. Students write prompts on paper; the teacher-only tool or curated set produces the outputs.

Facilitation steps:

1. Read the weak prompt and its stated goal.
2. Judge the v1 output against the goal and name one weakness.
3. Write a v2 prompt that targets that one weakness.
4. Predict how the output should change and why.
5. Compare with the revealed v2 output and log the result.

Unplugged fallback: Unplugged: use prompt cards and pre-generated outputs. Pairs match each v2 change to the output

improvement it caused, with no live tool needed.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Grade 5 Advanced: refines prompts to improve output quality and explains why

• Check: Prompt-refinement log: v1 to v2 with a reasoned justification

• Success criteria: Makes a refinement that measurably improves the output and explains the causal reason.

- Evidence: Before/after prompt log.
- Feedback / adjustment: For trial-and-error without reasoning, prompt 'name the change and its effect'; extend

to a two-variable prompt. Tool use stays teacher-mediated.

Differentiation

Representation: Show the refinement loop as a labelled cycle diagram, a spoken model, and a written v1/v2 example.

Action & expression: Students may draft v2 by writing, by choosing from prompt-part cards, or by dictating to a scribe.

Engagement: Use goals students care about (a poster for a school date festival, a message to a younger class) so the goal is meaningful.

People of Determination: Offer prompt-part cards (audience, format, constraint) to assemble a v2 without heavy writing. Provide the loop as a checklist for working memory support.

EAL: Bilingual glossary for audience, format, constraint, ambiguous. Stems: 'My change was _____. It should help because _____.'

Stretch / next tier: Ask secure students to write a v3 that improves the output on a second dimension without losing the first gain.

Answer key / model responses

Task / item Model response Marking note

Name the weakness in

'write about Al Ain'.

The task is ambiguous and has no audience, format, or length, so the model produces a vague general paragraph.

Accept any correctly named weakness that matches the output problem.

Write a targeted v2. 'Write a 4-sentence caption for Grade 3 students about why Al Ain's oasis matters, in simple words.' This adds audience, format, and length.

Look for a change that directly targets the named weakness.

Explain why the change worked.

Naming the audience and length removed the model's need to guess, so the output became focused and age-appropriate.

Reward a causal explanation, not just 'it is better'.

Assessment rubric (observable look-fors)

Criterion Mastery Partial Not yet

Improves the prompt toward the goal

v2 clearly targets a named

weakness and produces an output that better meets the goal.

v2 changes the prompt but only partly addresses the weakness or goal.

v2 changes little or moves away from the goal.

Explains why the change worked

Gives a clear causal reason linking the specific change to the output improvement.

Gives a general reason with a weak causal link.

States it is better with no reason.

Resources & links

- Weak-prompt refinement cards - Cards with a weak prompt, a clear goal, and space for v2, tied to UAE contexts. (teacher-prepared)

- v1/v2 output sets - Pre-generated before-and-after outputs the teacher reveals to keep use account-free. (teacher-prepared)

Week 2 - Lab | Prompt Refinement Workshop (Lab)

Strand: AI Solution Design & Development | Framework ref(s): AIF·SD·G5·A | 1 x 45-minute period (Lab)

Learning outcome (framework): Refines prompts to improve output quality and explains why.

Objectives & success criteria

- Iterate a prompt across several versions to raise output quality, logging each change and its effect.

- Refine for two goals at once (clarity and accuracy) and discuss the trade-off.

Student-facing success criteria:

- I can iterate a prompt across several versions and log each change and its effect.

- I can refine for two goals at once and explain the trade-off between them.

- I can use my log as evidence for which change helped the most.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Core lesson you made a single reasoned refinement (v1 to v2) and explained why it worked. In this Lab you

run a full workshop: you iterate across several versions, track each change against the goal, and then juggle two goals at

once, noticing when improving one costs the other.

Activate: Re-show the Core refinement loop and one strong v1-to-v2 example. Set the Lab

challenge: today we do not stop at v2 -

we iterate to v3 and v4, and we chase two goals at once.

Teach: Introduce iterative logging: each version records the change, the reason, and the effect, so the log becomes evidence. Then introduce two-goal refinement - clarity and accuracy - and the idea of a trade-off, where a change that helps one goal can hurt the other.

Apply: Model one iteration cycle and note it in the log, then point out where a change for clarity might reduce accuracy.

Check: Ask: what makes your log useful as evidence? Listen for 'it shows which change caused which effect'.

Critical & analytical thinking focus

Create & evaluate - iterate and manage a trade-off across prompt versions (Create/Evaluate | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

Iterative prompt refinement treats prompting as a design cycle rather than a single attempt: you make a change, observe

the effect, and feed what you learned into the next version. Keeping a version log turns this into evidence-based work - by

changing one thing per version you can attribute each output change to a cause, and the log reveals which single change

gave the biggest gain. The deeper Grade 6 idea is optimising for more than one goal at once.

Real outputs are often judged

on several dimensions - here, clarity (is it easy to understand) and accuracy (is it correct and on-topic) - and these can pull

in different directions. A change that makes an output shorter and clearer might drop an important accurate detail, creating

a trade-off you must weigh against your purpose. Recognising and managing trade-offs is a core systems-thinking and

design skill: there is rarely one perfect answer, only a best balance for the goal. As always, strong prompting gains its

specificity from task, audience, format, and constraints, never from personal or identifying data, and all tool interaction

stays teacher-mediated and account-free. By the end you can show, from your own log, not just that your prompt improved

but which change caused which improvement and what it cost.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Starter

6 min

Re-activate the refinement loop and introduce iterative logging and the two goals: clarity and accuracy. Show the workshop log template.

Recall the loop and set up the log. Note the two goals

to be balanced.

- Why keep a version log instead of just remembering?
- What are our two goals today, and could they ever conflict?

Iterate

18 min

Give each team a weak prompt and goal. Teams iterate v1 to v4, changing one thing per version, logging the change, reason, and revealed effect. Circulate to enforce

In teams, iterate the prompt across up to four versions, logging each change, its reason, and the revealed

- Which version gave the biggest jump in quality, and why?
 - How do you know that change, not another, caused it?
- one-change-at-a-time and reasoned prediction.

output effect.

Two goals

12 min

Introduce the second goal so teams now refine for clarity and accuracy together. Support them to spot a trade-off where improving one costs the other and to choose a balanced version for the purpose.

Refine for both goals, identify a trade-off in the log, and choose the best-balanced version, justifying it against the purpose.

- Where did helping clarity hurt accuracy, or the reverse?
- Which version is the best balance for this purpose, and why?

Plenary

5 min

Compare teams' best versions and their trade-off decisions. Reinforce that a strong justification cites the log. Restate

no-personal-data and teacher-mediated-tool reminders.

Present your best-balanced version and defend it using your log. Record the class's insight on trade-offs.

- Whose balance is best justified by their log?

- What did managing two goals teach you about design choices?

Anticipated misconceptions

Misconception How to address it

You can improve every goal at once with no cost. Goals often trade off; improving clarity can drop an accurate detail.

Use the log to show a real trade-off so students expect them.

The final version is best just because it is last. The best version is the best balance for the goal, which may be v3, not

v4. Have teams compare versions on evidence, not order.

A longer log means better work. Quality is in the reasoning per version, not the number of rows.

Reward clear cause-and-effect entries over volume.

Hands-on activity guide

Overview: A substantial iteration workshop: teams refine a weak prompt across up to four versions with a one-change-per-version log, then add a second goal and manage the clarity-accuracy trade-off, choosing and justifying a best-balanced version.

Grouping / timing: Teams of three with roles (prompt-writer, logger, evaluator), rotated once.

Tool: Weak-prompt starters with stated goals and pre-generated outputs for each iteration (teacher-mediated, account-free) or a school-approved child-safe tool operated by the teacher only. - Teams write each version on paper; the teacher runs the curated/approved tool or reveals the matching pre-generated output. No student accounts, no personal data.

Facilitation steps:

1. Set the goal and judge the v1 output against it.
2. Change one thing to make v2; predict and log the effect, then check the revealed output.
3. Repeat to v3 and v4, one change per version, logging cause and effect.
4. Add the second goal (accuracy alongside clarity).
5. Find a version where helping one goal hurt the other - record the trade-off.
6. Choose the best-balanced version for the purpose and justify it from the log.

Unplugged fallback: Unplugged: prompt cards plus pre-generated outputs for each version.

Teams sequence the cards, annotate

each with its change and effect, and mark the trade-off point, with no live tool needed.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Grade 5 Advanced: refines prompts to improve output quality and explains why

- Check: Iteration log across several versions plus a two-goal trade-off justification
- Success criteria: Iterates a prompt with a reasoned log and chooses a best-balanced version, explaining the trade-off with evidence.
- Evidence: Multi-version refinement log with trade-off note.
- Feedback / adjustment: For teams that only lengthen prompts, prompt 'which change caused which effect?'; extend to naming the exact cost of each gain.

Differentiation

Representation: Provide the log template, a worked iteration example, and a spoken model so students meet the process in several forms.

Action & expression: Teams can assemble versions from prompt-part cards or write them; logging can be scribed, typed, or card-based.

Engagement: A meaningful task (a clear, accurate notice for a school date-harvest event) makes the two goals feel real and worth balancing.

People of Determination: Offer prompt-part cards and a pre-structured log so students with writing or memory needs can focus on the reasoning. Assign the evaluator role to students who prefer judging over writing.

EAL: Bilingual terms for iterate, trade-off, balance, accuracy. Stems: 'Version ____ helped ____ but hurt ____, so the best balance is ____.'

Stretch / next tier: Ask top teams to add a third goal (tone) and describe how balancing three goals differs from two.

Answer key / model responses

Task / item Model response Marking note

Log one iteration. v2 added the audience 'Grade 3', reason: remove guessing, effect: output became simpler and on-level.

Look for a single change with a clear cause-effect entry.

Identify a trade-off. v3 cut the notice to two sentences for clarity, but that dropped the accurate detail about the event time, hurting accuracy.

Reward a specific place where one goal cost the other.

Choose a best-balanced version.

v4 keeps the time and place (accuracy) in simple short sentences (clarity), so it best balances both for the purpose.

Accept any version if justified from the log against the goal.

Assessment rubric (observable look-fors)

Criterion Mastery Partial Not yet

Iterates with a reasoned log Logs several versions, each with one change and a clear cause-effect entry.

Logs versions but some changes bundle several edits or lack reasons.

Changes prompts without a usable log of cause and effect.

Manages the two-goal trade-off

Names a specific trade-off and justifies a best-balanced version using log evidence.

Notices a trade-off but the justification is partial.

Does not recognise a trade-off or picks a version without reason.

Resources & links

- Iteration workshop log - A multi-version template capturing change, reason, and effect per version. (teacher-prepared)
- Iteration output packs - Pre-generated outputs for each prompt version so the workshop stays account-free. (teacher-prepared)

Week 3 - Core | How Data Shapes a Solution's Outcome

Strand: Critical Evaluation & Informed Interaction | Framework ref(s): AIF·CE·G4·A | 1 x 45-minute period (Core)

Learning outcome (framework): Explains how data influences the outcome of a solution.

Objectives & success criteria

- Explain how the data used changes the outcome of a given solution.
- Predict how a change in the data would change the solution's result.

Student-facing success criteria:

- I can explain, in my own words, how the data fed into a solution shapes the outcome it produces.
- I can predict what would happen to the outcome if the data were changed, and give a reason.
- I can point to a specific part of the data that most affects the result.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In Grade 5 you learned that AI systems learn patterns from examples rather than following rules a person wrote

by hand. You also sorted inputs and outputs. This lesson bridges the Grade 4 Advanced Critical Evaluation skill, which was

paused in Grade 5, so we return to it now with a Grade 6 depth: not just that data matters, but exactly how a change in the data changes the answer.

Activate: Show two photo-sorting apps side by side on the board. One was shown many pictures of desert foxes; the other was shown mostly pictures of cats. Ask: if both apps see the same new picture of a fox, will they answer the same way? Why not?

Teach: Explain that a solution is the whole system - the data plus the rule or model built from it - and the outcome is the answer it gives. Change the data and you change what the system learned, so you change the outcome. The rule did not change; the examples did.

Apply: In pairs, students trace one example: 'This app was trained on 100 sunny-day photos and 3 rainy-day photos. It is shown a rainy street.' They predict the outcome and say which data caused it.

Check: Cold-call three pairs for their prediction and the data reason. Listen for 'because it mostly saw sunny examples' rather than 'because it is broken'.

Critical & analytical thinking focus

Analyse - trace data's influence on a solution (Analyse | DOK 2-3 | SOLO: Relational)

The technical concept (teacher background)

A machine-learning solution is not a fixed set of instructions a programmer typed out; it is a model built from data. The

model looks across many examples and captures the patterns it finds there. Because the pattern comes from the

examples, the examples quietly decide what the model believes is normal. If the training data is mostly one kind of thing,

the model treats that kind as the default and does worse on the kinds it rarely saw. This is why two solutions built from the

same code but different data can give opposite answers to the same question. Data influence works in two directions: the

amount of data for each category, and the quality or balance of that data. A small change - adding many examples of a rare

category, or removing a biased batch - can move the outcome a lot. Tracing this influence means following the path from a

piece of data, through what the model learned, to the answer on the screen. Being able to trace that path is the heart of

evaluating whether you should trust a solution's outcome.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook - two apps,
one photo

7 min

Present the two photo-sorting apps

trained on different data and pose the fox question. Record student predictions on the board without judging them yet.

Predict which app labels the fox correctly and explain your reasoning to a partner.

- Why might two apps with the same code give different answers?
- What is different between the two apps if the code is the same?

Direct teaching -

solution, data, outcome

Define solution, data, and outcome using the flow visual G6-W3-01. Model tracing one example aloud, pointing from the data

Copy the data-to-outcome flow into notes and label the three parts in your own

- Which part of a solution can we change without touching the code?
- Where in the flow does the data

10 min pile to the learned pattern to the answer. words. do its work?

Guided practice

- trace a worked solution

12 min

Hand out the worked solution card (a fruit-ripeness sorter trained on data listed on the card). Model tracing how the data leads to its outcome on a sample fruit, then release students to trace a second fruit.

In pairs, trace the data through the sorter for your assigned fruit and write one sentence explaining the outcome from the data.

- Which examples in the data explain this outcome?
- What did the model rarely see, and how does that show in the answer?

What-if probe -

predict a data

change

10 min

Pose the change: 'We add 50 photos of unripe green dates to the training data.'

Facilitate predictions and push for the data reason behind each prediction.

Predict how the outcome

changes after the data

change and justify it by

pointing to the specific data

added or removed.

- Which single data change would most alter the outcome?

- Why would that change move the answer more than another change?

Consolidate -

explanation slip

6 min

Collect the explanation slip and prediction. Name-check strong tracings and the responsible-use reminder that we never explain a solution using anyone's personal data.

Write your final data-to-outcome explanation and your what-if prediction on the exit slip.

- How would you convince a friend that the data, not luck, caused this outcome?

Anticipated misconceptions

Misconception How to address it

The code is what decides the answer, so the data is not that important.

Show two solutions with identical code but different data giving opposite answers. The learned pattern lives in the data; the code is only the machinery that reads it.

If a solution gets something wrong, it must be broken.

Reframe a wrong answer as a data story: the model probably saw few or unbalanced examples of that case. It is doing exactly what its data taught it.

Adding any data at all improves the outcome. Explain that adding more of what it already has changes little, while adding examples of a rare or missing category can shift the outcome a lot. Balance matters, not just amount.

Hands-on activity guide

Overview: A short teacher-mediated trace: on the board the teacher runs a curated demo of the fruit-ripeness sorter, and

students trace on their own cards how each listed data batch leads to the outcome shown.

Grouping / timing: Pairs, then whole-class share.

Tool: Board demo of a school-approved child-safe image sorter, run by the teacher only (no student accounts); simulated-

AI outcome cards as the account-free path. - Teacher-run on the classroom board; students never log in and never enter

any personal or identifying data. Curated outputs are pre-selected so no live searching is needed.

Facilitation steps:

1. Read the data list printed on your worked solution card and say what the sorter mostly saw.
2. Watch the teacher run one fruit through the sorter on the board and note the outcome.
3. Trace: draw an arrow from the data batch you think caused the outcome to the answer.
4. Write one sentence: 'The outcome is X because the data mostly showed Y.'
5. Verify: check your reason against a second fruit the teacher runs - does your data story still hold?
6. Risk check: name one way this data could make the sorter unfair, and one non-AI way to double-check a fruit by hand.

Unplugged fallback: Unplugged: worked solution cards where students swap data cards into the sorter and trace the outcome by hand, no device needed.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Grade 4 Advanced: explains how data influences the outcome of a solution

• Check: Data-to-outcome explanation on a worked solution (with a 'what if the data changed' probe)

• Success criteria: Correctly explains the data's influence and predicts the effect of a data change.

• Evidence: Explanation slip + prediction.

• Feedback / adjustment: Bridges the Grade 5 gap explicitly; model tracing the data through the solution for students who stall.

Differentiation

Representation: Offer the flow visual, a spoken walkthrough, and a physical card version so the same idea arrives three ways.

Action & expression: Let students respond by drawing arrows, speaking to a partner, or writing a sentence.

Engagement: Use local, familiar contexts - dates, fruit markets in Al Ain - so the data feels real.

People of Determination: Provide a partly completed trace with the arrows started and one sentence stem filled in; allow a scribe or verbal answer instead of writing.

EAL: Pre-teach data, outcome, pattern, balance with picture cards; give the sentence frame

'The outcome is ____ because the data mostly showed ____.'

Stretch / next tier: Ask students to design a data change that would flip the outcome and explain the smallest change that would do it.

Answer key / model responses

Task / item Model response Marking note

Explain the sorter's outcome from its data.

The sorter labelled the date 'ripe' because its training data was mostly ripe brown dates, so it treats brown as the normal ripe look.

Look for a named data feature causing the outcome, not just 'the data did it'.

Predict the effect of adding 50 unripe green date photos.

It would start labelling more green dates 'unripe' correctly, because it now has examples of that category it was missing.

Reward reasoning that links the added category to the shifted outcome.

Identify which single data change most alters the outcome.

Adding the missing green-date examples changes it most, because the sorter had almost none, so those examples fill the biggest gap.

The key idea is that filling a gap moves the outcome more than adding to a full category.

Assessment rubric (observable look-fors)

Criterion Mastery Partial Not yet

Explains data's influence on the outcome

Traces a clear path from specific data to the outcome with correct reasoning.

States that data affects the outcome but the link is general or partly correct.

Attributes the outcome to luck, the code, or something being broken.

Predicts effect of a data change

Predicts the outcome shift and justifies it by pointing to the exact data changed.

Predicts a plausible shift but the justification is vague.

No prediction, or a prediction unconnected to the data change.

Resources & links

- Worked solution cards - fruit-ripeness sorter - Printable cards listing the training data and showing sample outcomes for tracing. (teacher-prepared)
- ADEK K-12 AI Fluency Framework - Reference for the Critical Evaluation strand and the Grade 4 Advanced skill this lesson bridges. (<https://www.adek.gov.ae>)

Week 3 - Lab | Trace-the-Data Investigation (Lab)

Strand: Critical Evaluation & Informed Interaction | Framework ref(s): AIF·CE·G4·A | 1 x 45-minute period (Lab)

Learning outcome (framework): Explains how data influences the outcome of a solution.

Objectives & success criteria

- Change the data in a worked solution and observe and record the effect on the outcome.
- Identify which single data change most alters the outcome, and explain why.

Student-facing success criteria:

- I can run a controlled data change and record the outcome before and after.
- I can compare data changes and identify which one moved the outcome the most.
- I can explain, using my recorded evidence, why that change had the biggest effect.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Core lesson you traced how data shapes an outcome and predicted the effect of a data change. In this

Lab you test those predictions by actually swapping the data and watching what happens, then judging which change matters most.

Activate: Recap the Core prediction: adding green date photos should improve unripe detection.

Ask students to hold onto that prediction - today we test it.

Teach: Introduce the idea of a fair test: change one thing at a time so you know what caused the outcome shift. This is a controlled data experiment.

Apply: Model one swap on the board: run a date before, add the green batch, run it again, and record both outcomes side by side.

Check: Ask students what would go wrong if they changed two data batches at once. Listen for 'you would not know which one caused the change'.

Critical & analytical thinking focus

Analyse - trace data's influence on a solution (Analyse | DOK 2-3 | SOLO: Relational)

The technical concept (teacher background)

A controlled data experiment is how you turn a prediction about influence into evidence. You fix everything except one data

batch, measure the outcome before, make the single change, and measure the outcome after.

The difference between

before and after is the influence of that batch, isolated cleanly. If you change several things at once you get a shift you

cannot attribute to any one cause, so you learn nothing about influence. Scientists call this changing one variable at a time,

and it works exactly the same way for data. Recording matters as much as changing: without a before-and-after log,

memory blurs and you cannot compare batches fairly. When you run several swaps, the batch that produces the biggest

before-to-after gap is the most influential, and usually it is the batch that filled the largest gap in what the model had already

seen. This is the practical, hands-on core of evaluating a solution: not guessing about data, but measuring its effect.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Set up the fair

test

8 min

Review controlled change - one data

batch at a time. Distribute the

investigation log and the deck of data

cards for the ripeness sorter.

Write your Core prediction

at the top of the log so you

can check it against

evidence later.

- Why must we change only one data batch at a time?

- What will we record before and after each change?

Run swaps and

record

16 min

Run each requested swap on the board demo (or hand groups the simulated-AI outcome cards). Confirm each group records before and after outcomes accurately.

For each data card you swap in, record the outcome before and after and calculate how much the outcome moved.

- Did the outcome move as you predicted?
- How big was the shift compared with the last swap?

Compare and rank the

Facilitate ranking: which swap produced the biggest before-to-after gap? Prompt Rank your swaps from most to least influential and mark

- Which change most altered the outcome?
- changes

10 min

groups to explain the winner using their log.

the single most powerful change.

- Why did that change beat the others?

Verify and risk-check

7 min

Guide a verification step: re-run the winning swap on a fresh fruit to confirm the effect repeats. Prompt a basic risk check on unfair data.

Re-test the top change on a new fruit and note whether the effect holds; write one risk this data could create.

- Does the effect repeat, or was it a one-off?
- Could this data change make the

sorter unfair to some fruit?

Conclude from

evidence

4 min

Collect the investigation log and conclusion. Reinforce that no personal or identifying data ever goes into the sorter.

Write a one-sentence

conclusion naming the most

influential data change and

your evidence for it.

- What is your evidence, not just your opinion, that this change mattered most?

Anticipated misconceptions

Misconception How to address it

You can change several data batches at once and still know what caused the outcome shift.

Demonstrate a two-change swap and ask students to attribute the result - they cannot. Only one change at a time gives clean evidence.

A bigger pile of data always causes a bigger outcome shift.

Show that a small batch filling a missing category can outmove a large batch of what the model already had. Gaps matter more than size.

If the outcome changed once, that proves the data caused it.

Require a re-test on a fresh example. An effect that repeats is evidence; a one-off might be coincidence.

Hands-on activity guide

Overview: A structured controlled experiment: groups request data swaps, the teacher runs each on the board sorter (or groups use simulated-AI outcome cards), and every group logs before and after outcomes to find and verify the most influential change.

Grouping / timing: Groups of three with roles - swapper, recorder, checker.

Tool: Board demo of a school-approved child-safe image sorter run by the teacher only, plus simulated-AI outcome cards

for the account-free path. - Teacher-operated on the classroom board; no student logins, no personal data entered.

Outcome cards are curated in advance so students investigate without live searching.

Facilitation steps:

1. Write your prediction from the Core lesson at the top of the log.
2. Choose one data card to swap in - just one.

3. Record the outcome BEFORE the swap.
4. Ask the teacher to run the swap, then record the outcome AFTER.
5. Calculate the shift: how far did the outcome move (none, a little, a lot)?
6. Repeat for three different single swaps, one at a time.
7. Rank the swaps and mark the most influential.
8. Verify: re-run the winning swap on a fresh fruit and check the effect repeats.
9. Risk check: name one way the winning data change could make the sorter unfair, plus a non-AI way to double-check by hand.

Unplugged fallback: Unplugged: worked solution cards where students physically swap data cards into the sorter and trace each outcome by hand, keeping the same before-and-after log.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Grade 4 Advanced: explains how data influences the outcome of a solution

• Check: Investigation log showing before/after outcomes for several single data swaps, plus a ranked conclusion

conclusion

• Success criteria: Records outcomes for controlled single swaps and identifies, with evidence, the most influential change.

influential change.

• Evidence: Completed investigation log + one-sentence evidence-based conclusion.

• Feedback / adjustment: Reward isolating one change at a time; coach students who changed several batches

to redo one cleanly.

Differentiation

Representation: Provide the log as a table, a spoken protocol, and a card-based physical version.

Action & expression: Students can record with ticks and arrows, numbers, or short sentences.

Engagement: Give each group a role card so every student has a clear job in the experiment.

People of Determination: Offer a pre-structured log with the before/after columns labelled and one row modelled; allow the

checker role to respond verbally.

EAL: Pre-teach swap, before, after, shift, influential; provide the frame 'The ____ swap moved the outcome the most because ____.'

Stretch / next tier: Challenge students to design a swap that produces zero outcome change and explain why some data has no influence.

Answer key / model responses

Task / item Model response Marking note

Record before and after

for the green-date swap.

Before: brown date 'ripe' and green date 'ripe' (wrong). After adding

green batch: green date now 'unripe' (correct). Shift: a lot.

Check the before column is

filled before the swap, not after.

Rank three single swaps by influence.

Green-date batch moved it a lot; extra brown-date batch moved it none; blurry-photo batch moved it a little. Winner: green-date batch.

The winner should be the batch that filled the biggest data gap.

Explain why the winning swap beat the others.

It filled a category the sorter barely had, so it changed what the model treated as normal, while adding more brown dates only repeated what it already knew.

Look for the gap-filling idea, backed by the logged shifts.

Verify the winning effect. Re-running on a fresh green date also gave 'unripe', so the effect repeated - it is evidence, not a one-off.

Accept any clear re-test that confirms repeatability.

Assessment rubric (observable look-fors)

Criterion Mastery Partial Not yet

Runs a controlled data change

Changes one batch at a time and records clean before/after outcomes for every swap.

Mostly isolates changes but one swap is muddled or a before value is missing.

Changes several batches at once so outcomes cannot be attributed.

Identifies the most influential change with evidence

Ranks swaps and justifies the winner using the logged before/after shifts.

Names a winner but the justification leans on opinion more than the log.

No clear winner, or a choice unsupported by any

recorded evidence.

Resources & links

- Trace-the-data investigation log - Printable before/after recording sheet with a ranking box. (teacher-prepared)
- Simulated-AI outcome card deck - Account-free cards showing the sorter's outcome for each data configuration. (teacher-prepared)

Week 4 - Core | Misinformation Risks and Verification

Strand: Ethics & Governance | Framework ref(s): AIF·GE·G5·A | 1 x 45-minute period (Core)

Learning outcome (framework): Explains misinformation risks and proposes verification strategies.

Objectives & success criteria

- Explain how an AI output could spread misinformation.
- Propose at least one concrete verification strategy and explain why it helps.

Student-facing success criteria:

- I can explain a specific way an AI output could be wrong and spread misinformation.
- I can propose a concrete verification step and say why it makes the claim more trustworthy.
- I can move from 'it might be wrong' to naming the exact risk and the exact check.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: This bridges the Grade 5 Advanced Ethics and Governance skill. You have learned that AI outputs are built from

patterns in data, not from a person checking the facts. That means an output can sound confident and still be wrong. Today

you learn to name the risk precisely and propose a check that reduces it.

Activate: Show a curated AI output on the board that states a confident but false claim about a UAE landmark's height. Ask: does sounding sure make it true?

Teach: Explain misinformation: false information that spreads, whether or not anyone meant to deceive. An AI can generate it because it predicts fluent text, not verified truth.

Apply: In pairs, students name exactly what is risky about the shown output - not 'it is bad' but 'the number could be invented'.

Check: Cold-call for the specific risk. Push from vague 'it could be wrong' to 'the height figure might be made up'.

Critical & analytical thinking focus

Evaluate - reason about risk and verification (Evaluate | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

An AI language system generates text by predicting what words are likely to come next, based on patterns in the data it

learned from. It is optimised to sound fluent and plausible, not to be true, so it can state a wrong fact with the same

confidence as a right one. This is why AI output is a misinformation risk: a made-up statistic, a wrong date, or an invented

quote can look exactly as trustworthy as a real one and spread when people repeat it. The risk grows because the output carries no built-in flag for which parts are reliable. Verification is the response: an independent check against a trusted source that was not produced by the same system. Good verification is specific - it targets the exact claim at risk, such as a number or a name, and confirms it against something authoritative like an official website or an encyclopaedia. A verification strategy is stronger when it is layered, using more than one independent source, because a single source can also be wrong. Learning to name the precise risk and pair it with a precise check is the core skill of responsible AI use.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook - confident

but wrong

7 min

Display the curated false-height claim about a UAE landmark. Invite reactions, then reveal the correct figure from an official source.

Decide whether the output is trustworthy and explain your first reaction to a partner.

- Does sounding confident make a claim true?
- Which exact part of this output might be wrong?

Direct teaching -

why AI can

mislead

10 min

Explain next-word prediction versus fact-checking using visual G6-W4-01. Define misinformation and stress AI has no built-in truth flag.

Note in your own words why fluent text is not the same as true text.

- What is the AI actually optimised to do?
- Why does that create a misinformation risk?

Guided practice

- name the exact risk

Hand out three curated outputs. Model naming the specific risk in one, then release students to name the risk in the

For each output, write the specific risk - the exact claim that might be false

- Is the risk a number, a name, a date, or a source?
- What makes that particular part risky?

Propose a verification strategy

11 min

Introduce the risk-and-verify card. Model pairing a risk with a concrete check against a trusted source and explaining why it helps.

Complete a risk-and-verify card: name the risk, propose a concrete check, and give the reason it helps.

- Which trusted source would settle this claim?
- Why does that check reduce the risk?

Consolidate - responsible use
6 min

Collect cards. Reinforce that we never paste personal or identifying data into an AI tool and that verifying before sharing stops misinformation spreading.

Finalise your risk-and-verify card and add one sentence on why verifying before sharing matters.

- What is your responsibility before you share an AI answer?

Anticipated misconceptions

Misconception How to address it

If an AI sounds confident and detailed, the answer is probably correct.

Show a confident, detailed, and false output beside the real figure.

Fluency is produced by prediction, not by checking, so confidence is not evidence.

Misinformation only counts if someone meant to lie. Clarify that misinformation is false information that spreads regardless of intent. An AI with no intent can still generate and spread it.

Checking one source is always enough to verify a claim.

Explain that a single source can also be wrong; layered verification with two independent trusted sources is more reliable.

Hands-on activity guide

Overview: A short guided evaluation: students examine three curated AI outputs, name the specific misinformation risk in

each, and draft one risk-and-verify card pairing a risk with a concrete check.

Grouping / timing: Pairs, then whole-class share of the strongest cards.

Tool: Board display of curated AI outputs; risk-and-verify cards. Any live tool is operated by the teacher only using a school-approved child-safe tool; students use no accounts. - Teacher-mediated on the board; students never log in and never enter personal or identifying data. Outputs are curated in advance so the class evaluates, not generates.

Facilitation steps:

1. Read each curated output carefully and underline the exact claim you doubt.
2. Name the specific risk: is it an invented number, name, date, or source?
3. Choose a trusted source that could settle the claim.
4. Write the concrete verification step and why it helps.
5. Verify one claim live against the trusted source with your teacher and record the verdict.
6. Risk check: state what could go wrong if this output were shared unchecked, and a non-AI way to confirm it.

Unplugged fallback: Unplugged: curated suspect-output cards plus a trusted reference sheet; students match each risk to a check by hand.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Grade 5 Advanced: explains misinformation risks and proposes verification strategies

- Check: Risk-and-verify card: name the misinformation risk + a verification step

- Success criteria: Explains a plausible misinformation risk and proposes a workable verification strategy with a reason.

- Evidence: Risk-and-verify card.

- Feedback / adjustment: Move 'it could be wrong' toward a specific risk + check; extend to layered verification.

Differentiation

Representation: Present outputs on the board, on cards, and read aloud so the claim reaches every learner.

Action & expression: Students can underline, speak, or write their risk and check.

Engagement: Use UAE contexts students can care about - a landmark, a local record - to make the stakes real.

People of Determination: Provide a card with the risk-type options listed to circle and a sentence stem for the check; allow verbal responses.

EAL: Pre-teach misinformation, claim, source, verify, trustworthy with icons; give the frame 'The risk is _____. I would check it by _____ because _____.'

Stretch / next tier: Ask students to design a layered check using two independent sources and explain why two beats one.

Answer key / model responses

Task / item Model response Marking note

Name the specific misinformation risk in a curated output.

The height figure for the landmark could be invented, because the AI predicts plausible numbers and may not have the real one.

Reward a named risky element, not a general 'it might be wrong'.

Propose a concrete verification strategy.

Check the height on the landmark's official website or a reputable encyclopaedia, because those are maintained by people who verify facts.

The check must target the specific claim and name a trusted source.

Explain why the check helps.

An official source is independent of the AI and is kept accurate, so if it agrees the claim is far more trustworthy.

Look for the independence and reliability reasoning.

Assessment rubric (observable look-fors)

Criterion Mastery Partial Not yet

Explains a misinformation risk

Names the exact claim at risk and explains clearly why an AI could get it wrong.

Identifies a risk but keeps it general or the reason is thin.

Says only that it 'might be wrong' with no specific risk or reason.

Proposes a verification strategy with a reason

Proposes a concrete check against a named trusted source and explains why it reduces the risk.

Proposes a check but the source is vague or the reason is missing.

No workable check, or a check that would not address the risk.

Resources & links

- Curated suspect-output cards - Pre-selected AI outputs with an embedded misinformation risk for evaluation. (teacher-prepared)
- Trusted reference sheet - A short list of authoritative UAE and general sources for verifying claims. (teacher-prepared)

Week 4 - Lab | Verify and De-risk Workshop (Lab)

Strand: Ethics & Governance | Framework ref(s): AIF·GE·G5·A | 1 x 45-minute period (Lab)

Learning outcome (framework): Explains misinformation risks and proposes verification strategies.

Objectives & success criteria

- Apply a verification routine to a set of suspect outputs and record the verdict for each.
- Compare two verification strategies and judge which is more reliable.

Student-facing success criteria:

- I can run a verification routine on several suspect outputs and record a clear verdict for each.
- I can compare two verification strategies and judge which is more reliable, with a reason.
- I can back my verdicts with evidence from a trusted source.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Core lesson you named misinformation risks and proposed checks. In this Lab you apply a repeatable verification routine to a batch of suspect outputs, reach a verdict on each, and then compare two verification strategies to judge which is more reliable.

Activate: Recap the risk-and-verify card. Ask: if you had ten suspect outputs, how would you check them all fairly and quickly?

Teach: Introduce a verification routine - a fixed set of steps you apply the same way to every output, ending in a verdict: verified, false, or unclear.

Apply: Model running the routine on one output end to end, reaching a verdict against a trusted source.

Check: Ask what a fixed routine gives you that judging each output differently does not. Listen for fairness and consistency.

Critical & analytical thinking focus

Evaluate - reason about risk and verification (Evaluate | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

A verification routine is a fixed sequence of steps you apply identically to every output, so your judgements are consistent

rather than swayed by how confident an output happens to sound. A typical routine isolates the specific claim, selects an

independent trusted source, compares the claim against it, and records a verdict such as verified, false, or unclear.

Applying the same routine to a batch lets you compare outputs fairly and spot patterns in where AI tends to fail. Not all

verification strategies are equally reliable, though. Checking against a single source is faster but fragile, because that

source can also be wrong or out of date. Cross-checking against two independent trusted sources is slower but far more

reliable, because two independent sources are unlikely to share the same error. Judging which strategy is more reliable

means weighing speed against confidence and reasoning about independence - whether the sources could fail together.

This is DOK 3 evaluation: not just doing a check, but comparing methods and defending which one you trust more, always

without ever entering personal or identifying data into any tool.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Set up the

routine

8 min

Present the four-step verification routine on the board and distribute the batch of curated suspect outputs and the trusted reference sheet.

Copy the routine steps onto your workshop sheet and read the trusted reference sheet.

- What are the fixed steps of our routine?

- What are the three possible verdicts?

Run the routine
on the batch

15 min

Circulate as groups apply the routine to each suspect output; the teacher performs any live source check on the board when a group requests it.

Apply the routine to each output, record the verdict and the evidence, and flag any output you cannot resolve.

- What is your verdict, and what evidence supports it?

- Which outputs were false, and how could you tell?

Compare two
strategies

Set the comparison: single-source check versus two-source cross-check. Prompt Verify one tricky output both ways and judge which

- Where did the single-source check let you down?

12 min groups to try both on the same tricky output and weigh them.
strategy is more reliable,
giving a reason.

- Why are two independent sources harder to fool?

Risk check and
de-risk

6 min

Guide a basic risk check: what harm could a false output cause if shared, and how would verification have prevented it?

For one false output, write the harm it could have caused and the check that de-risks it.

- What is the real-world cost of sharing this unchecked?
- Which check would have stopped it?

Conclude and reflect

4 min

Collect the workshop sheet. Reinforce responsible use: verify before sharing and never enter personal data into a tool.

Write your judgement on which verification strategy you would use and why.

- Which strategy earns your trust, and what is the trade-off?

Anticipated misconceptions

Misconception How to address it

A verdict is just a gut feeling about whether the output seems right.

Require each verdict to cite evidence from a trusted source. A routine replaces gut feeling with a consistent, checkable process.

The faster verification strategy is always the better one.

Compare speed against reliability: a single-source check is faster but can be fooled by one wrong source. Reliability, not speed alone, decides which to trust.

If two sources agree, the claim is guaranteed true. Explain that agreement raises confidence but is not proof, especially

if the sources are not truly independent. Judge whether they could fail together.

Hands-on activity guide

Overview: A full verification workshop: groups apply a fixed four-step routine to a batch of curated suspect outputs, record a verdict and evidence for each, then run a controlled comparison of single-source versus two-source checking to judge reliability.

Grouping / timing: Groups of three or four with roles - claim-isolator, source-checker, verdict-recorder, reporter.

Tool: Curated suspect-output cards and a trusted reference sheet; any live source lookup is done by the teacher on the board using a school-approved child-safe source. No student accounts. - Teacher-mediated; students verify against printed trusted references and request live checks from the teacher. No logins, no personal or identifying data entered anywhere.

Facilitation steps:

1. Copy the four routine steps onto your workshop sheet.
2. For each suspect output, isolate the specific claim to check.
3. Select an independent trusted source from the reference sheet.
4. Compare the claim against the source and record a verdict: verified, false, or unclear.
5. Write the evidence that supports each verdict.
6. Pick one tricky output and verify it two ways: single source, then two-source cross-check.
7. Judge which strategy was more reliable and explain why, weighing speed against confidence.
8. Risk check: for one false output, write the harm it could cause if shared and the check that de-risks it.
9. Note a non-AI fallback: how you would confirm the same fact without any AI tool.

Unplugged fallback: Unplugged: curated suspect-output cards plus a trusted reference sheet; groups run the whole routine and comparison on paper, no device needed.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Grade 5 Advanced: explains misinformation risks and proposes verification strategies

• Check: Completed verification workshop sheet with verdicts and evidence for a batch, plus a reasoned

comparison of two strategies

• Success criteria: Applies the routine to reach evidence-based verdicts and judges which verification strategy is more reliable with a reason.

• Evidence: Workshop sheet with per-output verdicts + strategy comparison and judgement.

• Feedback / adjustment: Reward evidence-backed verdicts; push students who prefer speed to weigh reliability and source independence.

Differentiation

Representation: Provide the routine as a numbered list, a flowchart, and a spoken walkthrough.

Action & expression: Students can record verdicts with symbols, short phrases, or full sentences.

Engagement: Assign clear group roles so every student owns part of the workshop.

People of Determination: Offer a workshop sheet with the routine steps pre-printed and verdict options to circle; allow the reporter role to answer verbally.

EAL: Pre-teach routine, verdict, evidence, independent, reliable; give the frame 'My verdict is ____ because the trusted source ____.'

Stretch / next tier: Ask students to design a three-layer verification for a high-stakes claim and justify when the extra effort is worth it.

Answer key / model responses

Task / item Model response Marking note

Reach a verdict on a

suspect output with
evidence.

Verdict: false. The output said the tower was 900 metres; the official source lists a different height, so the claim does not match a trusted source.

Every verdict must cite the
trusted source, not a feeling.

Flag an unresolved
output.

Verdict: unclear. No trusted source on the sheet covered this claim, so I could not confirm or reject it and would need another source.

Accept 'unclear' as a valid,
honest verdict when evidence
is missing.

Compare single-source
and two-source
strategies.

Single-source was faster but the one source was out of date; the two-source cross-check caught the error, so it is more reliable.

Look for reasoning about
source independence, not
just which was quicker.

De-risk a false output. If shared, the false height could mislead classmates in a project; a two-source check before sharing would have stopped it.

Reward linking a concrete
harm to the specific check
that prevents it.

Assessment rubric (observable look-fors)

Criterion Mastery Partial Not yet

Applies the verification
routine to reach evidence-
based verdicts

Reaches a correct verdict for
each output, each backed by
cited trusted-source evidence.

Reaches verdicts for most
outputs but some lack cited
evidence.

Verdicts are guesses with
no trusted-source evidence.

Compares two strategies
and judges reliability

Judges which strategy is more
reliable and defends it by

reasoning about source
independence and trade-offs.
Picks a strategy but the
reasoning is thin or leans only
on speed.

No comparison, or a choice
with no reasoning.

Resources & links

- Verification workshop sheet - Printable sheet with the routine steps, a verdict log, and a strategy-comparison box. (teacher-prepared)
- Curated suspect-output cards + trusted reference sheet - Account-free batch of outputs to verify against a short list of authoritative sources. (teacher-prepared)

Week 5 - Core | Trustworthy AI Helper - Design & Test the Data (Part 1 Core)

Strand: Integration - all four strands | Framework ref(s): AIF·CU·G5·A AIF·SD·G5·A
AIF·CE·G4·A AIF·GE·G5·A | 1 x 45-minute period (Core)

Objectives & success criteria

- Design a small prompt-driven AI helper: state the task, the structured prompt, and the data it will use.
- Test how the quality of the data changes the helper's output, and explain the effect.

Student-facing success criteria:

- I can design a helper with a clear task, a structured prompt, and a data set.
- I can test how data quality changes the output and explain why.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: Bridging Weeks 1-2 (data quality affects outputs; refining prompts). The helper needs good data and a clear prompt.

Activate: Show two versions of a small 'recommend a book' helper - one fed clean data, one fed noisy data - and ask which to trust.

Teach: Explain that a helper is a small solution: a task + a structured prompt + data. Its output is only as good as the data and the prompt.

Apply: Pairs name the task their helper will do and one thing that must be true about its data.

Check: Quick check: 'What three parts does a helper need?' (Task, structured prompt, good data.)

Critical & analytical thinking focus

Analyse & create - design a solution and test data-quality effects (Analyse/Create | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

A prompt-driven AI helper is a small solution built from three parts: the task it must complete, the structured prompt that

instructs the tool, and the data it uses. Because the output is produced from the data and the prompt, poor-quality data

(inaccurate, unbalanced, or incomplete) produces a poor output even when the prompt is good. Testing means feeding the helper clean data and then degraded data and comparing the outputs, so you can see and explain the effect of data quality.

Designing the helper well now - a clear task, a precise structured prompt, and checked data - is what makes it trustworthy later. In Part 1 you design and test the data; the Lab builds the prototype; Part 2 traces the data's influence, verifies the output, and checks risks. All AI use is teacher-mediated or simulated, account-free, and uses no personal data.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Starter /

retrieval

5 min

Recap Weeks 1-2. Show the task brief:

build a trustworthy AI helper.

Recall that data quality and a clear prompt shape the output.

- What makes an output trustworthy?

- What could make a helper unreliable?

Direct teaching

(I do)

10 min

Model designing a helper: task, structured prompt, data. Then feed clean vs noisy data (teacher-mediated/simulated) and compare outputs.

Watch how the three parts fit and how data quality changes the output.

- Which part is the task, prompt, and data?

- How did the noisy data change the output?

Guided practice

(we do)

10 min

With the class, draft one helper design and predict how a data-quality change will affect it.

Help draft the design and predict the effect of degraded data.

- What is our helper's task?
- What will happen if the data is incomplete?

Independent
(you do)

15 min

Pairs design their helper on the planner and run one clean-vs-noisy data test, recording the effect. Circulate; press for the data-quality-to-output link.

Design the helper and test how data quality changes the output.

- Is your prompt structured and precise?
- How did data quality change your output?

Plenary / exit Pairs state their task and one data-quality finding; note the Lab will build the

Explain the helper's task • What is your helper's task?

5 min prototype. and one data-quality effect. • Why does data quality matter for it?

Anticipated misconceptions

Misconception How to address it

A good prompt fixes bad data. Show a strong prompt on noisy data giving a weak output. Both the prompt and the data must be good.

The helper 'knows' the answer. It produces the output from the data and prompt; it has no knowledge. Trace one output to prove it.

Hands-on activity guide

Overview: Pairs design a small prompt-driven AI helper (task + structured prompt + data) on a planner, then run one clean-vs-noisy data test and record how the output changes.

Grouping / timing: Pairs (mixed readiness).

Tool: Unplugged helper planner + a clean and a noisy data card; AI generation is teacher-mediated or simulated. - Teacher-printed; account-free; no personal data.

Facilitation steps:

1. Write your helper's task.
2. Write a structured prompt (goal + details + form).
3. Prepare the data it will use.

4. Run it on clean data (teacher-mediated/simulated) and note the output.

5. Run it on noisy data and note the output.

6. Explain how data quality changed the output.

Unplugged fallback: Compare teacher-provided clean-data and noisy-data outputs on cards; the design and analysis steps are identical.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Integrated mastery - CU (data quality affects outputs) and SD (design a prompt-driven solution)

- Check: Helper design planner + clean-vs-noisy data test reviewed against the mastery criteria
- Success criteria: Designs a helper with a clear task, structured prompt and data, and explains how data quality changed the output.

- Evidence: Completed helper planner (Trustworthy AI Helper Log, Part 1).

- Feedback / adjustment: Press the data-quality-to-output link; ensure the prompt is structured.

Record CU and

SD as mastery/partial/not-yet.

Differentiation

Representation: Planner with a worked example; data shown as table and chart.

Action & expression: Design on cards, annotate, or dictate.

Engagement: Let pairs choose a helper task they care about (recommend a book, sort recycling, suggest a healthy snack).

People of Determination: Reduced data set, sentence frames, extra time.

EAL: Frames 'My helper's task is ...', 'Noisy data changed the output by ...'; pre-teach task, structured prompt, data quality.

Stretch / next tier: Predict how much noise it takes before the output becomes unusable, and explain.

Answer key / model responses

Task / item Model response Marking note

Helper design Task: recommend one library book. Prompt: 'Using the class survey, suggest one book most pupils would enjoy, in one sentence with a

Accept any clear task +

structured prompt + named

reason.' Data: the balanced reading survey. data.

Data-quality test On clean data it suggested a popular adventure book; on noisy data (a duplicated entry) it over-suggested one title. Poorer data gave a less fair output.

Look for a clear data-quality-to-output effect.

Assessment rubric (observable look-fors)

Criterion Mastery Partial Not yet

Design the helper (SD) Clear task, structured prompt, and named data.

Two of three, or one unclear. Design incomplete.

Test data quality (CU) Runs a clean-vs-noisy test and explains the effect.

Runs a test but explanation is vague.

No test or effect.

Resources & links

- Trustworthy AI Helper kit (printable) - Planner, clean/noisy data cards, and the two-part Log. (school-printed (no account))

Week 5 - Lab | Trustworthy AI Helper - Build & Refine the Prompt (Part 1 Lab)

Strand: Integration - all four strands | Framework ref(s): AIF·CU·G5·A AIF·SD·G5·A

AIF·CE·G4·A AIF·GE·G5·A | 1 x 45-minute period (Lab)

Objectives & success criteria

- Build a working prototype of the helper using the structured prompt and improved data.
- Refine the prompt across versions to raise output quality, logging each change and its effect.

Student-facing success criteria:

- I can build a working prototype of my helper.
- I can refine the prompt across versions and explain what improved.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: Week 5 Core: the helper design and the data-quality test.

Activate: Pairs re-read their W5 Core design and pick the first prompt version to build.

Teach: Explain that building means running the prompt, checking the output, and refining the prompt one change at a time.

Apply: Pairs state prompt v1 and one thing they might improve.

Check: Quick check: 'Why refine one change at a time?' (So you can tell what helped.)

Critical & analytical thinking focus

Create & evaluate - build and iterate a solution (Create/Evaluate | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

Building the prototype means turning the design into a working helper and improving it by iterating the prompt. You run the

structured prompt on the improved data, read the output, judge it against the task, then change ONE part of the prompt and

run it again. Changing one variable at a time lets you attribute any improvement to that change.

A prompt-refinement log

records version 1, the change, and the effect, building the evidence-based habit real developers use. The Lab's extra period

gives time to reach a genuinely better output. All AI use is teacher-mediated or simulated, account-free, and uses no

personal data.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Set up

6 min

Pairs collect their design and the data; the

teacher opens the mediated/simulated routine.

Prepare prompt v1 and the data.

- What is your prompt v1?
- What will you check in the output?

Build & run (I do / we do)

10 min

Model running prompt v1, judging the output against the task, and making one refinement.

Run prompt v1 and judge the output against the task.

- Did the output meet the task?
- What one change might improve it?

Refine loop (you do)

22 min

Pairs iterate the prompt across versions, logging each change and its effect; circulate and press for one-change-at-a-time reasoning.

Refine the prompt across versions; log each change and its effect.

- What did this change improve, and why?
- Are you changing one thing at a time?

Close

7 min

Pairs record their best prompt version and note what improved; preview Part 2 (verify and de-risk).

Record the best version and the biggest improvement.

- Which version was best, and why?
- What will you check in Part 2?

Anticipated misconceptions

Misconception How to address it

Change lots of the prompt at once to improve faster. Show that multiple changes hide which one helped. Refine one change, then run again.

A longer prompt is always better. Show a bloated prompt that confuses the output. Precise beats long.

Hands-on activity guide

Overview: Pairs build a working prototype of their helper and iterate the prompt across several versions, logging each change and its effect, to reach a genuinely better output.

Grouping / timing: Pairs at a mediated station or with simulated-AI cards.

Tool: Teacher-mediated AI demo or simulated-AI cards + the prompt-refinement log. -

Teacher-mediated; account-free; no

personal data. A school-approved child-safe class tool may be used by the teacher only.

Facilitation steps:

1. Run prompt v1 on the improved data.
2. Judge the output against the task.
3. Change ONE part of the prompt; run again.
4. Log the change and its effect.
5. Repeat until the output clearly meets the task.
6. Record your best prompt version.

Unplugged fallback: Use prompt cards + teacher-provided outputs for each version; the run-judge-refine-log steps are identical.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Integrated mastery - SD (refine prompts to improve output quality and explain why)

- Check: Prompt-refinement log (v1 to best) with a reasoned justification for the key change
- Success criteria: Iterates the prompt to a measurably better output and explains the causal reason for the improvement.

- Evidence: Prompt-refinement log (Trustworthy AI Helper Log, Part 1 Lab).

- Feedback / adjustment: For trial-and-error, prompt 'name the change and its effect'. Record SD as mastery/partial/not-yet.

Differentiation

Representation: Prompt frames and a worked v1-to-v2 example.

Action & expression: Iterate on cards, on screen (mediated), or by dictating.

Engagement: Celebrate each improvement; pairs choose their helper context.

People of Determination: Fewer required versions, sentence frames, extra time.

EAL: Frames 'I changed ... and the output ...'; pre-teach refine, version, effect.

Stretch / next tier: Refine for two goals at once (accuracy + clarity) and discuss the trade-off.

Answer key / model responses

Task / item Model response Marking note

Prompt-refinement log v1 'suggest a book' gave a vague list. v2 added 'one book, with a reason, using the survey' - the output became specific and justified.

The detail added is what improved it.

Reward a one-change improvement with a causal reason.

Assessment rubric (observable look-fors)

Criterion Mastery Partial Not yet

Build a prototype (SD) Working prototype that runs the task.

Partly working. Not yet working.

Refine with reasoning (SD) Iterates to a better output and explains the causal change.

Improves but reason unclear. No reasoned refinement.

Resources & links

- Prompt-refinement log - Version, change, and effect columns for the build. (school-printed (no account))

Week 6 - Core | Trustworthy AI Helper - Trace the Data & Plan Checks (Part 2 Core)

Strand: Integration - all four strands | Framework ref(s): AIF·CU·G5·A AIF·SD·G5·A

AIF·CE·G4·A AIF·GE·G5·A | 1 x 45-minute period (Core)

Objectives & success criteria

- Explain how the data used influences the helper's outcome, and predict a data change's effect.
- Plan how you will verify the output and check for basic risks before trusting it.

Student-facing success criteria:

- I can explain how the data shapes my helper's outcome and predict a change.
- I can plan a verification and a risk check for my helper.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: Week 3 (data influences a solution's outcome) and Week 4 (misinformation risks and verification), plus the W5 prototype.

Activate: Show a helper whose output changes when one data entry is swapped, and ask why.

Teach: Explain that the data drives the outcome, so before trusting a helper you plan a verification (check a claim vs a trusted

source) and a risk check (could it be wrong, unsafe, biased?).

Apply: Pairs name one claim to verify and one risk to check.

Check: Quick check: 'Why plan verification and a risk check?' (An output can be wrong or unsafe; we check before trusting.)

Critical & analytical thinking focus

Analyse & evaluate - trace data influence and plan reliability checks (Analyse/Evaluate | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

A helper's outcome is driven by its data: change the data and the outcome changes. Tracing that influence - naming which

data drives which part of the output - is how you understand and trust a solution. Before relying on the helper you plan two

checks. Verification means testing a claim in the output against a trusted, independent source, not by re-asking the tool. A

risk check asks whether the output could be wrong, unsafe, biased, or a privacy problem, and what would reduce that risk.

Planning these checks in the Core sets up the Lab, where you carry them out, refine if needed, and present. This systematic evaluation and risk awareness is the heart of being an AI Solution Builder. All AI use is teacher-mediated or simulated, account-free, and uses no personal data.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Starter

5 min

Recap Weeks 3-4. Pairs collect their W5 prototype.

Recall how data influences an outcome and how to verify.

- How does the data drive the outcome?
- How do we check a claim?

Direct teaching

(I do)

10 min

Model tracing one output back to the data, then plan a verification and a risk check on a demo helper.

Follow the trace and the two planned checks.

- Which data drove this part of the output?
- What claim will we verify, and how?

Guided (we do)

10 min

Pairs trace one part of their output to the data and draft their verification and risk-check plan.

Trace the data's influence; draft the two checks.

- What data change would alter your outcome?
- What is your biggest risk?

Independent

(you do)

15 min

Pairs complete the trace and the check plan on the Log; circulate and press for a trusted-source verification.

Complete the data trace and plan verification + risk check.

- Is your verification against a trusted source?

- What would reduce your top risk?

Plenary / exit Pairs share one data-influence finding and Share a data-influence • How does data shape your

5 min one planned check; the Lab will run them. finding and one check. outcome?

- What will you verify in the Lab?

Anticipated misconceptions

Misconception How to address it

Verifying means asking the AI again. Verify against a trusted, independent source (data, a reference), not the same tool.

If it sounds confident it is safe. Show a confident but risky output; plan a risk check regardless of tone.

Hands-on activity guide

Overview: Pairs trace how their helper's data influences its outcome, then plan a verification (claim vs trusted source) and a basic risk check before trusting it.

Grouping / timing: Pairs.

Tool: The W5 prototype + the Trustworthy AI Helper Log (Part 2) + a trusted reference sheet. -

Teacher-printed; AI use

teacher-mediated/simulated; no personal data.

Facilitation steps:

1. Pick one part of your output and name the data that drives it.
2. Predict how a data change would alter the outcome.
3. Choose one claim to verify against a trusted source.
4. Name one risk (wrong, unsafe, biased, privacy) and a way to reduce it.
5. Write your verification and risk-check plan.

Unplugged fallback: Use the worked-solution cards to trace data-to-outcome; plan the checks on paper.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Integrated mastery - CE (data influences the outcome) and planning for GE

(verification and risk)

- Check: Data-influence trace + verification/risk plan on the Log, reviewed against the mastery criteria

- Success criteria: Explains the data's influence on the outcome, predicts a data change, and plans a trusted-source verification and a risk check.

- Evidence: Trustworthy AI Helper Log, Part 2 (Core section).

- Feedback / adjustment: Ensure verification uses a trusted source; press the data-to-outcome link. Record CE

as mastery/partial/not-yet.

Differentiation

Representation: Annotated solution diagram; verification and risk planners with examples.

Action & expression: Trace on cards, annotate, or dictate.

Engagement: Real contexts where trust matters (a study helper, a recycling sorter).

People of Determination: Reduced trace, sentence frames, extra time.

EAL: Frames 'The data ... drives the outcome ...', 'I will verify ... against ...'; pre-teach trace, verify, risk.

Stretch / next tier: Identify the single data change with the largest effect and justify.

Answer key / model responses

Task / item Model response Marking note

Data-influence trace The survey's most-picked genre drove the book suggestion; if we swapped in a class that prefers science, the suggestion would change to a science book.

Look for a specific data-to-outcome link and a prediction.

Verification & risk plan Verify 'most pupils prefer adventure' against the survey totals. Risk: a Verification must use a duplicate could bias the suggestion; reduce it by de-duplicating the data.

trusted source; risk must have a mitigation.

Assessment rubric (observable look-fors)

Criterion Mastery Partial Not yet

Trace data influence (CE) Explains the data's influence and predicts a change.

Partial explanation. No clear link.

Plan checks (GE) Plans a trusted-source verification and a risk check with a mitigation.

Plans one check. No workable plan.

Resources & links

- Verification & risk planner - Planner for the verification and the risk check. (school-printed (no account))

Week 6 - Lab | Trustworthy AI Helper - Verify, De-risk & Showcase (Part 2 Lab)

Strand: Integration - all four strands | Framework ref(s): AIF·CU·G5·A AIF·SD·G5·A

AIF·CE·G4·A AIF·GE·G5·A | 1 x 45-minute period (Lab)

Objectives & success criteria

- Carry out the verification and risk check, refine the helper if a check fails, and confirm it can be trusted.
- Present the Trustworthy AI Helper and reflect on the thinking used across all four skills.

Student-facing success criteria:

- I can verify my helper's output and reduce a risk, refining if needed.
- I can present my helper and reflect on my thinking.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: Week 6 Core: the data trace and the verification and risk-check plan.

Activate: Pairs re-read their check plan and set up the trusted reference.

Teach: Explain that you now run the checks, and if one fails you refine the data or prompt and re-verify.

Apply: Pairs state the claim they will verify first.

Check: Quick check: 'What do you do if a check fails?' (Refine and re-verify.)

Critical & analytical thinking focus

Evaluate & justify - verify, de-risk, and reflect (Evaluate/Create | DOK 3 | SOLO: Relational -> Extended abstract)

The technical concept (teacher background)

Confirming trust means carrying out the checks you planned. You verify a claim in the output against a trusted, independent

source; if it fails, you refine the data or the prompt and re-verify. You run your risk check and apply a mitigation - for example

de-duplicating data to reduce bias, or adding a safe-use note. A helper is trustworthy only when it survives these checks,

and you can say why. Presenting the helper and reflecting across the four skills - data quality, prompt refinement, tracing

data influence, and verification/risk - consolidates the transfer that confirms readiness for the Grade 6 unit. All AI use is

teacher-mediated or simulated, account-free, and uses no personal data.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Set up

6 min

Pairs collect their prototype and check plan; the teacher sets up the mediated/simulated routine and trusted references.

Prepare the first verification. • Which claim will you verify first?

- What is your trusted source?

Verify & de-risk

(you do)

22 min

Pairs run their verification and risk check, refine and re-verify if a check fails;

circulate and press for trusted-source checks.

Verify a claim, apply a risk mitigation, refine and re-verify if needed.

- Did the claim hold?
- What risk did you reduce, and how?

Prepare
showcase

9 min

Pairs prepare a one-minute showcase: the task, one improvement, and why it can be trusted.

Prepare the showcase and the trust argument.

- Why can your helper be trusted?
- What is your strongest evidence?

Showcase &
reflect

8 min

Two or three pairs present; class notes one strong verification and one risk mitigation. Students reflect across the four skills.

Present and reflect across the four skills.

- What did you verify and find?
- Which of the four skills was hardest, and how did you work through it?

Anticipated misconceptions

Misconception How to address it

One check is enough. If a check fails, refine and re-verify; check the new output too.

A risk you name but do not act on is handled. A risk is only reduced when you apply a mitigation and re-check.

Hands-on activity guide

Overview: Pairs carry out their verification and risk check, refine the helper if a check fails, confirm it can be trusted with evidence, present a one-minute showcase, and reflect across the four skills.

Grouping / timing: Pairs, then a showcase carousel.

Tool: The prototype + Trustworthy AI Helper Log (Part 2 Lab) + trusted reference sheet; AI use teacher-mediated/simulated.

- Teacher-mediated; account-free; no personal data. Non-AI fallback: verify curated outputs against the reference sheet.

Facilitation steps:

1. Verify your chosen claim against a trusted source.
2. If it fails, refine the data or prompt and re-verify.
3. Run your risk check and apply one mitigation.
4. Confirm your helper can be trusted, with evidence.
5. Prepare a one-minute showcase.
6. Present and complete your reflection.

Unplugged fallback: Where generation is not possible, verify teacher-provided outputs against the reference sheet; the verify-de-risk-reflect steps are identical.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Integrated mastery confirmation - GE (verification and risk) with CU, SD, CE from earlier, plus reflection

• Check: Completed verification + risk mitigation on the Log, a one-minute showcase, and the four-skill reflection

• Success criteria: Verifies a claim against a trusted source, applies a risk mitigation (refining if needed), argues trust with evidence, presents, and reflects across the four skills.

• Evidence: Trustworthy AI Helper Log (Part 2 Lab) and observed showcase.

• Feedback / adjustment: In-course mastery confirmation across all four strands; any not-yet becomes a targeted Prerequisite Primer in the relevant Week 7+ lesson.

Differentiation

Representation: Verification flowchart and risk matrix with icons; showcase can be spoken, shown, or drawn.

Action & expression: Verify on paper, aloud to a scribe, or by annotating; present orally, with a poster, or via a partner.

Engagement: Celebrate trustworthy helpers; pairs choose how to showcase.

People of Determination: Reduced checklist, extra time, frames for reflection, a scribe.

EAL: Frames 'I verified ... it held/failed because ...', 'I reduced the risk of ... by ...'; pre-teach verify, mitigate, trustworthy.

Stretch / next tier: Layer two verification strategies and judge which is more reliable; weigh a fairness/safety trade-off.

Answer key / model responses

Task / item Model response Marking note

Verification Claim: 'adventure is the most-picked genre.' Verified against the survey totals - it held.

Trusted source, not re-asking the tool.

Risk mitigation Risk: a duplicate entry biased the suggestion. Mitigation: removed the

duplicate and re-ran; the suggestion became fairer.

Reward an applied mitigation
with a re-check.

Reflection (four skills) Refining the prompt was easiest; verifying was hardest because I had to find a trusted source. My helper is trustworthy because it passed both checks.

Reflect across data quality,
prompt refinement, data-
influence, verification/risk.

Assessment rubric (observable look-fors)

Criterion Mastery Partial Not yet

Verify (GE) Verifies a claim against a trusted
source, refining if it fails.

Verifies partially or without a
trusted source.

Does not verify.

De-risk (GE) Applies a risk mitigation and re-
checks.

Names a risk but does not act. No risk handling.

Reflection & transfer Reflects across all four skills and
names a next step.

Reflects on some skills only. Little or no reflection.

Resources & links

- Verification flowchart + risk matrix (printable) - A claim-check flowchart and a simple likelihood x impact risk matrix. (school-printed (no account))

- Curated output set (fallback) - Teacher-provided outputs to verify where generation is not possible. (teacher-prepared)

Appendix A - Glossary of Key Terms

Teacher reference and student-facing definitions. Introduce each term where it first appears.

Data - information a system collects and uses to make a decision.

Label - a name or tag given to an example so a system can sort or learn it.

Rule - an instruction a system follows, often IF ... THEN ...

Outcome / output - the result a system produces after processing its input.

Fair / unfair - fair treats similar people or things equally; unfair wrongly favours or leaves out some.

Bias - a slant in a system's outputs, usually from unbalanced, skewed or low-quality data.

Personal information - data about a real person (name, photo, home, age) that must be kept private.

Rule-based system - a system that follows fixed IF-THEN rules a person wrote.

Learning system / model - a system that learns patterns from many examples (training data) instead of fixed rules.

Training data - the examples a learning system learns from.

Prompt - the words we give an AI system to tell it what to do.

Data quality / noisy data - how complete and correct data is; noisy data is messy or error-filled.

Trustworthy AI - an AI system whose outputs are reliable, fair and safe to use.

Appendix B - Understanding Bias: How Data Shapes Outputs

How learning systems use training data

At Grade 6 students meet learning systems (models). Unlike a rule-based system that follows fixed IF-THEN

rules a person wrote, a learning system finds patterns in many examples called training data and uses those

patterns to decide. So the training data shapes what the system does: if the examples are unbalanced, noisy or

low quality, the model can be biased or wrong. This is why 'AI is just rules' is only half the story - modern AI also

learns from data.

A concrete, age-appropriate example

The Trustworthy Helper. If we build a helper from noisy (messy, error-filled) data, it gives unreliable answers.

Cleaning the data - fixing errors and filling gaps - makes the helper more trustworthy. Data quality drives

trustworthy outcomes.

Misconceptions to name and correct (say them aloud, then correct)

- 'AI is just fixed rules.' -> Rule-based systems follow rules; learning systems also learn patterns from

training data. Grade 6 studies both.

- 'Data is neutral / always true.' -> Training data is chosen by people; unbalanced or noisy data biases the model.

- 'AI understands like a person.' -> A model matches learned patterns; it has no understanding.

Use as the core of the fairness/data-quality lesson and as a one-line check in every hands-on activity: 'What

data did the system learn from? Who might be left out?'

Appendix C - Rule-Based vs Learning Systems (comparison)

Rule-based system Learning system (model)

How it decides Follows fixed IF-THEN rules a person wrote. Finds patterns in training data and uses them to decide.

Where it can go

wrong

A wrong or unfair rule. Unbalanced, noisy or low-quality training data.

Example A sorting robot with IF-THEN cards. An image labeller trained on many photos.

To improve it Fix or rewrite the rule. Improve the data (more balanced, cleaner).

Appendix D - Recommended Tools & Simulations + Unplugged Alternatives (UAE-accessible)

- Primary (teacher-mediated, account-free): Teachable Machine (teacher-run, no student logins) for a rule-

vs-learning demo; Machine Learning for Kids (teacher-mediated); a prompt-based helper on the teacher screen.

- Fully unplugged core: printed clean-vs-noisy dataset cards, prompt-refinement logs and a rule-vs-learning comparison card - every lesson runs with no device.
- Alternatives for restricted environments: (1) paper-only clean-vs-noisy data sorts and prompt-refinement activities; (2) a single teacher-projected simulation or a pre-recorded demo played offline.
- Device requirements: one teacher device + projector is sufficient. Offline option: all demos can be pre-recorded; no student accounts or internet needed for the unplugged path. Confirm every tool against your school's approved-apps list (UAE).

• UDL notes travel with every lesson (multiple means of representation, action & expression, and engagement).

Appendix E - Criteria Checklist (accuracy, fairness, usefulness, reliability)

Use when judging any output. Print one prompt per card as 'criteria cards' for pairs.

- Accuracy: did the system give the right answer on the examples we tested?
- Coverage: did we test examples from every group?
- Fairness: does any group get the wrong output more often? Who might be left out?
- Usefulness: does the output actually help the task?
- Reliability: does it give consistent answers when we try again?
- Data check: was the data balanced, complete and clean? What was missing or noisy?
- Fix: what one change to the data, prompt or rule would improve it?

Appendix F - Privacy: Why We Protect Personal Information

Rule: we never put personal information (real names, photos, home address, age) into a system; we use synthetic (made-up) data only.

• Why - ethical: it is each person's right to keep their information private; sharing it can hurt or embarrass someone.

• Why - technical: once data enters a system it can be stored, copied and reused in ways we cannot take back, and small pieces of data can be combined to identify a person. Synthetic data removes this risk.

Appendix G - Printable Resources (templates)

Prompt frames

Sentence starters students use to instruct a helper: 'Please [task] using [these examples]. Show your steps. If

unsure, say so.' Print as cards.

Clean vs noisy dataset cards

Example Clean or noisy? Problem to fix

Prompt-refinement log

Version Prompt we tried What the output missed Improved prompt

1

2

Improvement / iteration log

Version What we tested What went wrong Change we made Better?

1

2

Appendix H - Real-World & Cross-Curricular Connections

- Real AI: data-and-model decisions power photo tagging, recommendation and image/voice recognition;

helper prompting is how people use assistants and chat tools responsibly.

- UAE example: a school library or an app store that sorts and recommends by labels and past choices - a

real 'uses data to decide' system students can see.

- Cross-curricular anchors: Maths (data, proportion, sets); Science (classification, evidence);

Humanities /

Moral education (fairness, privacy, responsible use); Literacy (clear prompts and success criteria).

- Deployment norms: real teams test a system for accuracy and fairness before people rely on it - exactly

what the criteria checklist rehearses.

Appendix I - Looking Ahead

This bridging unit prepares you for the Grade 6 main course and later grades, where data quality and trustworthy

AI become central.

STUDENT WORKSHEETS

Grade 6 (Year 7) | Phase 3 / Cycle 2 | AI Solution Builder

Main Course, Term 1 - Rule-Based vs Learning Systems

(ML Foundations)

Emerging tier | 10 weeks (Core + Lab)

Field Detail

Program Grade 6 Main Course, Term 1 (Emerging tier)

Unit theme Rule-Based vs Learning Systems - ML Foundations

Signature activity A structured, prompt-driven AI solution to a simple task, evaluated for accuracy and

risk

Strands AI Conceptual Understanding | AI Solution Design & Development | Critical

Evaluation & Informed Interaction | Ethics & Governance

Delivery model 2 x 45-minute periods per week - a Core lesson + a paired Lab lesson each week

Lessons in this pack 20 lessons (10 Core + 10 Lab)

Week 1 - Core - Worksheet | Unit Launch - Rule-Based vs Learning Systems

Name: _____ Class: _____ Date: _____

Sort three systems

Write each system in the correct column and give one reason.

This follows fixed rules because ____

This learns from data because ____

My two questions

Write two genuine questions about rule-based and learning systems for the question wall.

System sort record

Complete the table for the three systems you sorted.

System Fixed rules or learns from data My reason

Exit ticket

In one sentence, how is a learning system different from a rule-based system?

Challenge (stretch)

Invent your own system card and swap with a partner - can they classify and justify it?

Week 1 - Lab - Worksheet | Rule vs Learning - Hands-On Exploration (Lab)

Name: _____ Class: _____ Date: _____

Station comparison

Log what each station did for the same kind of input.

Station A always ____

Station B changed when ____

Design a test

Write a test that would tell whether a system follows rules or learns, and what each result would mean.

Difference log

Record the differences you observed between the two stations.

What we tried Station A did Station B did

Exit ticket

What one input would you give a mystery machine to find out if it learns? Why?

Challenge (stretch)

Add an unfair set of example cards to Station B and explain how it changed the pattern the group learned.

Week 2 - Core - Worksheet | Rules vs Learning - How Each System Works

Name: _____ Class: _____ Date: _____

Feature chart

Complete the chart with a feature and an example for each system type.

Rule-based: logic comes from ____

Learning: logic comes from ____

Classify and justify

Classify two systems and underline the deciding feature.

This is a ____ system because ____

Classification record

Record each system, your classification, and the deciding feature.

System Rule-based or learning Deciding feature

Exit ticket

Classify two systems your teacher names and give the feature that separates them.

Challenge (stretch)

Find a system that is genuinely hard to classify and argue both sides using features.

Week 2 - Lab - Worksheet | Train a Simple Model (Lab)

Name: _____ Class: _____ Date: _____

Build and test log

Record your categories, then predict and test each held-back example.

I predict this will be classified as ____

The model actually classified it as ____

Data-safety and risk check

Tick the safety checks and write one risk and one improvement.

My data was safe because ____

One risk was ____

One improvement is ____

Training and testing record

Log each testing example, your prediction, the result, and whether it was correct.

Testing example Predicted Model result Correct?

Exit ticket

How did your training examples change what the model learned? Give one example.

Challenge (stretch)

Design a 'trick' testing example, predict whether the model will fail, then test it and explain the result.

Week 3 - Core - Worksheet | Structured Prompts to Complete a Task

Name: _____ Class: _____ Date: _____

Write a structured prompt

Fill in each part of the prompt frame for your task.

Goal: ____

Context: ____

Format: ____

Evaluate and adjust

Judge completion and write one adjustment.

Task complete? ____ because ____

My one adjustment: ____

Task-completion log

Record your prompt, the output, whether the task was completed, and your adjustment.

Structured prompt Output summary Task complete? One adjustment

Exit ticket

What are the three parts of a structured prompt, and which helped you most today?

Challenge (stretch)

Add an 'Audience' part to your prompt and describe how it changed the output.

Week 3 - Lab - Worksheet | Prompt-to-Task Workshop (Lab)

Name: _____ Class: _____ Date: _____

Compare two structures

Run two structures on the task and mark which requirements each met.

Structure A met ____ of 3

Structure B met ____ of 3

The stronger structure was ____ because ____

Design a reusable template

Write your reusable template with labelled slots and a verify step.

Goal: ____

Context (no personal data): ____

Format: ____

Verify: ____

Structure comparison record

Record each structure, requirements met, accuracy check, and which won.

Structure Requirements met Accurate? Stronger?

Exit ticket

Write your reusable template in one line and say how it checks accuracy.

Challenge (stretch)

Test your template on a genuinely different task and note where it needed changing.

Week 4 - Core - Worksheet | Judging Outputs - Correct, Incorrect, Unclear

Name: _____ Class: _____ Date: _____

Triage three outputs

For each output, circle a verdict and write a checkable reason.

This output is ... because when I checked it against ...

I cannot verify this because ...

Signal and action

Name one signal that an output cannot be trusted, and the action you would take.

A warning signal is ...

My action would be ...

Output-triage record

Record your verdict, reason, and action for each output.

Output Verdict Reason (checkable) Action

Exit ticket

Flag these three outputs correct / incorrect / unclear and give a reason for each.

Challenge (stretch)

Find the single detail in a 'correct-looking' output that actually makes it incorrect, and explain how you spotted it.

Week 4 - Lab - Worksheet | Output Triage Clinic (Lab)

Name: _____ Class: _____ Date: _____

Clinic diagnoses

For each station, record the verdict, a checkable reason, and a verification action.

Verdict:

Reason:

Verification action:

Defend your ladder

Explain why your top output is more trustworthy than your bottom output.

Top is more trustworthy because ...

Bottom is least trustworthy because ...

Triage clinic record

Complete one row per station output.

Station Verdict Reason Verification action Ladder rank

Exit ticket

Name one risk of using the least trustworthy output as-is, and who it could affect.

Challenge (stretch)

Redesign one unclear output's prompt so its result could be triaged as clearly correct or incorrect.

Week 5 - Core - Worksheet | Risks and Responsible Use

Name: _____ Class: _____ Date: _____

Risk, who, rule

For two scenarios, complete the chain: risk, who is affected, responsible-use rule.

The risk is ...

It could affect ...

A rule that reduces it is ...

Spot the private data

Underline any personal data in a sample prompt and rewrite it safely.

Personal data found:

Safe rewrite:

Risk register

Record each risk you find with its who and rule.

Scenario Risk family Who is affected Responsible-use rule

Exit ticket

Name two genuine risks of using AI tools and one responsible-use rule that reduces one of them.

Challenge (stretch)

Write a class responsible-use charter of three rules, each matched to a specific risk it reduces.

Week 5 - Lab - Worksheet | Class AI Risk Audit (Lab)

Name: _____ Class: _____ Date: _____

Audit and flag

For each log entry, note whether it meets the rules and flag any gap.

Meets rules? Yes / No

Gap found:

Risk and who:

Prioritise and fix

Explain why your top risk is highest priority and write a rule to fix it.

Top risk because ...

Mitigation rule:

Class risk audit sheet

Record each flagged risk with its matrix placement and priority.

Log entry Gap / risk Likelihood Impact Priority

Exit ticket

State the highest-priority risk your group found and the one rule you would add to reduce it.

Challenge (stretch)

Design a one-page class responsible-use charter that closes every gap your audit found, ordered by priority.

Week 6 - Core - Worksheet | Bringing the Strands Together

Name: _____ Class: _____ Date: _____

Run the chain

Complete each of the four steps for the guided challenge.

System type:

My prompt:

Triage verdict and reason:

Risk check and choice:

Explain the links

Write two sentences linking system, prompt, and output.

Because the system was ...

The prompt made the output ... which means ...

Integration challenge record

Record your result for each strand.

Strand What I did Result Link to next strand

Exit ticket

Explain how the system type and prompt shaped your output, and name one responsible-use choice you made.

Challenge (stretch)

Redesign the prompt to lower the risk further, and predict how the output and triage would change.

Week 6 - Lab - Worksheet | Integrated Build-and-Check (Lab)

Name: _____ Class: _____ Date: _____

Build-and-check log

Record your evidence at each of the four stations.

System type + why:

Prompt:

Accuracy verdict + reason:

Risk verdict + mitigation:

Decide and diagnose

Make your use decision and name your weakest strand with a reason.

Decision: use / do not use / use-with-changes because ...

Weakest strand:

Action to improve:

End-to-end build-and-check record

Complete one row per station and finish with your decision.

Station / strand Evidence produced Verdict Note for signature solution

Exit ticket

State your use decision with a reason, and name the strand you most need to improve before the signature solution.

Challenge (stretch)

Rewrite your prompt so the output would earn a lower risk verdict, and predict how the use decision would change.

Week 7 - Core - Worksheet | Signature Solution - Planning

Name: _____ Class: _____ Date: _____

Name the five parts

Using your task card, complete each part of the plan in one line.

The goal is to ____.

The AI acts as a ____ (role).

The task is to ____.

One rule is ____.

The inputs are ____ (no personal data).

I will check accuracy by ____.

One risk is ____; my plan reduces it by ____.

Spot the weak plan

Read the two sample plans provided. Tick the stronger one and write one sentence saying which part makes it stronger.

The stronger plan is ____ because ____.

[VISUAL G6-W7-03 | Type: planner] • Purpose: Compare a complete plan against an incomplete one.

• Content & layout: Two side-by-side plan cards; Plan A has all five parts ticked, Plan B is missing the evaluation and risk rows.

• Labels & text: ['Plan A (complete)', 'Plan B (missing checks)']

• Style: greyscale line-art, clear age 11-12, UAE-inclusive

• Size & orientation: medium

• Alt-text: Two plan cards side by side, one complete and one missing evaluation and risk rows.

• Caption: A missing check is a hidden weak spot.

My plan review record

Fill one row for each of the five parts and mark whether it is ready.

Plan part What I wrote Ready? (Y/N)

Exit ticket

Write your goal sentence and the one risk your plan reduces.

Challenge (stretch)

Add a second rule to your prompt structure and explain which kind of wrong answer it is designed to stop.

Week 7 - Lab - Worksheet | Prototype and Pressure-Test the Plan (Lab)

Name: _____ Class: _____ Date: _____

Predict then test

Before the board run, predict how your prompt will handle one tricky input. After the run, record what actually happened.

I predict the output will be ____.

The actual output was ____.

The error type was ____ (wrong / unclear / missing).

Write your fallback branch

Turn your exposed failure into a safe fallback using the if-then frame.

If ____ then ____ and ask a person.

[VISUAL G6-W7-06 | Type: planner] • Purpose: Scaffold an if-then fallback branch.

• Content & layout: A branch template with an 'If check fails' node splitting into a 'then do' box and an 'ask a person' box.

• Labels & text: ['If check fails', 'Then do', 'Ask a person']

• Style: greyscale line-art, clear age 11-12, UAE-inclusive

• Size & orientation: medium

• Alt-text: A branch template splitting a failed check into a then-do action and an ask-a-person step.

• Caption: Every fallback keeps a person in the loop.

Prototype build/test log

Record each run: the input used, the output, whether it met the goal, and the error type if any.

Input (curated) Output Met goal? (Y/N) Error type

Exit ticket

Write the weak spot you found and the fallback branch you designed for it.

Challenge (stretch)

Design a second fallback for a different error type and say which failure is more likely and why.

Week 8 - Core - Worksheet | Signature Solution - Building

Name: _____ Class: _____ Date: _____

Flag the errors

Read the five curated outputs provided. Mark each correct, unclear, or wrong and flag the ones that fail the check.

Output ____ is ____ (correct / unclear / wrong) because ____.

[VISUAL G6-W8-03 | Type: build/test log] • Purpose: Practise evaluating and flagging outputs.

- Content & layout: A five-row table of inputs and outputs with an empty judgement column and a flag column;

two rows contain seeded errors.

- Labels & text: ['Input', 'Output', 'Judgement', 'Flag?']

- Style: greyscale line-art, clear age 11-12, UAE-inclusive

- Size & orientation: medium

- Alt-text: A five-row evaluation table with judgement and flag columns and two seeded errors.

- Caption: Evaluate every row - errors hide among correct-looking ones.

Log one improvement

Choose one flagged output and record your improvement using the loop frame.

Evaluate: ____.

Change: ____.

Result: ____.

Build and improvement log

Record each output, your judgement, whether you flagged it, and the improvement you made.

Input (curated) Output Judgement Improvement made

Exit ticket

Write your one improvement as evaluate -> change -> result.

Challenge (stretch)

Make a second improvement and use your log to say which change raised accuracy more, with evidence.

Week 8 - Lab - Worksheet | Extended Build and Evaluation (Lab)

Name: _____ Class: _____ Date: _____

Run and count

Record the result of each of the six curated test inputs, mark correct or not, and compute your accuracy count.

Input ____ output ____ is ____ (correct / not).

My accuracy is ____ out of 6.

[VISUAL G6-W8-06 | Type: build/test log] • Purpose: Scaffold the test-set run and accuracy count.

- Content & layout: A six-row grid for inputs, outputs and a correct column, with a boxed accuracy total at the bottom.

- Labels & text: ['Input', 'Output', 'Correct?', 'Total /6']

- Style: greyscale line-art, clear age 11-12, UAE-inclusive

- Size & orientation: medium

- Alt-text: A six-row grid with a boxed accuracy total out of six.

- Caption: Count every case to state accuracy honestly.

Document the fix and risk check

Record how your fallback handled a failure, the fix you made, and your three-question risk check.

The fallback handled ____ by ____.

My fix was ____ and the result was ____.

Risk check - accurate? ____ clear? ____ personal data used? no.

Test-set evaluation and risk record

Fill one row per test input with its result, then complete the risk-check summary below.

Input (curated) Output Correct? (Y/N) Note / fix

Exit ticket

Write your accuracy count and confirm your risk check found no personal data.

Challenge (stretch)

Add a seventh harder edge case, predict the result, test it, and say whether your solution generalised and why.

Week 9 - Core - Worksheet | Signature Solution - Refining & Finishing

Name: _____ Class: _____ Date: _____

Three-lens evaluation

Judge your solution on each lens. Give a score or a yes/no and the evidence behind it.

Accuracy: I tested ... and got ...

Usefulness: it helps my user because ...

Risk: I checked ... so ...

One improvement, proven

Describe the single change you made and show before and after evidence.

The change I made was ...

Before: ... After: ...

Nothing else broke because ...

Refinement log

Record each check, your judgement and the evidence.

Lens Judgement Evidence Action taken

Exit ticket

What one change improved your solution most, and how do you know?

Challenge (stretch)

Compare two possible improvements and justify, with evidence, why you chose one over the other.

Week 9 - Lab - Worksheet | Final Iteration and Risk Review (Lab)

Name: _____ Class: _____ Date: _____

Risk matrix

List three risks and place each by likelihood and seriousness, then decide an action.

Risk 1: ... likelihood ... seriousness ... action ...

Risk 2: ...

Risk 3: ...

Final iteration evidence

Record your last change and the before-and-after test results.

Final change: ...

Before: ... After: ...

Safe-and-fair statement: ...

Final iteration log

Log each risk, its matrix position, the action, and confirm privacy.

Risk Likelihood Seriousness Action / privacy check

Exit ticket

Write your safe-and-fair statement and one risk you managed with evidence.

Challenge (stretch)

Redesign one part of your solution so a high risk moves to a lower box on the matrix, and justify the change.

Week 10 - Core - Worksheet | Showcase, Assessment & Reflection

Name: _____ Class: _____ Date: _____

My two justified choices

Write two design or evaluation choices and give a reason and evidence for each.

Choice 1: ... because ... evidence ...

Choice 2: ... because ... evidence ...

Term-1 assessment task

Answer the four short prompts, one for each strand, at Emerging level.

Concepts: ...

Design & build: ...

Evaluate & use: ...

Wider effects: ...

Strand evidence tracker

For each strand, note your evidence and your confidence.

Strand My evidence Confidence Next step

Exit ticket

One thing I learned this term is ... and my goal for next term is ...

Challenge (stretch)

Explain a trade-off you made between two choices and why it was worth it.

Week 10 - Lab - Worksheet | Showcase Carousel and Reflection (Lab)

Name: _____ Class: _____ Date: _____

Carousel feedback slips

For two peers, write two stars and a wish, each tagged to a strand.

Star (strand): ...

Star (strand): ...

Wish (strand): ...

Four-strand Emerging summary and goal

Write one Emerging statement per strand, then set one Term-2 goal.

Concepts: I can ...

Design & build: I can ...

Evaluate & use: I can ...

Wider effects: I can ...

Term-2 goal: ...

End-of-term reflection tracker

For each strand, record your Emerging evidence and next-term focus.

Strand Emerging evidence Feedback received Term-2 focus

Exit ticket

My proudest strand this term is ... and my Term-2 goal toward Proficient is ...

Challenge (stretch)

Combine feedback from several peers into one improved design idea to try in Term 2, and explain your choice.

Word bank

Rule-based system - follows fixed IF-THEN rules a person wrote.

Learning system (model) - learns patterns from many examples (training data) instead of fixed rules.

Training data - the examples a learning system learns from.

Prompt - the words we give an AI system to tell it what to do.

Data quality / noisy data - how complete and correct data is; noisy data is messy or has errors.

Bias - when a model is unfair because its training data was unbalanced or noisy.

Trustworthy AI - an AI whose answers are reliable, fair and safe to use.

Personal information - facts about a real person that we keep private.

Two kinds of AI

A rule-based system follows fixed rules a person wrote (IF ... THEN ...). A learning system (model) is

different: it learns patterns from lots of examples called training data. If the training data is unbalanced or

messy, the model can be biased - even though no one wrote a wrong rule.

Fair or unfair? A quick story

We train a helper with many cat photos but very few dogs. Shown a new dog, it says 'cat'. No one wrote a

wrong rule - the training data was unbalanced, so the model learned a biased pattern.

Balanced, clean

training data gives fairer, more accurate answers.

My check

- Is the training data balanced?

- Is the data clean or noisy?

- Did I test different kinds of examples?

- Who might be left out or treated unfairly?

- What one change to the data would make it more trustworthy?

Keep it private

We never type a real name, photo, home or age into a tool. We use made-up (synthetic) data only. Once

information goes into a system it is hard to take back.

TEACHER PACK

Grade 6 (Year 7) | Phase 3 / Cycle 2 | AI Solution Builder

Main Course, Term 1 - Rule-Based vs Learning Systems

(ML Foundations)

Emerging tier | 10 weeks (Core + Lab)

Field Detail

Program Grade 6 Main Course, Term 1 (Emerging tier)

Unit theme Rule-Based vs Learning Systems - ML Foundations

Signature activity A structured, prompt-driven AI solution to a simple task, evaluated for accuracy and risk

Strands AI Conceptual Understanding | AI Solution Design & Development | Critical Evaluation & Informed Interaction | Ethics & Governance

Delivery model 2 x 45-minute periods per week - a Core lesson + a paired Lab lesson each week

Lessons in this pack 20 lessons (10 Core + 10 Lab)

1. About this Teacher Pack

This pack contains everything needed to teach the Grade 6 main course, Term 1: ten weeks of Emerging-tier

lessons in which students distinguish rule-based from learning systems, meet machine-learning foundations

(training and testing data), use structured prompts to complete tasks, judge outputs, weigh risks, and build a

structured, prompt-driven AI solution evaluated for accuracy and risk. It opens Phase 3 (AI Solution Builder) and

steps up from Grade 5 in technical depth, cognitive demand, and safeguarding. Each week is a Core lesson

(teaches the outcome) and a paired Lab lesson (deeper hands-on build/investigation on the same outcome).

Weeks 1-5 teach one strand focus each; Week 6 integrates them; Weeks 7-10 plan, build, refine and showcase

the signature solution.

How to use this pack

- Teach each week as Core (period 1) then Lab (period 2) on the same outcome, across the term.
- Keep all AI/ML use teacher-mediated and account-free; train any model only on non-personal data (objects/drawings/sounds); never enter personal data.
- Build the evaluate-verify-risk habit into every AI activity; provide the unplugged fallback whenever devices or approval are unavailable.
- Use the per-lesson formative check to confirm the outcome before moving on.
- Pair this Teacher Pack with the Student Lesson Content and Student Worksheets (same order).
- Weeks 7-10 run the signature solution end-to-end: plan (approve-to-build gate), build, refine, showcase and assess.
- Strand-naming note: the ethics strand is labelled 'Responsible AI Use' (Grade 4), 'Ethics & Responsible Use'

(Grade 5), and 'Ethics & Governance' (Grade 6); the data strand is labelled 'Data Awareness' (Grade 4), 'Data Literacy' (Grade 5), and folded into AI Conceptual Understanding from Grade 6 onward - these track the same

strand across grades, not separate strands.

Safeguarding & responsible use

Grade 6 adds introductory ML on top of AI interaction. Keep all AI and ML use teacher-mediated and account-free for

students; prefer board demonstrations, simulated-AI cards, or curated output sets, or a teacher-managed school-

approved child-safe assistant. Train any classifier only on non-personal data - objects, drawings, shapes, or sounds -

never faces, voices, names, or any personal or sensitive data. Never enter personal data into a prompt. Build

evaluation, verification, and risk checks into every activity. Keep all activity supervised, use only school-approved

devices, post nothing online, and follow ADEK safeguarding and responsible-use guidance.

Contexts and names are

UAE-appropriate and inclusive.

Materials & tools (UAE-accessible)

Tool / material What it is & why Access route Unplugged fallback

Rule vs learning demos

+ system cards

(unplugged)

A rule-card machine and a learn-from-examples

card game, plus real-system sort cards (W1-W2).

School-printed; teacher-

run; no account.

Discuss familiar

systems and sort on the board.

Teachable classifier

(introductory ML,

teacher-mediated)

An age-appropriate image/sound/pose

classifier trained on NON-personal data

(objects, drawings, shapes, sounds) to show how a model learns (W2).

Browser-based, teacher-

managed device; no

student accounts; no

faces/voices/names.

Unplugged paper-

classifier: students
hand-sort training and
testing cards.

Simulated-AI /
teacher-mediated AI
demo

Board demonstrations and simulated-AI cards
for structured prompting, task completion, and
output triage (W3-W9).

Account-free; teacher-
run; no personal data.

Curated pre-generated
output sets.

Output-evaluation,
verification & risk
sheets (printable)

Triage cards, reliability checklists, verification
flowcharts, and a risk matrix (W4-W10).

School-printed. Record in an exercise
book.

Signature-solution
planner & rubric
(printable)

Solution-architecture planner, project success
criteria, build/test logs, and the Emerging
analytic rubric (W6-W10).

School-printed. Plan and log on paper.

School-approved AI
assistant (optional,
teacher-managed)

A child-safe, school-approved assistant used
only by the teacher on the board for a live demo,
if permitted.

Teacher-managed class
account; no student sign-
in; no personal data.

Simulated-AI cards or
curated output sets.

Program at a glance

Lesson Strand Framework ref(s)

Week

1 -

Core

Unit Launch - Rule-Based vs Learning Systems Integration AIF·CU·G6·E AIF·SD·G6·E

AIF·CE·G6·E AIF·GE·G6·E

Week

1 - Lab

Rule vs Learning - Hands-On Exploration (Lab) Integration AIF·CU·G6·E AIF·SD·G6·E

AIF·CE·G6·E AIF·GE·G6·E

Week

2 -

Core

Rules vs Learning - How Each System Works AI Conceptual Understanding AIF·CU·G6·E

Week

2 - Lab

Train a Simple Model (Lab) AI Conceptual Understanding AIF·CU·G6·E

Week

3 -

Core

Structured Prompts to Complete a Task AI Solution Design & Development AIF·SD·G6·E

Week

3 - Lab

Prompt-to-Task Workshop (Lab) AI Solution Design & Development AIF·SD·G6·E

Week

4 -

Core

Judging Outputs - Correct, Incorrect, Unclear Critical Evaluation & Informed

Interaction

AIF·CE·G6·E

Week

4 - Lab

Output Triage Clinic (Lab) Critical Evaluation & Informed

Interaction

AIF·CE·G6·E

Week

5 -

Core

Risks and Responsible Use Ethics & Governance AIF·GE·G6·E

Week

5 - Lab

Class AI Risk Audit (Lab) Ethics & Governance AIF·GE·G6·E

Week

6 -

Core

Bringing the Strands Together Integration AIF·CU·G6·E AIF·SD·G6·E

AIF·CE·G6·E AIF·GE·G6·E

Week

6 - Lab

Integrated Build-and-Check (Lab) Integration AIF·CU·G6·E AIF·SD·G6·E
AIF·CE·G6·E AIF·GE·G6·E

Week

7 -

Core

Signature Solution - Planning Integration (Signature Solution -
Planning)

AIF·CU·G6·E AIF·SD·G6·E

AIF·CE·G6·E AIF·GE·G6·E

Week

7 - Lab

Prototype and Pressure-Test the Plan (Lab) Integration (Signature Solution -
Planning)

AIF·CU·G6·E AIF·SD·G6·E

AIF·CE·G6·E AIF·GE·G6·E

Week

8 -

Core

Signature Solution - Building Integration (Signature Solution -
Building)

AIF·CU·G6·E AIF·SD·G6·E

AIF·CE·G6·E AIF·GE·G6·E

Week

8 - Lab

Extended Build and Evaluation (Lab) Integration (Signature Solution -
Building)

AIF·CU·G6·E AIF·SD·G6·E

AIF·CE·G6·E AIF·GE·G6·E

Week

9 -

Core

Signature Solution - Refining & Finishing Integration (Signature Solution -
Refining & Finishing)

AIF·CU·G6·E AIF·SD·G6·E

AIF·CE·G6·E AIF·GE·G6·E

Week

9 - Lab

Final Iteration and Risk Review (Lab) Integration (Signature Solution -
Refining & Finishing)

AIF·CU·G6·E AIF·SD·G6·E

AIF·CE·G6·E AIF·GE·G6·E

Week

10 -

Core

Showcase, Assessment & Reflection Integration (Showcase, Assessment & Reflection)

AIF·CU·G6·E AIF·SD·G6·E

AIF·CE·G6·E AIF·GE·G6·E

Week

10 -

Lab

Showcase Carousel and Reflection (Lab) Integration (Showcase, Assessment & Reflection)

AIF·CU·G6·E AIF·SD·G6·E

AIF·CE·G6·E AIF·GE·G6·E

Week 1 - Core | Unit Launch - Rule-Based vs Learning Systems

Strand: Integration | Framework ref(s): AIF·CU·G6·E AIF·SD·G6·E AIF·CE·G6·E AIF·GE·G6·E |

1 x 45-minute period (Core)

Objectives & success criteria

- Generate at least two questions about how rule-based and learning systems differ.
- Classify three systems as 'follows fixed rules' or 'learns from data', justifying each.
- Co-construct the unit success criteria and the responsible-use norms for AI tools.

Student-facing success criteria:

- I can ask at least two genuine questions about rule-based and learning systems.
- I can sort three systems into 'follows fixed rules' or 'learns from data' and give a reason for each.
- I can help write our class responsible-use rules for AI tools.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In Grade 5 you explored what AI can and cannot do and practised giving clear instructions to a machine. This

unit builds on that by asking a deeper question: does the machine follow fixed rules a person wrote, or does it learn patterns

from examples? You do not need any coding for today.

Activate: Show two short board clips or cards: a vending machine giving change (fixed steps) and a photo app grouping pictures of

cats. Ask: which one was told exactly what to do, and which one seems to have figured something out?

Teach: Introduce the big idea of the whole unit - some systems follow rules written by people, and some systems learn patterns

from data. Give the plain definitions and one example of each.

Apply: In pairs, students place three system cards (traffic light timer, spam-email filter, spell-checker suggestions) under 'fixed rules' or 'learns from data' and write one reason.

Check: Each pair reads out one sorted card and its reason; you note which distinctions are secure and which are shaky for the

KWL.

Critical & analytical thinking focus

Question & analyse - frame the rule-vs-learning distinction (Understand/Analyse | DOK 1-2 | SOLO: Uni to Multistructural)

The technical concept (teacher background)

A rule-based system does exactly what a person has written into it, step by step. If a situation is not covered by a rule, the

system cannot handle it, because it has no rules to follow. A learning system is different - instead of being given every rule, it

is shown many examples and it finds patterns in that data. We call the pattern it builds a model.

The same learning system

can behave differently if you train it on different examples, because what it learns depends on the data it sees. This unit is

the start of Phase 3, where you begin to think like an AI solution builder, so you need to tell these two kinds of system apart.

Neither kind is magic - both are built and checked by people. Understanding which kind you are looking at helps you predict

when a system will work well and when it might fail.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook and

baseline

8 min

Show the vending machine and photo-grouping clips; hand out the baseline KWL sheet and ask students to write what they already Know and Want to know.

Watch, then jot one 'know'

and one 'want to know'

about how machines make decisions.

- Which machine was told every step?

- Which one seems to have worked something out for itself?

Teach the big

idea

10 min

Define rule-based system and learning system with the comparison visual; model one everyday example of each and think aloud about the difference.

Follow along, add the two definitions and one example

each to the student page.

- Where do the rules come from in

each case?

- What does a learning system need that a rule-based one does not?

Sort three

systems

Hand pairs the three system cards; circulate and press for reasons, not

Sort traffic-light timer, spam filter and spell-checker into

- What clue tells you it learned from examples?

12 min guesses. fixed-rules or learns-from-data; write a reason for each.

- Could a person write every rule for this task?

Question wall

8 min

Invite pairs to post at least two genuine questions on the class question wall; group similar questions live.

Write and post two questions you still have about rule-based versus learning systems.

- What is still puzzling about how a system learns?
- Which question would help us most this unit?

Co-construct norms

7 min

Lead a short discussion to agree the unit success criteria and a responsible-use charter (teacher-only tools, no personal data, always check outputs).

Suggest one class rule for using AI safely and fairly; record the agreed charter on the student page.

- Why should we never type someone's name or photo into an AI tool?

- How will we check if an AI output is correct?

Anticipated misconceptions

Misconception How to address it

A learning system just follows a big set of fixed rules that a programmer typed in.

Show that no person wrote a rule for every cat photo. The system was shown many examples and found the pattern itself. That pattern is the model, and it can change if the examples change.

If a machine gives an answer, it must understand the topic like a person.

Explain that a learning system matches patterns in data; it has no understanding or feelings. A confident-looking answer can still be wrong, which is why we always check.

Rule-based systems are old and useless, and learning systems are always better.

Both are useful. A traffic-light timer works perfectly with simple rules.

Learning systems are only better when the task is too varied for a person to write every rule.

Hands-on activity guide

Overview: An unplugged sorting and questioning launch: students compare a rule-card machine (fixed steps) with a set of

learn-from-examples cards, then sort three real systems and post questions. No devices or accounts are used by students.

Grouping / timing: Pairs for sorting, whole class for the question wall and charter.

Tool: Printed system cards and the baseline KWL sheet (teacher board demo only for the two clips). - Account-free for

students. Any AI clip is a curated, teacher-shown board demo using non-personal examples only.

Facilitation steps:

1. Watch the two short board clips and note which machine was told every step.
 2. Read the two definitions and add one example of each to your page.
 3. Sort the three system cards into 'fixed rules' or 'learns from data' with your partner.
 4. Write one reason under each card - what is your evidence?
 5. Post at least two genuine questions on the class question wall.
 6. Agree one responsible-use rule and add it to the class charter.
- Unplugged fallback: If no board clips are available, read the two printed scenario cards aloud and sort using only the printed cards - the lesson runs fully unplugged.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Baseline for the rule-based vs learning-system distinction and safe-use norms

- Check: Diagnostic KWL + question wall + rule/learning sort (not graded)

- Success criteria: Poses at least 2 genuine questions and sorts at least 2 of 3 systems with a reason.
- Evidence: Sorted systems + question wall on the baseline sheet.
- Feedback / adjustment: Cluster misconceptions to target Weeks 2-5 and plan the Primer; agree the responsible-use charter.

Differentiation

Representation: Pair each definition with an icon and a real photo-free example; colour-code the two sort columns.

Action & expression: Allow sorting by placing physical cards, pointing, or speaking the answer to a partner.

Engagement: Use familiar UAE examples (Salik toll gates as rule-based, a photo app grouping desert and beach pictures as learning).

People of Determination: Provide pre-cut large-print cards and a two-symbol choice board (gear = rules, seed = learns) for students who find writing hard; allow a scribe.

EAL: Give a bilingual glossary card for rule, learn, data, model, example; sentence frame 'This is ____ because ____'.

Stretch / next tier: Ask students to invent a fourth system card and challenge another pair to classify and justify it.

Answer key / model responses

Task / item Model response Marking note

Sort the traffic-light timer.

Fixed rules - it changes lights on a set timer that a person programmed.

Accept 'learns' only if the student describes a smart signal that adapts to traffic data.

Sort the spam-email filter.

Learns from data - it was trained on many example emails marked spam or not spam.

Key clue: it improves from examples, and no person wrote a rule for every email.

Write one question for the wall.

Example: 'How many examples does a system need before it learns well?'

Any genuine, on-topic question counts; reward

curiosity over neatness.

Suggest a responsible-use rule.

Example: 'We never type a real person's name, photo or voice into an AI tool.'

Look for a rule about privacy or checking outputs.

Assessment rubric (observable look-fors)

Criterion Emerging (expected) Approaching Not yet

Classifying systems Sorts at least 2 of 3 systems correctly with a sensible reason.

Sorts 1 system correctly or sorts without clear reasons.

Cannot yet sort or gives unrelated reasons.

Questioning Poses at least two genuine, on-topic questions.

Poses one on-topic question. No question posed yet.

Responsible-use thinking Suggests a clear safe-use rule and explains why.

Suggests a rule with little reason.

No safe-use idea offered yet.

Resources & links

- ADEK K-12 AI Fluency Framework - The Phase 3 outcomes this unit is built on; teacher reference. (<https://www.adek.gov.ae>)
- Class responsible-use charter template - Printable poster for co-constructing safe-use norms. (teacher-prepared handout)

Week 1 - Lab | Rule vs Learning - Hands-On Exploration (Lab)

Strand: Integration | Framework ref(s): AIF·CU·G6·E AIF·SD·G6·E AIF·CE·G6·E AIF·GE·G6·E | 1 x 45-minute period (Lab)

Objectives & success criteria

- Explore a rule-based demo and a simple learning demo and log the differences you observe.
- Devise a test that could tell whether a system follows rules or learns.
- Predict what would break each system and check your prediction.

Student-facing success criteria:

- I can run the rule-card machine and the learn-from-examples game and record what each does.
- I can log at least three differences I observed between them.
- I can design a fair test that would reveal whether a system follows rules or learns.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In this week's Core lesson you defined rule-based and learning systems and sorted examples. Now you

investigate them hands-on so the difference becomes something you can see and test.

Activate: Recap the two definitions in ten seconds using the gear and seed icons.

Teach: Explain today's two stations: a rule-card machine that always follows the same steps, and a learn-from-examples card

game where the group builds a pattern from cards.

Apply: Groups run both stations and log differences on the comparison sheet.

Check: Each group shares one difference and one prediction of what would break each system.

Critical & analytical thinking focus

Investigate & test - compare system behaviour (Analyse/Evaluate | DOK 2-3 | SOLO: Multistructural to Relational)

The technical concept (teacher background)

When you watch a rule-based system, its behaviour is completely predictable, because it always follows the same written

steps. If you give it something its rules do not cover, it either stops or gives a wrong result. A learning system behaves

differently - it was shown examples and built a model, so it can handle cases it has not seen exactly before, as long as they

resemble its examples. But it also fails in a different way: if you show it something very unlike its training examples, it

guesses badly. A good test to tell the two apart is to give the system a brand-new, unusual input and watch how it responds.

Today's stations let you feel this difference without any devices. The rule-card machine shows fixed, repeatable behaviour;

the card game shows a pattern being learned from examples the group chooses.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Set up stations

6 min

Explain the rule-card machine (a fixed if-then card stack) and the learn-from-examples game; assign groups and hand out the comparison log.

Read your role card and get your station materials ready.

- Which station will always do the same thing?
- Which one might change based on the cards we pick?

Run the rule-card machine

10 min

Guide groups to feed inputs through the fixed if-then cards and record outputs.

Feed several inputs,
including one strange one,
and log exactly what
happens each time.

- Did it ever change its behaviour?
- What happened with the strange input?

Play the learn-
from-examples
game

12 min

Run the card game: the group is shown
labelled example cards, then must
classify new cards using only the pattern
they saw.

Look at the example cards,
build a pattern as a group,
then classify the new cards
and log your results.

- How did more examples change your guesses?
- When did the group guess wrongly, and why?

Devise a test

10 min

Challenge each group to design one test
that would reveal whether an unknown
system follows rules or learns.

Write a test: what input
would you give, and what
result would show 'rules'

- What kind of input separates the two?
- How would the results look versus 'learns'? different?

Share and
predict

7 min

Collect one difference and one 'what
would break it' prediction from each
group; add strong ideas to the question
wall.

Share your best difference
and predict what would

break each system.

- Which system was more predictable?
- Which one adapts, and what is its weakness?

Anticipated misconceptions

Misconception How to address it

The learn-from-examples game is really just following hidden rules.

Point out that the group changed its pattern as new example cards appeared - a rule stack never changes. The pattern came from the examples, not from a fixed step list.

More example cards will always make the guesses better.

Show a case where adding a few odd or wrong cards made guesses worse. Quality and fairness of examples matter, not just how many there are.

A confident, fast guess in the game must be correct. Reveal a confidently wrong guess. Speed and confidence do not equal accuracy, which is why we test on new cards and check results.

Hands-on activity guide

Overview: A two-station unplugged investigation. Station A is a rule-card machine (fixed if-then cards) that always behaves the same way. Station B is a learn-from-examples card game where groups build a pattern from labelled non-personal cards (shapes, objects, drawings) and classify new ones. Students log differences, then design a test to tell rules from learning.

Grouping / timing: Groups of three with rotating roles: feeder, recorder, tester.

Tool: Printed rule-card stack, labelled example cards (shapes/objects/drawings - no faces, names or voices), and the comparison log. - Fully unplugged and account-free. No student devices; no personal data is ever used.

Facilitation steps:

1. Assign roles: feeder, recorder, tester.
2. At Station A, feed five inputs through the rule cards, including one unusual input, and log each output.
3. Note whether Station A ever changed its behaviour.
4. At Station B, study the labelled example cards and agree a pattern as a group.
5. Classify five new cards using only the pattern you built; log your guesses and the correct answers.
6. Record at least three differences between the two stations.
7. Design one test that would reveal whether an unknown system follows rules or learns.
8. Predict what input would break each station and note it.

Unplugged fallback: If printed cards are unavailable, run both stations verbally: the teacher reads fixed if-then rules for Station A, and describes example items aloud for Station B while students classify by ear.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Checks whether students can observe and articulate the rule-based vs learning

distinction through investigation

- Check: Comparison log with at least three logged differences and one devised test
- Success criteria: Logs at least three real differences and writes a test that would actually separate rules from

learning.

- Evidence: Completed comparison log and the group's devised test.

- Feedback / adjustment: If tests do not separate the two, model one strong test (give an unseen, unusual input) and have groups revise.

Differentiation

Representation: Offer the log as a table with icons; provide a worked row as a model.

Action & expression: Roles let students contribute by feeding cards, recording, or testing - writing is not the only route.

Engagement: Use UAE-themed non-personal cards (dates, camels drawn as line-art, geometric Islamic patterns) for the learning game.

People of Determination: Give large-print cards, a card holder, and a partner scribe; allow the tester role to respond by pointing.

EAL: Provide sentence frames: 'Station A always ____', 'Station B changed when ____'; bilingual labels on cards.

Stretch / next tier: Ask groups to add a deliberately unfair set of example cards and explain how it changed the learned pattern.

Answer key / model responses

Task / item Model response Marking note

Log a difference between the stations.

Station A always gave the same output; Station B's guesses changed as we saw more example cards.

Any accurate observed difference is valid.

What happened with the unusual input at Station

A?

It had no rule for it, so it stopped or gave a wrong answer. Reinforces that rule-based systems fail on uncovered cases.

Design a test to tell rules from learning.

Give the system a brand-new, unusual input; if it always reacts the same way it is rule-based, if it adapts using patterns it likely learned.

Accept any test using a novel input to reveal adaptation.

Predict what would break Station B.

Showing it a card very unlike any example - it would guess badly. Links to data quality and coverage.

Assessment rubric (observable look-fors)

Criterion Emerging (expected) Approaching Not yet

Observation log Logs at least three accurate differences with detail.

Logs one or two differences. No clear differences logged yet.

Devising a test Designs a test that would truly separate rules from learning.

Designs a test that partly separates them.

No workable test yet.

Prediction Predicts what breaks each system with a sound reason.

Predicts for one system. No reasoned prediction yet.

Resources & links

- Rule-card machine pack - Printable if-then cards for Station A. (teacher-prepared handout)
- Learn-from-examples card set - Labelled non-personal cards (shapes, objects, patterns) for Station B. (teacher-prepared handout)

Week 2 - Core | Rules vs Learning - How Each System Works

Strand: AI Conceptual Understanding | Framework ref(s): AIF·CU·G6·E | 1 x 45-minute period (Core)

Learning outcome (framework): Recognises the difference between rule-based and learning systems.

Objectives & success criteria

- Distinguish a rule-based system from a learning system using two clear features.
- Give one everyday example of each and explain the difference.
- Use the terms rule-based, learning, data and model accurately.

Student-facing success criteria:

- I can name two features that separate rule-based and learning systems.
- I can give one real example of each and explain why it fits.
- I can use the words rule-based, learning, data and model correctly.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: Last week you launched the unit and investigated a rule-card machine and a learning game. Now you sharpen the distinction into clear, named features you can use to classify any system.

Activate: Flash four quick systems on the board; thumbs up for 'learns', thumbs down for 'rules'.
Teach: Build a two-column feature chart together: where rules come from, and how each system handles a new case.

Apply: Pairs complete the feature chart for two everyday UAE systems and name the distinguishing feature.

Check: Cold-call pairs to justify one classification using a feature, not a surface trait.

Critical & analytical thinking focus

Analyse & classify - contrast system types (Understand/Analyse | DOK 2 | SOLO: Multistructural)

The technical concept (teacher background)

A rule-based system and a learning system can do the same job yet work in completely different ways. In a rule-based

system, a person decides the logic in advance and writes it as fixed steps; the data flows through those steps unchanged. In

a learning system, a person does not write the logic - they collect data, label it, and let the system build a model that

captures the pattern. Two features separate them cleanly. First, where the logic comes from: written by a person versus

learned from data. Second, how they handle something new: a rule-based system can only handle cases its rules cover,

while a learning system can generalise to new cases that resemble its examples. The model is the heart of a learning system

- it is the pattern learned from the data, and it is only as good as the examples it was trained on.

Knowing these features lets

you classify almost any system you meet.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Warm-up sort

7 min

Flash four systems; students vote rules or learns and give a one-word reason.

Vote and share a quick reason for two of them.

- What made you vote that way?
- Was it a surface clue or a real feature?

Build the feature chart

12 min

Co-construct a two-column chart: source of logic, handling of new cases, need for data; model the language rule-based, learning, data, model.

Copy the feature chart and

add one example word to
each row.

- Who writes the logic in each
column?

- Which column needs lots of
data?

Classify
everyday
systems

12 min

Give pairs two UAE examples (a metro
ticket gate; a voice assistant that
improves with use) to classify using
features.

Classify each example and
name the feature that
decides it.

- Which feature is the strongest
evidence here?

- Could a person write every rule
for this?

Precision with
words

Run a quick 'use the word correctly' round
for data and model; correct loose usage.

Complete two sentences
using data and model

- Is the model the same as the
data?

8 min accurately. • What is the model built from?

Exit ticket

6 min

Issue the exit ticket: classify two systems
and give the distinguishing feature.

Complete the exit ticket
independently.

- What single feature best proves
your answer?

- Did you avoid a surface trait?

Anticipated misconceptions

Misconception How to address it

You can tell a learning system just because it is on a
phone or looks modern.

That is a surface trait. A calculator app is modern but rule-based.

Judge by the features: where the logic comes from and how it handles new cases.

The data and the model are the same thing. The data is the examples shown to the system; the model is the pattern it builds from that data. Different data can build a different model.

A learning system understands the meaning of what it classifies.

It matches patterns in data, it does not understand meaning. It can classify a cat photo without knowing what a cat is.

Hands-on activity guide

Overview: A structured classification activity: students build a feature chart, then use it to classify everyday UAE systems, defending each choice with a feature rather than a surface trait. Unplugged; any AI shown is a teacher board demo.

Grouping / timing: Pairs, with whole-class charting.

Tool: Feature-chart handout and everyday-system cards. - Account-free for students; teacher-only board demo for any live tool.

Facilitation steps:

1. Copy the two-column feature chart.
2. Add one example to each row with your partner.
3. Classify two everyday systems using the chart.
4. For each, underline the feature that decided your answer.
5. Write two sentences using data and model correctly.

Unplugged fallback: Runs fully unplugged using only the printed chart and cards; no demo required.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Directly checks 'recognises the difference between rule-based and learning systems'

- Check: Exit ticket: classify two systems and give the feature that distinguishes them
- Success criteria: Correctly classifies both and names a distinguishing feature (not a surface trait).
- Evidence: Exit slips sorted secure / partial / not-yet.
- Feedback / adjustment: Re-teach with a two-column feature chart if under 70 percent distinguish correctly;

Primer next lesson.

Differentiation

Representation: Provide the feature chart pre-filled with icons for lower-literacy readers.

Action & expression: Let students classify by placing a token in a column or by writing.

Engagement: Use systems students meet daily in the UAE (metro gate, weather app, voice assistant).

People of Determination: Offer a two-choice token board and reduce the exit ticket to one system with support; allow verbal answers.

EAL: Bilingual glossary; sentence frame 'This is a ____ system because the logic is ____'.

Stretch / next tier: Ask students to find a system that is hard to classify and argue both sides using features.

Answer key / model responses

Task / item Model response Marking note

Classify a metro ticket gate.

Rule-based - a person set fixed rules (valid card opens the gate). Distinguishing feature: logic written by a person.

Classify a voice assistant that improves with use.

Learning - it builds a model from lots of voice and text data and handles new phrases.

Distinguishing feature: handles new cases by generalising.

Use the word model in a sentence.

Example: 'The model is the pattern the system learned from many example photos.'

Check model is not confused with data.

Name two features that separate the systems.

Where the logic comes from, and how each handles a new case. Both features must be non-surface.

Assessment rubric (observable look-fors)

Criterion Emerging (expected) Approaching Not yet

Distinguishing features Names two clear, non-surface features that separate the systems.

Names one feature or mixes in a surface trait.

No valid feature yet.

Classifying with justification Classifies both examples correctly with a feature reason.

Classifies one correctly. Cannot yet classify with a reason.

Using vocabulary Uses data and model accurately and distinctly.

Uses one term accurately. Terms confused or unused.

Resources & links

- Feature-chart handout - Two-column chart comparing rule-based and learning systems. (teacher-prepared handout)
- ADEK K-12 AI Fluency Framework - Outcome reference for AI Conceptual Understanding. (<https://www.adek.gov.ae>)

Week 2 - Lab | Train a Simple Model (Lab)

Strand: AI Conceptual Understanding | Framework ref(s): AIF·CU·G6·E | 1 x 45-minute period (Lab)

Learning outcome (framework): Recognises the difference between rule-based and learning systems.

Objectives & success criteria

- Train a simple classifier on non-personal examples (objects/drawings/sounds) and test it on new examples.
- Explain how the training examples shaped what the model learned.
- Run a basic risk and data-safety check on the training set.

Student-facing success criteria:

- I can sort examples into training and testing sets and train a simple classifier.
- I can test the model on new examples and record how it did.
- I can explain how my training examples changed what the model learned, and name one data-safety rule.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In this week's Core lesson you learned that a learning system builds a model from data. Now you build one

yourself - either with a teachable image classifier on non-personal objects or with an unplugged paper-classifier - and see

how the training examples shape it.

Activate: Show the training-to-testing flow diagram and recap: examples in, model built, new examples tested.

Teach: Explain the two golden rules: keep training and testing examples separate, and never use personal data (no faces, voices or names).

Apply: Groups build a training set of non-personal examples, train, then test on held-back examples and log results.

Check: Each group reports one result and one change that would improve the model.

Critical & analytical thinking focus

Build & investigate - train and test a model (Apply/Analyse/Evaluate | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

Training a model means showing a learning system many labelled examples so it can find the pattern that separates the

categories. We split our examples into two sets on purpose. The training set is what the model learns from. The testing set is

examples the model has never seen, which we use to check whether it really learned the pattern or just memorised. If a model does well on training examples but poorly on new ones, it did not truly learn - this is why testing on fresh data matters. The examples you choose shape the model directly: if all your 'apple' pictures are red, the model may fail on a green apple, because it learned 'red' instead of 'apple'. That is why data quality, variety and fairness matter as much as quantity. A basic risk check asks two questions: are the examples safe and non-personal, and are they varied enough to be fair?

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Set the golden

rules

6 min

Introduce the training/testing split and the data-safety rule (objects, drawings, shapes or sounds only - never faces, voices or names); show the flow diagram.

Note the two golden rules

on your log.

- Why hold back some examples for testing?

- Why never use a real face or name?

Build the

training set

10 min

Guide groups to collect and label 8-10 non-personal examples per category and set aside 3 for testing.

Collect and label your training examples; keep your testing examples separate and hidden.

- Are your examples varied enough?

- Did any personal data sneak in?

Train the model

8 min

Run the teachable classifier on the board (teacher-operated) with each group's set, or have groups build the paper-classifier

pattern.

Watch or build your model
from the training examples
only.

- What pattern is the model picking up?
- Did you train on testing examples by mistake?

Test on new
examples

12 min

Feed each group's held-back testing
examples and record correct/incorrect;
discuss surprises.

Predict, then test each held-
back example and log
whether the model got it
right.

- Did it work on examples it had never seen?
- Which examples fooled it, and why?

Risk and
improve

9 min

Lead a basic risk check and ask each
group for one improvement to the training
data.

Complete the data-safety
and risk check; write one
way to improve your training
set.

- Was your data safe and fair?
- What example would make the model stronger?

Anticipated misconceptions

Misconception How to address it

Testing the model on the same examples it trained
on proves it works.

That only shows it memorised. Real proof comes from testing on new,
held-back examples the model has never seen.

Adding more examples always makes the model
better.

Only if they are varied and fair. Ten red apples will not teach 'apple' -

they teach 'red'. Variety and quality matter, not just quantity.

The model understands what an apple is. It only found a pattern in pixels or sounds. It has no understanding,

which is why an unusual example can fool it.

Hands-on activity guide

Overview: A substantial build-and-test lab. Groups create a training set of non-personal examples (objects, simple

drawings, shapes or sounds), the teacher trains a school-approved child-safe teachable classifier on the board (or groups

build an unplugged paper-classifier), then everyone tests the model on held-back examples and runs a data-safety and risk

check. The training-to-testing flow and the data-safety rule are central.

Grouping / timing: Groups of three: data collector, labeller, tester.

Tool: A school-approved child-safe teachable image/sound classifier, operated by the teacher only, on non-personal data;

OR an unplugged paper-classifier kit. - Account-free for students. If a digital classifier is used, only the teacher operates it,

only on non-personal objects/drawings/shapes/sounds - never faces, voices or names.

Facilitation steps:

1. Choose two categories of non-personal items (for example, dates versus dried lemons, or circles versus triangles).
2. Collect and label 8-10 clear training examples per category.
3. Set aside 3 testing examples per category and keep them hidden.
4. Train the model (teacher-operated classifier or your paper-classifier) using training examples only.
5. Predict how each testing example will be classified.
6. Test each held-back example and log correct or incorrect.
7. Explain how your training examples shaped what the model learned.
8. Run the data-safety and risk check, then write one improvement to your training set.

Unplugged fallback: Unplugged paper-classifier: groups sort labelled cards into a decision chart built only from training cards,

then classify held-back testing cards by hand - identical learning, no devices.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Deepens 'recognises the difference between rule-based and learning systems' by

building a learning system and seeing how data shapes it

• Check: Training-and-testing log with results and an explanation of how the data shaped the model

• Success criteria: Trains and tests a model on separate example sets and explains how the training examples changed what it learned.

• Evidence: Completed training/testing log plus the data-safety and risk check.

• Feedback / adjustment: If groups train and test on the same examples, re-model the split and have them re-run

on held-back data.

Differentiation

Representation: Provide the flow diagram and a worked example of a labelled training set.

Action & expression: Roles allow contribution through collecting, labelling or testing; verbal logging accepted.

Engagement: Use appealing UAE non-personal items (dates, geometric patterns, drawn camels, everyday classroom objects).

People of Determination: Provide pre-labelled example cards, large images, and a partner scribe; allow the tester to signal correct/incorrect by card.

EAL: Bilingual labels on categories; sentence frame 'The model learned ____ because my examples were ____'.

Stretch / next tier: Ask groups to design a 'trick' testing example and predict whether the model will fail and why.

Answer key / model responses

Task / item Model response Marking note

Why keep testing

examples separate?

So we can check the model on things it never saw and know it truly learned, not memorised.

Core training/testing idea.

Explain how your training examples shaped the model.

Example: all our circles were large, so the model missed a small circle - it learned 'large' as well as 'round'.

Reward any accurate cause-and-effect from data to model.

Run the data-safety check.

All examples were objects or drawings, no faces, voices or names - safe.

Must confirm non-personal data.

Suggest one improvement.

Add varied examples (different sizes, colours, angles) so the model learns the real category.

Links quality and variety to accuracy.

Assessment rubric (observable look-fors)

Criterion Emerging (expected) Approaching Not yet

Training and testing Keeps sets separate, trains, and

tests on new examples correctly.

Trains and tests but blurs the two sets.

Does not yet separate training and testing.

Explaining data effect Clearly explains how the examples shaped the model.

Partial explanation. No link between data and model yet.

Data safety and risk Completes the safety check and names a real risk or improvement.

Partial safety check. Safety check not completed.

Resources & links

- School-approved teachable classifier - Child-safe image/sound trainer, teacher-operated on non-personal data only. (school-approved tool (teacher account))
- Paper-classifier kit - Unplugged decision-chart and labelled non-personal cards. (teacher-prepared handout)

Week 3 - Core | Structured Prompts to Complete a Task

Strand: AI Solution Design & Development | Framework ref(s): AIF·SD·G6·E | 1 x 45-minute period (Core)

Learning outcome (framework): Uses AI tools with structured prompts to complete tasks.

Objectives & success criteria

- Use a structured prompt to complete a defined task with an AI tool.
- Describe the input structure used and whether the task was completed.
- Adjust the prompt once to complete the task more fully.

Student-facing success criteria:

- I can write a structured prompt with a clear goal, context and format.
- I can say whether the task was completed and why.
- I can improve my prompt once to complete the task more fully.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: You now know the difference between rule-based and learning systems. Many AI tools are learning systems, so the way you word your request - the prompt - changes what you get back. Today you learn to structure a prompt to complete a real task.

Activate: Show two prompts for the same task - a vague one and a structured one - and the two curated outputs they produced.

Teach: Introduce a simple prompt structure: Goal, Context, Format. Model completing one class task with it.

Apply: Pairs write a structured prompt for a defined task; the teacher runs it on the approved tool and shows the output.

Check: Pairs judge whether the task was completed and write one adjustment.

Critical & analytical thinking focus

Apply & evaluate - structured prompting to a goal (Apply/Evaluate | DOK 2-3 | SOLO: Relational)

The technical concept (teacher background)

A prompt is the instruction you give an AI tool, and because many AI tools are learning systems, the wording strongly shapes

the result. A structured prompt states three things clearly: the goal (what you want done), the context (the details the tool

needs), and the format (how the answer should look). A vague prompt leaves the tool guessing, so it often returns

something off-target or incomplete. A structured prompt narrows the possibilities and makes the task far more likely to be

completed. Prompting is not a one-shot skill - you evaluate the output against the task, then adjust the prompt to close the

gap. This is the start of designing AI solutions: define the task, structure the input, check the output, and improve. Because

the tool can produce confident but wrong answers, you always verify the result and never include personal data in the

prompt.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Two prompts,

two outputs

8 min

Show a vague prompt and a structured

prompt for the same task with their

curated outputs; ask which completed

the task.

Compare the two outputs

and note why one worked

better.

- What did the vague prompt leave out?

- Which output actually finished the task?

Teach Goal-

Context-Format

10 min

Introduce the three-part structure; model

writing one prompt for a class task (for

example, summarise a short paragraph

about the UAE for a Grade 4 reader in

three bullet points).

Copy the structure and label
the three parts in the model
prompt.

- What is the goal here?
- What context does the tool need?
- What format did we ask for?

Write a
structured
prompt

12 min

Give pairs a defined task; circulate as they
draft a Goal-Context-Format prompt; run
selected prompts on the approved tool at
the board.

Write a structured prompt
for your task using all three
parts.

- Have you stated the goal clearly?
- Is the format specific?

Evaluate and
adjust

9 min

Display outputs; guide pairs to judge task
completion and make one improving
adjustment; rerun one.

Decide if the task was
completed, then adjust your
prompt once to complete it
more fully.

- What part of the task is missing?
- Which single change would fix it?

Verify and log

6 min

Model a quick verification and risk check;
students complete the task-completion
log.

Verify the output against the
task and finish your
completion log.

- Is the output actually correct?
- Did any personal data go into the
prompt? It should not.

Anticipated misconceptions

Misconception How to address it

A longer prompt is automatically a better prompt. Structure beats length. A short prompt with a clear goal, context and format often beats a long, rambling one. Add detail only where it helps the task.

If the AI gives a confident, polished answer, the task is complete and correct.

A confident output can still be wrong or incomplete. Always compare it against the task and verify before trusting it.

You should put lots of personal details in a prompt so the AI knows who you are.

Never put personal or identifying data into a prompt. Give only the task details the tool needs; keep names, photos and private facts out.

Hands-on activity guide

Overview: A guided structured-prompting session. Pairs write Goal-Context-Format prompts for a defined, non-personal task; the teacher runs approved prompts on a school-approved child-safe tool and displays outputs. Pairs evaluate completion, make one adjustment, verify, and run a risk check.

Grouping / timing: Pairs, with teacher-mediated tool use at the board.

Tool: A school-approved child-safe AI tool, operated by the teacher only; students write prompts on paper first. - Account-free for students. All tool use is teacher-mediated on the board; no student logins; no personal data in any prompt.

Facilitation steps:

1. Read the defined task and underline what 'complete' means.
2. Write a structured prompt with Goal, Context and Format.
3. Hand your prompt to the teacher to run on the approved tool.
4. Judge whether the output completed the task.
5. Make one adjustment to your prompt to complete it more fully.
6. Verify the output against the task and complete a quick risk check.

Unplugged fallback: Use simulated-AI cards: pre-printed outputs for a vague prompt and a structured prompt so pairs practise structuring, evaluating and adjusting with no live tool.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Directly checks 'uses AI tools with structured prompts to complete tasks'

- Check: Task-completion log: structured prompt, output, 'task complete?' + one adjustment
- Success criteria: Completes the task using a structured prompt and records one adjustment that improved completion.
- Evidence: Task-completion log.
- Feedback / adjustment: Provide a stronger prompt structure for incomplete tasks; extend to multi-part prompts. Tool use is teacher-mediated and safe.

Differentiation

Representation: Provide a Goal-Context-Format prompt frame with an example filled in.

Action & expression: Students can dictate their prompt to a scribe or write it; evaluation can be spoken.

Engagement: Use tasks rooted in UAE topics students care about (a short summary about Sheikh Zayed's tree-planting, a packing list for a desert trip).

People of Determination: Offer a cloze prompt frame and reduce to completing one part at a time; allow verbal responses.

EAL: Bilingual labels for Goal, Context, Format; sentence frames for the evaluation.

Stretch / next tier: Ask students to add a fourth element (Audience) and show how it improved the output.

Answer key / model responses

Task / item Model response Marking note

Identify the three parts of a structured prompt.

Goal (what to do), Context (details the tool needs), Format (how the answer should look).

All three must be present.

Say whether a task was completed.

Example: 'Not fully - it gave four bullets but I asked for three, and one was off-topic.'

Look for judgement against the task, not just 'it looks good'.

Adjust the prompt once. Example: add 'Give exactly three bullet points, each under 12 words.'

One clear change that improves completion.

Run a risk check. No personal data was included; I verified the facts against the source before trusting them.

Both privacy and verification should appear.

Assessment rubric (observable look-fors)

Criterion Emerging (expected) Approaching Not yet

Structured prompt Writes a prompt with a clear goal, context and format.

Includes two of the three parts. Prompt is vague or unstructured.

Evaluating completion Judges completion accurately against the task.

Judges partly or vaguely. No clear judgement yet.

Adjusting Makes one adjustment that improves completion.

Makes a change with little effect.

No useful adjustment yet.

Resources & links

- Goal-Context-Format prompt frame - Printable scaffold for structuring prompts. (teacher-prepared handout)
- School-approved child-safe AI tool - Teacher-operated tool for board demonstrations. (school-approved tool (teacher account))

Week 3 - Lab | Prompt-to-Task Workshop (Lab)

Strand: AI Solution Design & Development | Framework ref(s): AIF·SD·G6·E | 1 x 45-minute period (Lab)

Learning outcome (framework): Uses AI tools with structured prompts to complete tasks.

Objectives & success criteria

- Complete a second, richer task and compare which prompt structure worked better.
- Design a reusable prompt structure for a class of tasks.
- Evaluate the outputs for accuracy and run a basic risk check.

Student-facing success criteria:

- I can complete a richer task with a structured prompt and compare two structures.
- I can design a reusable prompt template for similar tasks.
- I can check an output for accuracy and complete a risk check.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In this week's Core lesson you learned the Goal-Context-Format structure and completed one task. Now you

tackle a richer task, compare prompt structures, and design a reusable template - the first step of the signature activity in later weeks.

Activate: Recap Goal-Context-Format and show two structured prompts that completed a task to different degrees.

Teach: Introduce comparing structures and designing a reusable template for a class of tasks (for example, 'summarise any short text for a younger reader').

Apply: Groups complete a richer task with two prompt structures, compare completion and accuracy, then draft a reusable template.

Check: Each group shares its template and one accuracy check it ran.

Critical & analytical thinking focus

Design & evaluate - build a reusable prompt structure (Create/Evaluate | DOK 3 | SOLO: Relational to Extended abstract)

The technical concept (teacher background)

Once you can structure a single prompt, the next design step is to compare structures and reuse what works. A richer task - one with several requirements - shows the difference between prompt structures clearly, because a weak structure leaves

parts of the task unfinished. By running two structures on the same task and comparing completion and accuracy, you gather evidence about which design is better, rather than guessing. From that evidence you can build a reusable template: a prompt skeleton with labelled slots that works for a whole class of similar tasks. This is real solution design - you are not just wording one request, you are building a tool others can reuse. Because outputs can be confidently wrong, every reusable template includes a verification step and a risk check, and no template ever asks for personal data. This workshop is the foundation for the signature activity, where you will evaluate a prompt-driven solution for accuracy and risk.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Recap and set
the task

6 min

Recap Goal-Context-Format; present a richer task with three requirements and explain the compare-and-template goal.

Read the richer task and list its three requirements.

- What are the separate requirements here?
- What would 'fully complete' look like?

Run two
structures

13 min

Have groups draft two structured prompts (for example, one with format first, one with context first); run selected prompts on the approved tool and display outputs.

Draft two structured prompts for the same task and predict which will complete it better.

- Which structure covers all three requirements?
- Where might each fall short?

Compare
completion and
accuracy

Guide groups to score each output for

completion and verify accuracy against a source.

Compare the two outputs,
mark which requirements
were met, and check the

- Which structure completed more of the task?
- Was every claim in the output 10 min facts. accurate?

Design a
reusable
template
10 min

Support groups to turn the winning structure into a labelled reusable template with a verify-and-risk step built in.

Write a reusable prompt
template with labelled slots
plus a verification and risk
check.

- Which slots stay the same for every task?
- Where does the verify step belong?

Share and risk
check
6 min

Collect templates; run a quick class risk check on personal-data and accuracy.

Share your template and complete the risk check.

- Could this template leak personal data? It must not.
- How does it force a check on accuracy?

Anticipated misconceptions

Misconception How to address it

Once a template completes one task, it will work perfectly for every task without checking.

A template is a strong starting point, not a guarantee. Each new task still needs the output verified for completion and accuracy.

The structure that gives the longest answer is the best structure.

Judge by how fully and accurately it completes the task's requirements, not by length. A shorter, complete answer wins. Accuracy checking is optional if the answer sounds right.

Sounding right is not being right. Every template must include a verification step against a reliable source.

Hands-on activity guide

Overview: A design workshop and the on-ramp to the signature activity. Groups complete a richer, multi-requirement task

using two prompt structures, compare completion and accuracy, then design a reusable prompt template with a built-in

verification and risk check. Tool use is teacher-mediated; an unplugged fallback uses prompt cards and curated outputs.

Grouping / timing: Groups of three: prompt writer, evaluator, template designer.

Tool: A school-approved child-safe AI tool operated by the teacher; prompt cards and curated outputs for the fallback. -

Account-free for students. All live tool use is teacher-mediated on the board; no personal data in any prompt.

Facilitation steps:

1. List the three requirements of the richer task.
2. Draft two structured prompts that order Goal, Context and Format differently.
3. Predict which structure will complete the task more fully.
4. Have the teacher run both prompts; record the outputs.
5. Mark which requirements each output met and verify the facts.
6. Choose the stronger structure and turn it into a reusable template with labelled slots.
7. Build a verification step and a risk check into the template.
8. Test your template mentally on a second, similar task.

Unplugged fallback: Unplugged prompt-card workshop: groups arrange Goal-Context-Format cards into two structures and match them to curated printed outputs, then design their template on paper - no live tool needed.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Extends 'uses AI tools with structured prompts to complete tasks' toward reusable, evaluated solutions

- Check: A reusable prompt template plus a comparison of two structures with accuracy checks
- Success criteria: Compares two structures on completion and accuracy and designs a reusable template with a verification step.
- Evidence: Comparison notes and the reusable template with its risk check.
- Feedback / adjustment: If templates omit verification, model a simple check-against-source step and have groups add it.

Differentiation

Representation: Provide a template skeleton with labelled empty slots and a worked example.
Action & expression: Roles let students contribute by writing prompts, evaluating, or designing the template; verbal contribution accepted.

Engagement: Use motivating UAE tasks (a visitor guide to a local souq, a short recycling notice for the class).

People of Determination: Offer a partly built template and a checklist for the comparison; allow the evaluator to respond by ticking a chart.

EAL: Bilingual slot labels; sentence frame 'Structure A completed ____ requirements; Structure B completed ____'.

Stretch / next tier: Ask groups to test their template on a genuinely different task type and report where it needed changing.

Answer key / model responses

Task / item Model response Marking note

List the requirements of the richer task.

Example for a souq visitor guide: three tips, each one sentence, friendly tone.

Requirements must be specific and checkable.

Compare the two structures.

Example: 'Structure A met all three requirements; Structure B missed the tone, so A was stronger.'

Comparison must be on completion or accuracy, not length.

Design a reusable template.

Goal: [task]; Context: [details, no personal data]; Format: [exact format]; then Verify: check facts against a source.

Template must include a verify/risk step.

Run the risk check. No personal data requested; output verified against a reliable source before use.

Both privacy and accuracy must appear.

Assessment rubric (observable look-fors)

Criterion Emerging (expected) Approaching Not yet

Comparing structures Compares two structures on completion and accuracy with evidence.

Compares loosely or on one

measure.

No real comparison yet.

Reusable template Designs a clear reusable template with labelled slots.

Designs a partial template. No workable template yet.

Accuracy and risk Builds in a verification step and completes the risk check.

Includes one of verification or risk check.

Neither included yet.

Resources & links

- Reusable template skeleton - Printable prompt template with labelled slots and a verify step. (teacher-prepared handout)
- Prompt-card and curated-output pack - Unplugged fallback materials for the workshop. (teacher-prepared handout)

Week 4 - Core | Judging Outputs - Correct, Incorrect, Unclear

Strand: Critical Evaluation & Informed Interaction | Framework ref(s): AIF·CE·G6·E | 1 x 45-minute period (Core)

Learning outcome (framework): Identifies when outputs are incorrect or unclear.

Objectives & success criteria

- Identify, in a set of outputs, which are incorrect or unclear and say why.
- Explain one signal that an output cannot be trusted as-is.
- Recommend one action to take when an output is unclear.

Student-facing success criteria:

- I can sort outputs into correct, incorrect, and unclear.
- I can give a reason for each judgement I make.
- I can name one action to take when an output cannot be trusted.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In Grade 5 you learned that AI tools can be confidently wrong. In earlier Grade 6 weeks you saw how a system

type and a prompt shape an output. Today you turn that into a habit: checking an output before you use it.

Activate: Show two answers to the same question - one right, one wrong but confident. Ask: which would you trust, and how could you tell?

Teach: Explain three verdicts an output can earn: correct (matches the task and a source), incorrect (clashes with what we can check), unclear (we cannot yet tell). Model checking one output aloud.

Apply: In pairs, students give a verdict to one board-displayed output and write the reason in six words or fewer.

Check: Cold-call three pairs: verdict plus reason. Correct any 'it just feels wrong' with a checkable reason.

Critical & analytical thinking focus

Evaluate - scrutinise outputs for correctness/clarity (Evaluate | DOK 2-3 | SOLO: Relational)

The technical concept (teacher background)

An AI language model produces text by predicting likely word patterns, not by looking up guaranteed facts. This means an

output can sound fluent and confident while still being wrong, a problem often called a hallucination. Because the model

does not know whether it is right, the job of judging correctness falls to you, the human user. A useful routine is to give every

output one of three verdicts: correct, incorrect, or unclear. Correct means it matches the task you set and agrees with a

source you trust. Incorrect means it clashes with something you can check, such as a fact, a number, or the instruction you

gave. Unclear means the output is vague, incomplete, or you simply cannot verify it yet, so it needs more checking before

use. Learning to separate 'sounds right' from 'is right' is the core skill of critical evaluation, and it protects you from acting on

a confident mistake.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook and

activate

7 min

Display two confident answers to 'How many bones are in the human body?' - one correct (206), one wrong (198). Ask students to vote and justify.

Vote, then explain to a partner how they could find out which answer is right.

- Which answer do you trust more, and why?

- Does sounding sure make an answer correct?

Teach the triage

routine

10 min

Introduce the three verdicts (correct / incorrect / unclear). Model a think-aloud: read the task, read the output, check against a source, assign a verdict with a reason.

Copy the three-verdict routine into their notes and follow one modelled example.

- What is the difference between incorrect and unclear?

- What could you check this against?

Guided practice

12 min

Show three curated outputs (a fact answer, a summary that adds a detail not in the source, a vague instruction). Guide

In pairs, assign verdicts and reasons to outputs 2 and 3 on a mini whiteboard.

- What signal here says 'do not trust as-is'?

- Which single detail makes this the class to triage the first together. incorrect?

Discuss signals and actions

9 min

Collect the signals students used (missing source, contradicts task, too vague, made-up detail). Link each unclear or incorrect verdict to an action: re-check, re-prompt, or ask a person.

Add one 'signal' and one matching 'action' to a class list.

- When an output is unclear, what is your next move?

- Who could you ask to verify?

Exit check

7 min

Hand out the output-triage exit card: three short outputs to flag correct / incorrect / unclear with a reason each.

Complete the exit card independently.

- Can you give a checkable reason, not just a feeling?

Anticipated misconceptions

Misconception How to address it

If an answer sounds confident and detailed, it must be correct.

Show a confident but wrong output. Fluent language is a writing style,

not proof. Ask: what did we check it against?

Incorrect and unclear are the same thing. Incorrect clashes with something we can check now; unclear means

we cannot tell yet and need more information. Sort one example into each.

If I cannot verify an output, I should just use it anyway.

Unclear means pause and check, not proceed. Model choosing an action: re-prompt, find a source, or ask a person.

Hands-on activity guide

Overview: Short paper triage: students judge three curated outputs and label each with a reason, building the routine used fully in the Lab.

Grouping / timing: Pairs, then independent exit card.

Tool: Teacher board demo only; students use printed output cards (account-free). - No student accounts. Teacher displays curated outputs; students work on paper.

Facilitation steps:

1. Read the task each output was meant to answer.
2. Read the output and check it against the task and any given source.
3. Assign a verdict: correct, incorrect, or unclear.
4. Write a six-word reason for the verdict.
5. Name one action for any unclear or incorrect output.

Unplugged fallback: If no board is available, distribute the same outputs as printed cards with a reference sheet.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Directly checks 'identifies when outputs are incorrect or unclear'

- Check: Output-triage exit card: flag correct / incorrect / unclear with a reason each
- Success criteria: Correctly triages at least 2 of 3 outputs and gives a valid reason for each judgement.

- Evidence: Output-triage card.

- Feedback / adjustment: Model 'check against the task and a source' for over-trusting students; extend to subtle inaccuracies.

Differentiation

Representation: Colour-code the three verdicts; pair each with an icon (tick, cross, question).

Action & expression: Allow verbal verdicts recorded by a partner instead of written reasons.

Engagement: Use a 'trust detective' framing so students hunt for the giveaway signal.

People of Determination: Provide a printed verdict-and-reason frame with the routine steps numbered; allow extra time and a scribe.

EAL: Give a bilingual glossary (correct, incorrect, unclear, source, verify) and sentence stems.

Stretch / next tier: Ask stretch students to find the single subtle detail that flips an output from correct to incorrect.

Answer key / model responses

Task / item Model response Marking note

Triage: 'The capital of the UAE is Abu Dhabi.'

Correct - matches a trusted source and the task. Reason must reference a check, not a feeling.

Triage: a summary that adds a prize not mentioned in the original text.

Incorrect - it states a detail that clashes with the source. Added detail is a classic hallucination signal.

Triage: 'Do the thing with the numbers to get the result.'

Unclear - too vague to verify; needs a clearer prompt. Action: re-prompt for specific steps.

Name one signal an output cannot be trusted as-is.

It contradicts the task, has no source, or contains a detail we cannot check.

Any one checkable signal is acceptable.

Assessment rubric (observable look-fors)

Criterion Emerging (expected) Approaching Not yet

Assigns correct verdicts Triages at least 2 of 3 outputs correctly.

Triages 1 of 3 correctly. Verdicts do not match the outputs.

Gives checkable reasons Each verdict has a reason tied to task or source.

Some reasons rely on feeling not evidence.

Reasons missing or unrelated.

Recommends an action Names a sensible action for an unclear output.

Names a vague action. No action given.

Resources & links

- Curated output-triage card set - Teacher-prepared correct/incorrect/unclear examples with sources. (teacher-prepared)
- ADEK K-12 AI Fluency Framework - Reference for the Critical Evaluation strand outcomes. (<https://www.adek.gov.ae>)

Week 4 - Lab | Output Triage Clinic (Lab)

Strand: Critical Evaluation & Informed Interaction | Framework ref(s): AIF·CE·G6·E | 1 x 45-minute period (Lab)

Learning outcome (framework): Identifies when outputs are incorrect or unclear.

Objectives & success criteria

- Apply a triage routine to a set of curated outputs and record correct/incorrect/unclear with reasons.
- Rank outputs from most to least trustworthy and defend the order.
- Choose a verification action for each untrustworthy output.

Student-facing success criteria:

- I can run the triage routine across a full set of outputs.
- I can rank outputs by trustworthiness and defend my order.
- I can choose a verification action for outputs I cannot trust.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Core lesson you learned the three-verdict routine and practised it on three outputs. In the Lab you run a

full clinic, triaging a larger set and ranking them.

Activate: Show a 'trust ladder' with two outputs already placed. Ask where a third belongs and why.

Teach: Explain the clinic stations and the record sheet columns; model triaging and placing one output on the ladder.

Apply: Groups triage one station output together as a warm-up.

Check: Each group reads out one verdict and its verification action.

Critical & analytical thinking focus

Evaluate - scrutinise outputs for correctness/clarity (Evaluate | DOK 2-3 | SOLO: Relational)

The technical concept (teacher background)

Verification is the practical side of critical evaluation: not just noticing an output might be wrong, but taking a concrete step

to check it. Different outputs need different checks - a factual claim can be compared to a reliable source, a summary can

be compared back to the original text, and a vague instruction may need a clearer prompt before it can be judged at all.

Ranking outputs by trustworthiness forces you to weigh these signals against one another rather than treating every answer

the same. In a clinic, you also practise a basic risk check: asking what could go wrong if someone used this output as-is.

Because the model cannot self-verify, this human checking layer is where safety and accuracy actually come from. The

routine you rehearse here becomes automatic and travels into every later task where you use an AI solution.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Set up the clinic

8 min

Explain stations, record sheet, and the trust ladder. Model triaging one output and placing it on the ladder with a spoken reason.

Form groups, collect record sheets, read the routine steps.

- What are the three verdicts?
- What will you check each output against?

Triage stations

15 min

Rotate between stations of curated outputs (fact claim, summary with added detail, unclear instruction, near-correct number). Prompt groups to name a verification action for each.

Triage each station output, record verdict, reason, and a verification action.

- What single detail decides this verdict?
- How would you verify it?

Rank on the trust ladder

10 min

Ask each group to order their outputs from most to least trustworthy and prepare a defence of the order.

Place outputs on the ladder and agree a group justification.

- Why is this one more trustworthy than that one?
- What would move it up the ladder?

Share and risk Two groups present their ladder; class adds a basic risk check: what could go

Present or critique one ranking and name one risk

- Who could be affected if this check

7 min

wrong if the least trustworthy output were

used?

of using an untrustworthy
output.

were used as-is?

Reflect

5 min

Collect record sheets; give one clinic each
a 'focus for next time'.

Write one verification action
they will reuse.

- Which check was most useful
today?

Anticipated misconceptions

Misconception How to address it

Ranking is just about which answer I like best. Ranking uses trustworthiness signals - sources, matches to task, checkability - not preference. Ask for the signal behind each placement.

Every output needs the same kind of check. A fact needs a source; a summary needs the original text; a vague instruction needs a clearer prompt. Match the check to the output type.

An unclear output is useless. Unclear means not yet judged; a verification action can often move it to correct or incorrect.

Hands-on activity guide

Overview: A rotating triage clinic where groups judge a curated set of outputs, choose a verification action for each, then rank all outputs on a trust ladder and defend the order.

Grouping / timing: Groups of 3-4 rotating through 4 stations.

Tool: Teacher-mediated board demo plus printed station cards; no student accounts. - Curated outputs prepared by the teacher and displayed or printed; students never enter data into a tool.

Facilitation steps:

1. At each station, read the task and the curated output.
2. Assign a verdict: correct, incorrect, or unclear.
3. Write a checkable reason for the verdict.
4. Choose a verification action (check a source, compare to original, re-prompt, ask a person).
5. After all stations, rank outputs on the trust ladder from most to least trustworthy.
6. Agree a group defence for the top and bottom placements.
7. Name one risk of using the least trustworthy output as-is.

Unplugged fallback: Unplugged: use printed output cards and a reference sheet; groups sort and rank on a paper ladder without any device.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Directly checks 'identifies when outputs are incorrect or unclear'

- Check: Completed clinic record sheet plus a defended trust-ladder ranking
- Success criteria: Triage the full set with valid reasons, chooses a verification action each, and defends a sensible ranking.
- Evidence: Clinic record sheet and trust ladder photo.
- Feedback / adjustment: For over-trusting groups, model matching a check to the output type; extend ranking to justify near-ties.

Differentiation

Representation: Provide a laminated routine card and colour-coded verdict tokens.

Action & expression: Allow groups to place physical tokens on the ladder instead of writing.

Engagement: Frame as a 'trust clinic' where each output is a patient needing a diagnosis.

People of Determination: Assign a clear role (reader, recorder, ranker) and give a numbered step card; permit extra time and a scribe.

EAL: Bilingual glossary and pre-labelled ladder rungs (most to least trustworthy) with icons.

Stretch / next tier: Require a written defence of one near-tie on the ladder and a proposed action to break the tie.

Answer key / model responses

Task / item Model response Marking note

Station A fact claim

about a UAE landmark.

Correct if it matches a trusted source; verification action = check official site.

Reason must cite the source used.

Station B summary with an added award.

Incorrect - added detail not in the original; action = compare to source text.

Signal: information appears that was never in the source.

Station C vague instruction.

Unclear; action = re-prompt for specific steps. Cannot be judged until clarified.

Rank the four outputs. Most trustworthy = verified fact; least = unverified added-detail summary.

Any defensible order with signal-based reasons is acceptable.

Assessment rubric (observable look-fors)

Criterion Emerging (expected) Approaching Not yet
Runs the triage routine Triages the full set with valid, checkable reasons.

Triages most, some reasons weak.

Routine not applied consistently.

Chooses verification actions Matches a suitable check to each output type.

Uses one generic check for all. No verification action chosen.

Defends the ranking Ranking justified with trustworthiness signals.

Ranking given with weak justification.

Ranking based on preference only.

Resources & links

- Triage clinic station pack - Four curated outputs with matched sources for verification. (teacher-prepared)

- Trust ladder template - Printable ladder for ranking outputs by trustworthiness. (teacher-prepared)

Week 5 - Core | Risks and Responsible Use

Strand: Ethics & Governance | Framework ref(s): AIF·GE·G6·E | 1 x 45-minute period (Core)

Learning outcome (framework): Recognises basic risks (e.g., incorrect outputs, unsafe use).

Objectives & success criteria

- Identify at least two basic risks of using AI tools (e.g., incorrect output, unsafe or private data use).
- Explain why each risk matters and who it could affect.
- State one responsible-use rule that reduces a risk.

Student-facing success criteria:

- I can name two genuine risks of using an AI tool.
- I can explain who each risk could affect.
- I can state a responsible-use rule that reduces a risk.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: Last week you learned to judge whether an output is correct, incorrect, or unclear.

This week you zoom out: even

a good-looking output carries risks, and responsible use means planning for them before they cause harm.

Activate: Show a short scenario where a student pastes a classmate's full name and address into an AI tool. Ask: what could go wrong here?

Teach: Introduce two risk families: accuracy risk (incorrect output used as-is) and safety/privacy risk (unsafe or private data use).

Give a clear example of each.

Apply: Pairs match three scenarios to a risk family and name who is affected.

Check: Cold-call: one risk, who it affects, and a rule that reduces it.

Critical & analytical thinking focus

Evaluate - identify and reason about risk (Evaluate | DOK 2 | SOLO: Relational)

The technical concept (teacher background)

Using an AI tool responsibly means thinking about risk before something goes wrong. Two basic risk families matter most at

this stage. The first is accuracy risk: because a model can produce confident but incorrect outputs, acting on one without

checking can spread mistakes into your schoolwork or advice you give others. The second is safety and privacy risk:

anything you type into a tool may be stored or used to train future systems, so entering personal or identifying data - names,

addresses, photos, ID numbers - can expose you or someone else. Every risk has a who: the person affected, which may be

you, a classmate, or your family. Responsible-use rules are simple, agreed habits that reduce a risk, such as 'always verify

before you use' or 'never enter personal data'. Governance is just the grown-up word for having and following such rules

together as a class, a school, or a country. Recognising risk early is what lets you enjoy the benefits of AI while keeping

people safe.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook and

activate

7 min

Present the 'pasted personal data' scenario and a 'used a wrong AI answer in homework' scenario. Ask what could go wrong in each.

Discuss with a partner and name one thing that could go wrong per scenario.

- What is the risk here?
- Who could be harmed?

Teach the two risk families

10 min

Define accuracy risk and safety/privacy risk. Give a clear example of each and name the person affected.

Note the two families and

one example each.

- Which family does each scenario belong to?

- Why does this risk matter?

Guided practice

- who is affected

12 min

Show three more scenarios. Guide the class to name the risk, the person affected, and a responsible-use rule.

In pairs, complete a risk-who-rule chain for two scenarios.

- Who is affected if nobody checks?

- What rule would have prevented this?

Build the

responsible-use

rules

9 min

Collect student rules into a short class charter (e.g., verify before use, never enter personal data, tell an adult if unsure).

Contribute one rule and match it to a risk it reduces.

- Which risk does your rule reduce?

- Is this rule realistic to follow?

Exit check

7 min

Hand out the risk-spotting exit card: name two risks plus one responsible-use rule.

Complete the exit card independently.

- Are your two risks genuine and different?

Anticipated misconceptions

Misconception How to address it

AI tools are either totally safe or totally dangerous. Risk is about specific situations, not the whole tool. Sort one safe use and one risky use of the same tool.

It is fine to type in anyone's personal details as long as it helps.

Personal data may be stored or used to train systems. Model replacing a real name with 'a student' in a prompt.

Saying 'it is bad' is enough to describe a risk. A real risk names what could go wrong and who it affects. Rewrite 'it is

bad' into a specific risk plus a person.

Hands-on activity guide

Overview: Scenario sorting: students match scenarios to a risk family, name who is affected, and attach a responsible-use

rule, previewing the full audit in the Lab.

Grouping / timing: Pairs, then independent exit card.

Tool: Teacher-mediated board demo; printed scenario cards; account-free. - No student accounts; teacher displays or prints scenarios.

Facilitation steps:

1. Read the scenario card.
2. Decide the risk family: accuracy or safety/privacy.
3. Name who could be affected.
4. Attach one responsible-use rule that reduces the risk.

Unplugged fallback: Use printed scenario cards and a rules reference sheet with no device.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Directly checks 'recognises basic risks (incorrect outputs, unsafe use)'

- Check: Risk-spotting exit card: name two risks + one responsible-use rule

- Success criteria: Names two genuine risks, explains who is affected, and states a matching responsible-use rule.

- Evidence: Risk-spotting card.

- Feedback / adjustment: Redirect vague 'it is bad' answers to a specific risk + who it affects; extend to privacy risks.

Differentiation

Representation: Use two clearly labelled risk-family icons (a cross for accuracy, a lock for privacy).

Action & expression: Allow students to sort physical scenario cards instead of writing.

Engagement: Frame as a 'risk spotter' role protecting the class.

People of Determination: Provide a risk-who-rule sentence frame and a printed rules bank; allow extra time.

EAL: Bilingual glossary (risk, privacy, personal data, responsible use, affect) and stems.

Stretch / next tier: Ask students to find a scenario that involves both risk families at once and explain the overlap.

Answer key / model responses

Task / item Model response Marking note

Name two basic risks of using AI tools.

Incorrect output used as-is (accuracy) and entering personal data (safety/privacy).

The two risks must be genuinely different.

Who is affected if a wrong AI answer is used in homework?

The student using it, and anyone who trusts their work. Naming a specific 'who' is required.

State a responsible-use rule that reduces the privacy risk.

Never enter personal or identifying data into an AI tool. Rule must match the named risk.

Why does the accuracy risk matter?

A confident wrong answer can spread mistakes if used without checking.

Should link risk to a consequence.

Assessment rubric (observable look-fors)

Criterion Emerging (expected) Approaching Not yet
Identifies genuine risks Names two different, genuine risks.

Names one genuine risk. Risks vague or not genuine.

Explains who is affected Names a specific person or group per risk.

Names who for one risk only. No 'who' given.

States a matching rule Rule clearly reduces a named risk.

Rule loosely related. No rule or unrelated rule.

Resources & links

- AI risk scenario cards - Teacher-prepared accuracy and privacy scenarios set in UAE school life. (teacher-prepared)
- ADEK responsible-use guidance - Reference for the Ethics & Governance strand and school rules. (<https://www.adek.gov.ae>)

Week 5 - Lab | Class AI Risk Audit (Lab)

Strand: Ethics & Governance | Framework ref(s): AIF·GE·G6·E | 1 x 45-minute period (Lab)

Learning outcome (framework): Recognises basic risks (e.g., incorrect outputs, unsafe use).

Objectives & success criteria

- Audit the class's own AI use against a responsible-use rule set and propose one improvement.
- Rank risks by likelihood and impact on a simple risk matrix and choose which to mitigate first.
- Write one responsible-use rule to close a gap found in the audit.

Student-facing success criteria:

- I can audit our AI use against a rule set and find a gap.
- I can place risks on a matrix by likelihood and impact.
- I can propose one rule to reduce the highest-priority risk.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Core lesson you learned two risk families and how to name who is affected. In the Lab you audit your own

class's AI use, rank the risks, and decide what to fix first.

Activate: Show an anonymised list of how the class has used AI demos so far. Ask: where might a risk be hiding?

Teach: Explain the audit checklist and the risk matrix; model auditing one use and placing its risk on the matrix.

Apply: Groups audit one class use together as a warm-up.

Check: Each group states one risk and where it sits on the matrix.

Critical & analytical thinking focus

Evaluate - identify and reason about risk (Evaluate | DOK 2 | SOLO: Relational)

The technical concept (teacher background)

Auditing means checking real practice against a set of rules to find gaps. In an AI risk audit you look at how AI has actually

been used - what was typed in, what outputs were trusted, whether any personal data appeared - and compare it to your

responsible-use rules. Not every risk deserves the same attention, so you weigh each one on two dimensions: likelihood

(how often it could happen) and impact (how much harm it would cause). Plotting risks on a matrix helps you prioritise: a

high-likelihood, high-impact risk should be mitigated before a rare, low-harm one. Mitigation means adding or strengthening

a rule to reduce a risk. This is exactly how organisations and governments approach governance, just at a class scale.

Because your audit uses only anonymised, non-identifying examples, it teaches the skill without creating the very privacy risk it studies.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Set up the audit

8 min

Present the anonymised class-use log and the audit checklist. Model auditing one entry and placing its risk on the matrix.

Form groups, read the checklist, note the two matrix axes.

- What are we comparing our use against?
- What do likelihood and impact

mean?

Run the audit

14 min

Groups work through the anonymised use log, flagging each entry against the rule set and identifying gaps.

Audit entries, record gaps found, and note the risk behind each gap.

- Which rule does this use break or stretch?
- Who could be affected?

Plot the risk

matrix

10 min

Guide groups to place their flagged risks on the likelihood-impact matrix and pick the highest priority.

Place each risk on the matrix and circle the one to mitigate first.

- Why is this risk higher priority than that one?
- What makes it high impact?

Propose a rule

8 min

Each group proposes one responsible-use rule to close its top-priority gap; class votes to add the strongest to the charter.

Write and pitch one mitigation rule.

- Will this rule actually reduce the risk?
- Is it realistic to follow?

Reflect

5 min

Collect audit sheets; give each group a 'focus for responsible use'.

Write one change they will make to their own AI use.

- What is your personal takeaway from the audit?

Anticipated misconceptions

Misconception How to address it

An audit is about blaming whoever made a mistake. The log is anonymised; audits fix systems, not people. Reframe a

flagged entry as a gap to close, not a person to blame.

All risks should be fixed at once and equally. Use the matrix to prioritise. Mitigate high-likelihood, high-impact risks

first; model placing two risks and choosing.

A rule counts even if nobody could realistically follow it.

Test each rule for realism. Rewrite an impractical rule into one the class can actually keep.

Hands-on activity guide

Overview: A structured, teacher-mediated audit of the class's own AI use against a responsible-use rule set, followed by

ranking the flagged risks on a likelihood-impact matrix and proposing one mitigation rule.

Grouping / timing: Groups of 3-4; whole-class charter vote at the end.

Tool: Unplugged and teacher-mediated: anonymised paper use-log and printed matrix; no student accounts. - Only

anonymised, non-identifying examples are used; students never enter real data anywhere.

Facilitation steps:

1. Read the anonymised class AI-use log.
2. Check each entry against the responsible-use rule set.
3. Flag any gap and name the risk and who it affects.
4. Place each flagged risk on the likelihood-impact matrix.
5. Circle the highest-priority risk to mitigate first.
6. Write one responsible-use rule that closes that gap.
7. Pitch the rule for the class charter.

Unplugged fallback: Fully unplugged already; if the log is unavailable, use pre-written anonymised scenario cards instead.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Directly checks 'recognises basic risks (incorrect outputs, unsafe use)'

- Check: Completed audit sheet with a matrix-placed priority risk and a proposed mitigation rule

- Success criteria: Finds a genuine gap, places risks sensibly on the matrix, and proposes a realistic rule that reduces the top risk.

- Evidence: Audit sheet and risk matrix photo.

- Feedback / adjustment: For blame-focused groups, redirect to fixing the system; extend by asking for a second-priority risk and rule.

Differentiation

Representation: Provide the matrix pre-drawn with labelled axes and a worked example placed.

Action & expression: Allow risks to be placed as sticky notes rather than written into cells.

Engagement: Frame groups as a 'class safety board' advising the school.

People of Determination: Assign roles (reader, flagger, plotter) and give a numbered step card; allow extra time and a scribe.

EAL: Bilingual glossary (audit, likelihood, impact, mitigate, gap) and matrix axis labels with icons.

Stretch / next tier: Ask groups to justify a near-tie placement and propose how a rule would move the risk down the matrix.

Answer key / model responses

Task / item Model response Marking note

Audit an entry where a demo prompt included a student's name.

Gap: personal data entered; risk = privacy exposure; who = that student.

Should map to the 'never enter personal data' rule.

Place the privacy risk on the matrix.

High impact; likelihood depends on habits - often high without a rule. Placement must be justified with a reason.

Choose which risk to mitigate first.

The high-likelihood, high-impact one (usually privacy). Priority follows matrix position.

Propose a mitigation rule.

Replace any real name with 'a student' before using a prompt. Rule must be realistic and reduce the named risk.

Assessment rubric (observable look-fors)

Criterion Emerging (expected) Approaching Not yet

Finds a genuine gap Identifies a real gap and its risk and who.

Identifies a gap but not the who.

No genuine gap found.

Uses the risk matrix Places risks sensibly and picks a justified priority.

Places risks but priority weakly justified.

Matrix not used to prioritise.

Proposes a mitigation rule Realistic rule that reduces the top risk.

Rule loosely related or hard to follow.

No workable rule proposed.

Resources & links

- Anonymised class AI-use log - Teacher-prepared, non-identifying record of how demos were used. (teacher-prepared)

- Likelihood-impact risk matrix template - Printable 2x2 matrix for ranking risks.

(teacher-prepared)

Week 6 - Core | Bringing the Strands Together

Strand: Integration | Framework ref(s): AIF·CU·G6·E AIF·SD·G6·E AIF·CE·G6·E AIF·GE·G6·E |

1 x 45-minute period (Core)

Objectives & success criteria

- Complete a guided challenge combining the concept, a prompted solution, an output check and a risk check.

- Explain how the system type and prompt shaped the output.

- Identify one responsible-use choice made.

Student-facing success criteria:

- I can complete a guided challenge that uses all four strands.

- I can explain how the system type and prompt shaped the output.

- I can name one responsible-use choice I made.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: Across Term 1 you learned four strands: rule-based versus learning systems

(concept), writing prompts (solution

design), judging outputs (critical evaluation), and spotting risks (ethics and governance). Today you weave all four into one

guided challenge.

Activate: Show four strand icons and ask students to match each to something they can now do.

Teach: Model the four-step chain on one mini task: name the system type, write a prompt, triage the output, run a risk check.

Apply: Pairs complete step one and two of a shared mini challenge on the board.

Check: Cold-call: how did the prompt change the output?

Critical & analytical thinking focus

Synthesise - connect all four strands (Analyse/Create | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

Real AI fluency is not four separate skills but one connected process. It starts with understanding the system: a rule-based

system follows fixed instructions, while a learning system finds patterns in data, and knowing which you face shapes what

you can expect. Next comes solution design: a clear, specific prompt guides the system toward a useful output, while a

vague prompt invites a vague or wrong one. Then comes critical evaluation: you triage the output as correct, incorrect, or

unclear and verify it against a source. Finally comes ethics and governance: you run a basic risk check and make a

responsible-use choice, such as keeping personal data out of the prompt. Each strand feeds the next - the system type

shapes the prompt, the prompt shapes the output, the output demands a check, and the whole process demands responsibility. Seeing these links is what turns isolated lessons into judgement you can use on any new AI task.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Recall the four strands

7 min

Display the four strand icons and a one-line reminder of each. Ask students to connect them into a chain.

Order the four strands into a process chain with a partner.

- Which strand comes first, and why?
- How does one strand feed the next?

Model the integrated chain

10 min

Take one guided challenge (e.g., 'summarise a short UAE history passage').

Model naming the system type, writing a prompt, triaging the output, and a risk check.

Follow the modelled chain and note each step.

- How did the prompt shape the output?
- What risk did we need to check?

Guided challenge

13 min

Give pairs a new guided challenge with a curated output. Guide them through all four steps against the project success criteria.

Complete the four-step chain and record each result.

- Is your output correct, incorrect, or unclear?

- What responsible-use choice did you make?

Connect and Ask pairs to explain how the system type Write a two-sentence • Which strand was strongest in

explain

8 min

and prompt shaped the output, and name one responsible-use choice.

explanation linking system, prompt, and output.

your work?

- Which needs more practice?

Exit and

feedforward

7 min

Review the integration task against success criteria; give each student one 'focus for planning' tied to a weaker strand.

Record their focus strand for the signature solution.

- Which strand is your focus before the big task?

Anticipated misconceptions

Misconception How to address it

The four strands are separate skills used one at a time.

They form a chain where each feeds the next. Trace the chain aloud on one task: system to prompt to output to risk.

Once the output looks good, the job is done. A good-looking output still needs a triage and a risk check. Model

adding both after a plausible output.

Responsible use is a separate add-on, not part of the task.

Show that the responsible-use choice (e.g., no personal data) is made during the prompt, not after.

Hands-on activity guide

Overview: A guided integration challenge where students run one task through all four strands and explain the links, previewing the signature solution.

Grouping / timing: Pairs, then independent explanation and exit.

Tool: Teacher-mediated board demo with a curated output; account-free. - No student accounts; teacher runs any demo and displays curated outputs.

Facilitation steps:

1. Name the system type behind the task.
2. Write or choose a clear prompt.
3. Triage the curated output as correct, incorrect, or unclear.
4. Run a basic risk check and note one responsible-use choice.
5. Explain how the system and prompt shaped the output.

Unplugged fallback: Use a fully unplugged version with printed task, prompt options, and a curated output card.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Integrates all Term-1 Emerging outcomes before the signature solution

- Check: Guided integration task reviewed against project success criteria; forward feedback
- Success criteria: Connects system+prompt to output, triages the output, and names a risk-reduction choice.
- Evidence: Integration task + feedback note.
- Feedback / adjustment: Each student gets one 'focus for planning' tied to a weaker strand.

Differentiation

Representation: Provide the four-strand chain as a labelled visual with icons.

Action & expression: Allow students to complete steps as a spoken chain recorded by a partner.

Engagement: Frame as 'putting on all four hats' of an AI builder.

People of Determination: Give a four-step chain organiser with sentence frames; allow extra time and a scribe.

EAL: Bilingual glossary (system type, prompt, triage, risk check) and linking stems (so, because, which means).

Stretch / next tier: Ask students to explain how changing the prompt would have changed both the output and the risk.

Answer key / model responses

Task / item Model response Marking note

Name the system type A learning system - it finds patterns in language data. Should contrast with rule-

for a text summariser. based.

Explain how the prompt shaped the output.

A specific prompt gave a focused, on-task summary; a vague one would drift.

Must link prompt wording to output quality.

Triage the curated output.

Correct, incorrect, or unclear with a checkable reason. Reason ties to task or source.

Name one responsible-use choice made.

Kept all personal data out of the prompt. Choice should be made during the task, not after.

Assessment rubric (observable look-fors)

Criterion Emerging (expected) Approaching Not yet
Connects the strands Links system and prompt to the
output clearly.

Links two strands only. Strands treated separately.

Triages the output Gives a verdict with a checkable
reason.

Gives a verdict without a clear
reason.

No triage attempted.

Names a responsible-use
choice

States a genuine risk-reduction
choice.

States a vague choice. No choice named.

Resources & links

- Four-strand integration challenge pack - Teacher-prepared task, prompt options, and curated output. (teacher-prepared)
- Project success criteria sheet - Criteria linking all four strands for the signature solution. (teacher-prepared)

Week 6 - Lab | Integrated Build-and-Check (Lab)

Strand: Integration | Framework ref(s): AIF·CU·G6·E AIF·SD·G6·E AIF·CE·G6·E AIF·GE·G6·E |
1 x 45-minute period (Lab)

Objectives & success criteria

- Run a small integrated build-and-check task end-to-end and record the checks made.
- Identify which strand most needs work before the signature solution.
- Evaluate the final output for accuracy and risk before deciding to use it.

Student-facing success criteria:

- I can run a build-and-check task through all four strands end-to-end.
- I can record the accuracy and risk checks I made.
- I can name which strand I most need to improve.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Core lesson you connected the four strands into one chain. In the Lab you run
a full build-and-check task

end-to-end, exactly like the signature solution you will meet next.

Activate: Show the build-and-check board: four stations matching the four strands. Ask what
evidence each station needs.

Teach: Explain the build-and-check record and how the final output earns an accuracy and risk
verdict; model one station.

Apply: Groups complete the first station (system type) as a warm-up.

Check: Each group states the evidence they will carry to the next station.

Critical & analytical thinking focus

Synthesise - connect all four strands (Analyse/Create | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

A build-and-check task mirrors how a real AI solution is produced and then defended. You build by choosing a system suited to the task and designing a prompt that steers it toward a useful output. You check by triaging that output for accuracy and by running a risk assessment before deciding whether to use it. The two halves are equally important: building without checking produces confident mistakes, while checking without building produces nothing. Running the task end-to-end, and recording the evidence at each stage, lets you see which strand is your weakest link - the one most likely to fail under the pressure of the signature solution. Because every step is teacher-mediated and uses curated or anonymised material, you rehearse the full workflow without ever exposing personal data. This is the rehearsal that turns four learned strands into one reliable habit you can trust on a task that matters.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Set up the build-and-check

8 min

Present four stations matching the strands and the build-and-check record.

Model station one and the evidence it produces.

Form groups, read the record, note the evidence needed at each station.

- What does build mean here, and what does check mean?
- What evidence proves each step?

Build - system and prompt

12 min

Guide groups through stations 1-2: name the system type, then design a clear prompt for the curated task.

Record the system type and their prompt, with a reason for each choice.

- Why is this system suited to the task?
- What makes your prompt specific?

Check -

accuracy and
risk

12 min

Guide groups through stations 3-4: triage
the curated output and run a risk
assessment, giving the output an
accuracy and a risk verdict.

Record a triage verdict with
reason and a risk verdict
with a mitigation.

- Is the output correct, incorrect, or unclear?
- What is the biggest risk, and how would you reduce it?

Decide and
diagnose

8 min

Ask each group to decide use / do not use
/ use-with-changes, then diagnose which
strand was their weakest link.

Make the use decision and
name the strand to improve.

- Would you use this output as-is?
- Which strand let you down most?

Reflect and
feedforward

5 min

Collect records; give each group one
target strand for the signature solution.

Write their focus strand and
one action to strengthen it.

- What will you practise before the big task?

Anticipated misconceptions

Misconception How to address it

Building the solution is the whole task. Half the marks are in the check. Model that an unchecked output

cannot be used; add the accuracy and risk verdicts.

A use decision is just yes or no. Show the third option: use-with-changes. Model turning a risky output

into a usable one with a fix.

My weakest strand does not matter if the output
looks fine.

The weakest link fails under pressure. Diagnose it now so the

signature solution does not expose it later.

Hands-on activity guide

Overview: A full end-to-end build-and-check task run across four stations, producing an accuracy verdict, a risk verdict, a use decision, and a diagnosis of the group's weakest strand - a direct rehearsal for the signature solution.

Grouping / timing: Groups of 3-4 moving through 4 stations; individual reflection at the end.

Tool: Teacher-mediated board demo with curated output; anonymised, account-free materials. -

Curated task and output

supplied by the teacher; students never enter personal or identifying data.

Facilitation steps:

1. Station 1: name the system type suited to the task and why.
2. Station 2: design a clear, specific prompt.
3. Station 3: triage the curated output for accuracy with a reason.
4. Station 4: run a risk assessment and choose a mitigation.
5. Give the output an accuracy verdict and a risk verdict.
6. Decide: use, do not use, or use-with-changes.
7. Diagnose the group's weakest strand before the signature solution.

Unplugged fallback: Unplugged: run all four stations with printed task, prompt options, curated output card, and a risk sheet - no device needed.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Integrates all Term-1 Emerging outcomes end-to-end as a rehearsal for the signature solution

• Check: Completed build-and-check record with accuracy verdict, risk verdict, use decision, and weakest-strand diagnosis

• Success criteria: Runs all four strands end-to-end, records both checks, makes a justified use decision, and names a weakest strand.

• Evidence: Build-and-check record.

• Feedback / adjustment: For build-heavy groups, weight attention to the checks; give each group one target strand for the signature solution.

Differentiation

Representation: Provide a station map with the four strands as labelled icons and a worked example.

Action & expression: Allow the record to be completed with tokens or short phrases rather than full sentences.

Engagement: Frame as a dress rehearsal for the signature solution.

People of Determination: Assign station roles and give a numbered evidence checklist; allow extra time and a scribe.

EAL: Bilingual glossary (build, check, accuracy, risk, mitigation, verdict) and decision stems.

Stretch / next tier: Require a written justification of the use-with-changes option and a redesigned prompt that lowers the risk verdict.

Answer key / model responses

Task / item Model response Marking note

Station 1: choose the

system type for a summarising task.

A learning system that finds language patterns; suited because summarising needs pattern recognition, not fixed rules.

Choice must be justified by the task.

Station 3: give an accuracy verdict.

Correct / incorrect / unclear with a reason checked against the source.

Reason must be checkable.

Station 4: give a risk verdict and mitigation.

Main risk = using an unverified detail; mitigation = verify before use. Mitigation must match the risk.

Diagnose the weakest strand.

Any strand, justified by where evidence was thinnest. Diagnosis should be honest and specific.

Assessment rubric (observable look-fors)

Criterion Emerging (expected) Approaching Not yet

Runs the task end-to-end Completes all four strands with recorded evidence.

Completes most strands with some evidence.

Task not run end-to-end.

Records both checks Gives a justified accuracy verdict and risk verdict.

Gives one check clearly, the other weakly.

Checks missing or unjustified.

Decides and diagnoses Makes a justified use decision and names a weakest strand.

Decides or diagnoses, not both.

Neither a decision nor a diagnosis.

Resources & links

- Build-and-check station pack - Teacher-prepared four-station task with a curated output and risk sheet. (teacher-prepared)
- Signature solution readiness checklist - Checklist mapping each strand to the evidence needed. (teacher-prepared)

Week 7 - Core | Signature Solution - Planning

Strand: Integration (Signature Solution - Planning) | Framework ref(s): AIF·CU·G6·E

AIF·SD·G6·E AIF·CE·G6·E AIF·GE·G6·E | 1 x 45-minute period (Core)

Objectives & success criteria

- Plan a structured AI solution: state the goal, the prompt structure, and the check steps.
- Identify the inputs/data and the output-evaluation method.
- Describe one risk and how the plan reduces it.

Student-facing success criteria:

- I can state the goal of my solution in one clear sentence.
- I can lay out a prompt structure with a role, a task, and one rule.
- I can name the inputs my solution needs and how I will check the output.
- I can describe one risk and one way my plan reduces it.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: You have used prompts before and know that AI systems can be rule-based or learning-based. In Grade 5 you improved a single prompt. Now you plan a whole solution: not just one prompt, but a structure with inputs, checks and a fallback.

Activate: Show two board demos side by side: a vague request ('Help with the trip') and a structured one ('You are a bus timetable helper. List the three earliest buses from Al Ain to Abu Dhabi after 7am. If a time is missing, say so.'). Ask which is easier to check and why.

Teach: Explain that a signature solution is a small, reusable design: a goal, a prompt structure (role + task + one rule), the inputs it needs, an evaluation step, and a risk check. Model naming each part on the board.

Apply: In pairs, students label the parts of the structured demo prompt (role, task, rule) and suggest one check step for its output.

Check: Cold-call two pairs: 'What is the goal in one sentence?' and 'How would you check the output is correct?'

Critical & analytical thinking focus

Design & plan - architect a prompted solution (Create (plan) | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

A structured, prompt-driven solution is more than a single question typed into a tool. It is a small system with parts that fit together. First comes the goal: one plain sentence that says exactly what the solution should do and for whom. Next comes

the prompt structure, which usually names a role for the AI, a clear task, and at least one rule or limit that keeps the output focused. The solution also needs defined inputs, which is the information the prompt works on, and a matching output, which is the result you can inspect. Because AI systems, whether rule-based or learning-based, can produce wrong or unclear answers, a good plan builds in an evaluation step that says how you will judge the output for accuracy. A strong plan also names one realistic risk, such as a confident but incorrect answer, and shows one way the design lowers that risk, like adding a verification rule or a human check. Planning this way is systems thinking: you decide how the pieces connect before you build, so that problems are cheaper to catch and fix.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Hook and goal-

setting

7 min

Present the vague vs structured board demo and define 'signature solution' as a reusable design with parts. Reveal the Emerging project criteria.

Compare the two demos and write, in one sentence, what a good solution goal should include.

- What makes a request easy or hard to check?
- What does a goal sentence need to be useful?

Model the plan

parts

10 min

Think aloud while building a sample plan for a simple task (e.g. sorting lost-property notices), naming goal, prompt structure, inputs, evaluation, and one risk.

Annotate the modelled plan on their sheet, marking each of the five parts.

- Which part is the role, task, and rule?
- What input does this prompt need to work?

Guided planning

12 min

Hand each pair a task card. Circulate and prompt for the evaluation and risk steps, which are the most-often missed.

Draft the goal, prompt structure, and inputs for their own task card.

- Is your goal one clear sentence?
- What exactly will you feed in as input?

Add evaluation and risk

11 min

Model one evaluation method (compare output to a known correct answer) and one risk mitigation (add a 'say if unsure' rule). Support POD and EAL pairs with sentence frames.

Write how they will check the output and name one risk plus one way the plan reduces it.

- How will you know the output is accurate?
- What could go wrong, and how does your plan help?

Plan review and approve-to-build gate

5 min

Run a quick plan review against the rubric; approve plans that are complete or return them with one targeted prompt.

Self-check the plan against the five-part list and swap for a peer glance before submitting.

- Which of the five parts is strongest?
- Which part still needs one more sentence?

Anticipated misconceptions

Misconception How to address it

A good prompt is just a longer question. Length is not structure. Show that a short prompt with a role, a task, and one rule is easier to check than a long, rambling one. Structure beats size.

If the AI sounds confident, the answer must be correct.

Explain that learning systems can be confidently wrong. That is exactly why the plan needs an evaluation step and a verification rule.

Planning is a waste of time - just start typing prompts.

Show that a plan makes mistakes cheaper to catch. Naming inputs, checks, and risks before building saves rework and keeps data safe.

Hands-on activity guide

Overview: This Core lesson is planning-focused rather than a full build; the hands-on work is producing an approved plan sheet for the signature solution.

Grouping / timing: Pairs, with a teacher-led plan-review gate at the end.

Tool: No live AI account for students. Teacher board demo plus curated example outputs only. - Teacher-mediated on the board using a school-approved child-safe tool; students do not log in.

Facilitation steps:

1. Read your task card and write the goal in one sentence.
2. Build the prompt structure: role, task, and one rule.
3. List the inputs the prompt needs (no personal or identifying data).
4. Write your output-evaluation method (how you will check accuracy).
5. Name one risk and one way your plan reduces it, then submit for the approve-to-build gate.

Unplugged fallback: If no tool is available, work entirely on planning cards using curated sample outputs pinned to the board; the plan is complete without any live AI.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Checks planning of the signature solution against the Emerging project criteria

- Check: Plan review against the project rubric (Emerging); approve-to-build gate
- Success criteria: Plan states goal, prompt structure, inputs, an evaluation method, and one risk mitigation.
- Evidence: Approved plan sheet.
- Feedback / adjustment: Plans missing an evaluation or risk step are returned with one targeted prompt.

Differentiation

Representation: Offer the plan as a labelled diagram and as a five-line checklist so learners can choose a visual or a text route.

Action & expression: Students may record the plan by writing, by placing pre-made card labels, or by dictating to a partner scribe.

Engagement: Task cards use familiar UAE contexts (school lost-property, majlis event list, dates-market stall) so the goal feels real.

People of Determination: Provide a partly completed plan frame and enlarged, high-contrast cards; allow extra think time and a quiet planning corner.

EAL: Give sentence frames: 'The goal is to ____.', 'The AI acts as a ____.', 'One rule is ____.', 'I will check the output by ____.', 'One risk is ____; my plan reduces it by ____.'

Stretch / next tier: Add a second rule and a second evaluation method, and explain which check catches which kind of error.

Answer key / model responses

Task / item Model response Marking note

Write the goal in one sentence for a lost-property sorter.

The goal is to sort short lost-property notices into 'clothing', 'devices', or 'other' so staff can return items faster.

A strong goal names the task and who it helps.

Build a prompt structure for that goal.

You are a lost-property sorting helper. Read each notice and label it clothing, devices, or other. If a notice fits none, label it 'other' and say why.

Role + task + one rule; no names or personal data included.

State the evaluation method.

I will compare ten labelled notices to a teacher answer key and count how many labels match.

Accuracy is checked against a known-correct reference.

Name one risk and a mitigation.

Risk: the AI labels an unclear notice confidently but wrongly.

Mitigation: the rule forces it to say 'other' and give a reason, and a person checks any 'other'.

The mitigation connects directly to the named risk.

Assessment rubric (observable look-fors)

Criterion Emerging (expected) Approaching Not yet
Completeness of plan (five

parts)

Plan clearly states goal, prompt structure, inputs, an evaluation method, and one risk mitigation.

Plan includes most parts but one (often evaluation or risk) is thin or missing.

Plan is a loose prompt with two or fewer parts named.

Quality of evaluation and risk thinking

Evaluation method is checkable and the mitigation clearly reduces the named risk.

Evaluation or mitigation is present but vague or only loosely linked to a risk.

No workable evaluation method or risk is identified.

Responsible-use and data safety

Plan uses no personal or identifying data and includes a human check.

Plan mostly avoids personal data but the human check is unclear.

Plan would put personal or identifying data into the prompt.

Resources & links

- Signature Solution Plan Sheet (Emerging) - One-page five-part planning frame with sentence frames and a self-check strip.

(None)

- Curated Output Cards - Set A - Teacher-selected safe sample outputs used for evaluation practice, no student data. (None)

Week 7 - Lab | Prototype and Pressure-Test the Plan (Lab)

Strand: Integration (Signature Solution - Planning) | Framework ref(s): AIF·CU·G6·E

AIF·SD·G6·E AIF·CE·G6·E AIF·GE·G6·E | 1 x 45-minute period (Lab)

Objectives & success criteria

- Prototype the prompt structure and pressure-test the plan against a tricky input.
- Add a fallback branch for when the output fails a check.
- Record what changed and why after the pressure test.

Student-facing success criteria:

- I can run my prompt structure on a normal input and read the output.
- I can pressure-test it with a tricky input and spot where it fails.
- I can add a fallback branch that says what to do when a check fails.
- I can record the change I made and the reason for it.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: You have an approved five-part plan from the Core lesson. Today you turn the plan into a working prototype on

the board and stress it with hard inputs, then add a fallback so failures are handled, not hidden.

Activate: Recap the approved plan structure. Show a curated output where a normal input works, then ask: 'What if the input is messy or missing?'

Teach: Define prototype (a first working version), pressure test (trying a tricky input on purpose), and fallback branch (a planned response when a check fails).

Apply: In pairs, students predict how their prompt would behave on one tricky input before it is tested on the board.

Check: Ask two pairs to state their tricky input and what a good fallback would say.

Critical & analytical thinking focus

Prototype & test - stress a design and harden it (Create/Evaluate | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

A prototype is the first working version of a solution, built quickly so you can see how it really behaves rather than how you

hoped it would. Pressure-testing means deliberately feeding a design its hardest inputs: a missing value, an unclear notice,

or two items in one line. These edge cases reveal weak spots that normal inputs hide. When an output fails a check, a well-

designed solution does not simply stop or give a wrong answer confidently; it follows a fallback branch, which is a planned

response such as 'if the item is unclear, label it other and ask a person'. Designing fallbacks is a form of defensive design

used in real engineering, from lifts to payment apps. In a teacher-mediated classroom, the AI runs only on the board using a

school-approved child-safe tool, and no personal or identifying data is ever entered, so pressure-testing uses safe, curated

inputs. Each test is logged so the class can see the pattern of failures and decide which single change will help the most.

This is systems thinking in action: you improve the design, not just one answer.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Recap and set

the build target

6 min

Recap the approved plan and set today's

target: prototype, pressure-test, add one fallback branch.

Re-read their plan and underline the one rule most likely to fail under a tricky input.

- What is your prompt supposed to do on a normal input?
- Where might it break?

Prototype on a normal input

10 min

Run two or three pairs' prompt structures on the board with a normal curated input; capture outputs on the build/test log.

Read the output and mark whether it met the goal, recording the result on their log.

- Did the output meet the goal?
- Is anything unclear or extra?

Pressure-test with tricky

Feed each prototype a curated tricky input (missing, messy, or double item) and lead

Record what failed and classify the error type

- What kind of error is this?
- Did the AI sound confident even inputs

14 min

evaluation of the failure. (wrong, unclear, or missing). when wrong?

Add a fallback branch

10 min

Model writing an 'if a check fails, then...' branch and adding it to the prompt structure; support POD/EAL with a branch frame.

Write a fallback branch for their own solution and re-run it on the tricky input if time allows.

- What should happen when the

check fails?

- Does your fallback keep a person in the loop?

Log the change

and reason

5 min

Collect build/test logs and confirm each shows one change with a reason.

Complete the 'change made

/ reason' line and self-rate

readiness to build.

- What one change did you make?
- Why will it help next week's build?

Anticipated misconceptions

Misconception How to address it

If it works on a normal input, the solution is finished. Normal inputs hide weak spots. Show a prototype that passes a

normal case but fails a missing value; the tricky input is the real test.

A fallback means the solution failed. A fallback is good design, not failure. It turns a silent wrong answer

into a safe, handled outcome with a person in the loop.

We can pressure-test using real names or student data to be realistic.

Never. Realism does not need personal data. Use curated, made-up inputs; entering identifying data is against the rules and unsafe.

Hands-on activity guide

Overview: A substantial prototyping session: students turn their approved plan into a working prompt structure, stress it

with curated tricky inputs on the board, and design a fallback branch, all logged.

Grouping / timing: Pairs at planning stations feeding the teacher-run board demo in a rotation.

Tool: School-approved child-safe AI tool run by the teacher on the board; simulated-AI cards as backup. - Teacher-

mediated only; no student logins. All inputs are curated and contain no personal or identifying data.

Facilitation steps:

1. Copy your approved prompt structure onto the prototype slot of your build/test log.
2. Run it on a normal curated input (teacher board) and record whether it met the goal.
3. Pressure-test with a tricky curated input; record what failed and the error type.
4. Write a fallback branch: 'If the check fails, then ____ and ask a person.'
5. Re-run on the tricky input if time allows and log the change you made and the reason.

Unplugged fallback: Unplugged version: lay the plan out on cards and use curated output cards as the 'AI response'; a partner

reads the card so pairs can pressure-test and design a fallback with no live tool.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Checks that the plan survives a tricky input and gains a fallback branch

- Check: Build/test log review: one normal run, one pressure test, one fallback branch, one logged change
- Success criteria: Prototype is run, a tricky input exposes a weak spot, and a fallback branch plus a logged reason are added.
- Evidence: Completed build/test log with fallback branch.
- Feedback / adjustment: Logs without a fallback or without a classified error type are returned with one targeted prompt.

Differentiation

Representation: Show the fallback as a branching flow arrow and as an 'if-then' sentence so both visual and text learners can follow.

Action & expression: Students can prototype by writing, by arranging card labels, or by narrating their run to a scribe.

Engagement: Tricky inputs use playful but safe UAE contexts (a smudged majlis guest note, a two-items-in-one lost-property line).

People of Determination: Provide a pre-drawn branch template and enlarged log; allow a quieter station and extra processing time.

EAL: Give the frame 'If ____ then ____ and ask a person.' plus an error-type word bank: wrong, unclear, missing.

Stretch / next tier: Design two fallback branches for two different failure types and explain which is more likely and why.

Answer key / model responses

Task / item Model response Marking note

Prototype run on a normal input.

Input 'blue school jumper found in hall' -> output 'clothing'. Meets the goal.

Normal case passes; this is the baseline, not proof of a finished solution.

Pressure test with a tricky input.

Input 'found: charger and a scarf' -> output 'devices' only. Error type: missing (it dropped the scarf).

The tricky input exposes a real weak spot to fix.

Write a fallback branch. If a notice names two items, then label each item separately, and if still unclear label 'other' and ask a person.

The branch handles the

exposed failure and keeps a human in the loop.

Log the change and reason.

Change: added a two-item rule and a fallback. Reason: the pressure test showed the prompt dropped the second item.

The reason connects directly to the observed failure.

Assessment rubric (observable look-fors)

Criterion Emerging (expected) Approaching Not yet

Prototyping and testing Runs the prototype on a normal input and pressure-tests it with a tricky input, recording both.

Runs one case but the pressure test or the recording is incomplete.

No working prototype run is recorded.

Fallback design Adds a clear if-then fallback branch that handles the exposed failure and keeps a person in the loop.

Adds a fallback but it is vague or does not match the failure.

No fallback branch is added.

Evidence and reasoning Logs the change with a reason that links to the observed failure and classifies the error type.

Logs a change but the reason or error type is missing.

No change or reasoning is logged.

Resources & links

- Build/Test Log (Lab) - Two-column log for prototype runs, pressure tests, fallback branch and logged change. (None)
- Curated Tricky-Input Cards - Teacher-selected safe edge-case inputs (missing, messy, double item), no personal data. (None)

Week 8 - Core | Signature Solution - Building

Strand: Integration (Signature Solution - Building) | Framework ref(s): AIF·CU·G6·E

AIF·SD·G6·E AIF·CE·G6·E AIF·GE·G6·E | 1 x 45-minute period (Core)

Objectives & success criteria

- Build the solution by following the plan and prompt structure.

- Evaluate outputs for accuracy and flag any that are incorrect or unclear.
- Make at least one improvement based on an output check.

Student-facing success criteria:

- I can build my solution by following my approved plan.
- I can evaluate outputs and flag ones that are wrong or unclear.
- I can make one improvement and explain what changed and why.
- I can keep personal data out of every prompt.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: You have an approved plan and a pressure-tested prototype with a fallback branch.

Today you build the real

solution by following that plan, evaluate the outputs, and make one improvement based on what you find.

Activate: Show a curated set of five outputs from a sample solution; two are subtly wrong. Ask students to flag the wrong ones.

Teach: Explain build discipline: follow the plan, run inputs, evaluate each output against the check, flag errors, then change one thing. Define 'evaluate -> change -> result'.

Apply: Pairs practise flagging by marking the two wrong sample outputs and saying why.

Check: Ask two pairs which output they flagged and how they knew it was wrong.

Critical & analytical thinking focus

Create & evaluate - build, evaluate, iterate (Create/Evaluate | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

Building a signature solution means following your plan with discipline rather than improvising new prompts each time. You

feed defined inputs into your prompt structure, capture the outputs, and evaluate each one against the check you designed:

does it match the known-correct answer, and is it clear? Because learning-based AI can be confidently wrong, evaluation is

not optional; it is the step that separates a working solution from a risky one. When you find an output that is incorrect or

unclear, you flag it, which means marking it for attention rather than quietly accepting it. Then you make one improvement,

usually a small change to a rule or the prompt structure, and you record the loop as 'evaluate -> change -> result'. This

iteration loop is how real systems mature: one measured change, one observed result.

Throughout the build you keep

personal and identifying data out of every prompt and keep a person in the loop for anything flagged. A single, well-

explained improvement backed by evidence is worth more than many untested changes.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Build launch

and gate check

6 min

Confirm each pair has an approved plan and fallback; set the build target and the evaluate-change-result expectation.

Set up their build/test log and restate their goal and check in one line.

- What is your check for a correct output?

- What is your fallback if it fails?

Build by

following the

plan

12 min

Run pairs' prompt structures on the board with defined inputs; keep the build faithful to the plan.

Record each output on the log without changing the prompt yet.

- Are you following the plan or improvising?

- Is the input free of personal data?

Evaluate and

flag

12 min

Lead evaluation of outputs against the check; model flagging an incorrect or unclear one.

Mark each output correct, unclear, or wrong and flag any that fail the check.

- Does this output match the known-correct answer?

- Is it clear enough to act on?

Make one

improvement

10 min

Model changing one rule in response to a flagged output and re-running it; support POD/EAL with an improvement frame.

Make one improvement, re-run, and log evaluate ->

change -> result.

- What one change addresses the

flag?

- Did the result actually improve?

Improvement

share and

check-in

5 min

Run a quick roving check against the rubric; pair stuck students with a buddy and a stronger prompt structure.

Share their one logged improvement with another pair.

- What did your change fix?
- What would you improve next?

Anticipated misconceptions

Misconception How to address it

Building means writing new prompts until something looks good.

That is guessing, not building. Following the approved plan and changing one thing at a time lets you see what actually caused an improvement.

If most outputs look right, I do not need to check each one.

Subtle errors hide among correct-looking outputs. Evaluate every output against the check; that is how you catch the confident wrong answer.

More changes at once means faster improvement. Many changes at once hide which one helped. One change plus one observed result gives evidence you can trust.

Hands-on activity guide

Overview: Core build session: students build the solution from their approved plan, evaluate outputs, flag errors, and log one evidence-based improvement.

Grouping / timing: Pairs feeding the teacher-run board demo in rotation, with a buddy system for stuck pairs.

Tool: School-approved child-safe AI tool run by the teacher on the board; simulated-AI cards as backup. - Teacher-

mediated only; students never log in and never enter personal or identifying data.

Facilitation steps:

1. Restate your goal and check on the build/test log.
2. Run your prompt structure on defined, curated inputs (teacher board) and record each output.
3. Evaluate each output against your check and flag any that are wrong or unclear.
4. Make one improvement to a rule or the structure in response to a flag.
5. Re-run and log the loop as evaluate -> change -> result.

Unplugged fallback: Unplugged: use curated output cards as the AI responses; pairs evaluate and flag on the cards, then rewrite one rule and predict the new result.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Monitors build against the plan; each student logs one evaluation-based

improvement

- Check: Build check-in against plan + one logged improvement ('evaluate -> change -> result')

- Success criteria: Solution follows the plan, outputs are evaluated, and one improvement is made and explained.

- Evidence: Build progress + improvement log.

- Feedback / adjustment: Roving feedback against the rubric; stuck students get a stronger prompt structure and a peer buddy.

Differentiation

Representation: Present the iteration loop as a cycle diagram and as a three-word phrase (evaluate, change, result).

Action & expression: Students may record outputs by writing, ticking a flag column, or narrating to a scribe.

Engagement: Inputs stay in familiar UAE settings (school events, souq stall, library returns) so evaluation feels purposeful.

People of Determination: Provide a pre-formatted log with a flag column and enlarged text; allow extra time and a calm build station.

EAL: Give frames: 'This output is correct / unclear / wrong because ____.' and 'I changed ____ so that ____; the result was ____.'

Stretch / next tier: Make a second improvement and compare which change gave the bigger accuracy gain, with evidence from the log.

Answer key / model responses

Task / item Model response Marking note

Evaluate an output against the check.

Output 'clothing' for 'blue jumper' matches the answer key, so it is correct.

Evaluation compares to a known-correct reference, not to a hunch.

Flag an incorrect output. Output 'other' for 'tablet charger' is wrong; it should be 'devices'. Flag it.

Flagging marks it for a change and a human check.

Make one improvement. Added a rule: 'chargers, cables and headphones count as devices'.

Re-ran and the charger was labelled correctly.

One change, then one
observed result.

Log evaluate -> change -
> result.

Evaluate: charger mislabelled. Change: added a devices examples
rule. Result: charger now 'devices', accuracy up.

The loop shows cause and
effect clearly.

Assessment rubric (observable look-fors)

Criterion Emerging (expected) Approaching Not yet

Faithful build Builds the solution by following
the approved plan and prompt
structure with defined inputs.

Builds but drifts from the plan
or improvises prompts in
places.

No plan-based build is
produced.

Evaluation and flagging Evaluates outputs against the
check and flags incorrect or
unclear ones accurately.

Evaluates some outputs but
misses or mis-flags errors.

Outputs are accepted
without evaluation.

Evidence-based
improvement

Makes and explains one
improvement with a clear
evaluate -> change -> result log.

Makes a change but the reason
or result is unclear.

No improvement or
explanation is logged.

Resources & links

- Build/Improvement Log (Core) - Log with input, output, flag, and an evaluate-change-result strip. (None)

- Curated Output Set - Two Errors - Teacher-selected outputs with two seeded errors for flagging practice, no student data.

(None)

Week 8 - Lab | Extended Build and Evaluation (Lab)

Strand: Integration (Signature Solution - Building) | Framework ref(s): AIF·CU·G6·E

AIF·SD·G6·E AIF·CE·G6·E AIF·GE·G6·E | 1 x 45-minute

period (Lab)

Objectives & success criteria

- Complete a working solution and evaluate its outputs across several test inputs.
- Handle an incorrect output gracefully and document the fix.
- Run a basic risk check and confirm no personal data is used.

Student-facing success criteria:

- I can complete a working solution that follows my plan.
- I can evaluate its outputs across several test inputs and measure accuracy.
- I can handle an incorrect output gracefully using my fallback.
- I can run a basic risk check and document the fix I made.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: You have built the first version and logged one improvement. Today you complete the solution, evaluate it

across several test inputs, handle an incorrect output with your fallback, and run a basic risk check before you present it.

Activate: Show a nearly finished sample solution and a test set of six curated inputs; ask how we would prove it is accurate.

Teach: Define a test set, an accuracy count (correct out of total), graceful handling (using the fallback), and a basic risk check (accuracy, clarity, no personal data).

Apply: Pairs decide what counts as a correct output before testing, so evaluation is fair.

Check: Ask two pairs to state their accuracy target and their risk-check questions.

Critical & analytical thinking focus

Complete & evaluate - finish, test across cases, harden (Create/Evaluate | DOK 4 | SOLO: Extended abstract)

The technical concept (teacher background)

Completing a solution means more than getting one good answer; it means proving the solution behaves well across a set

of test inputs. A test set is a small, curated collection of inputs, including a few edge cases, that you run through the

finished solution to measure accuracy as a simple count of correct outputs out of the total.

When an output is incorrect, a

completed solution handles it gracefully by following its fallback branch rather than presenting a confident wrong answer,

and you document the fix so the change is traceable. A basic risk check asks three questions: is the solution accurate

enough for its purpose, are the outputs clear enough to act on, and does any prompt contain personal or identifying data. In

a teacher-mediated setting the AI runs only on the board with curated inputs, so the data-safety answer must always be

that no personal data is used. This is DOK 4 work because you are integrating build, evaluation across cases, fallback

handling, and a risk judgement into one coherent product, and reasoning about how the parts affect each other. A solution

you can measure and defend is far more trustworthy than one that merely looks finished.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Set the

completion

target and test

set

6 min

Confirm each pair's fallback works;

introduce the curated test set of six inputs

and define a correct output.

Write their accuracy target

and what counts as a

correct output.

- What does a correct output look like?

- How many correct out of six is good enough?

Complete the

build

10 min

Run each pair's solution to a finished

state on the board, ensuring the fallback

branch is wired in.

Confirm their solution

follows the plan and the

fallback is present,

recording readiness.

- Is the fallback branch actually in the prompt?

- Does the build match the plan?

Evaluate across Run all six curated inputs; lead a count of Record each result, mark • Where did accuracy drop?

the test set

13 min

correct outputs and identification of any failure.

correct or not, and compute accuracy (correct out of six).

- Did an edge case trigger the fallback correctly?

Handle a failure

and run the risk

check

11 min

Model graceful handling of one incorrect output via the fallback, then lead the three-question risk check; support POD/EAL with frames.

Document the fix and complete the accuracy / clarity / no-personal-data risk check.

- Did the fallback handle the failure safely?
- Is any prompt free of personal data?

Showcase and reflect

5 min

Invite two pairs to show their accuracy count and risk check; confirm evidence against the rubric.

Share their accuracy result and one documented fix.

- How accurate is your solution and how do you know?
- What is the biggest remaining risk?

Anticipated misconceptions

Misconception How to address it

One good output means the solution is accurate. Accuracy is about the whole test set, not one lucky case. Count correct outputs across several inputs to make a fair claim.

Handling a failure means deleting the wrong output and moving on.

Graceful handling means the fallback catches it and a person is involved, and the fix is documented so it is traceable.

A risk check is only about accuracy. A basic risk check also asks about clarity and data safety. A prompt

with personal data is a serious risk even if the answer is accurate.

Hands-on activity guide

Overview: Extended build and evaluation: students complete the solution, run it across a curated six-input test set, measure accuracy, handle a failure with the fallback, document the fix, and complete a basic risk check.

Grouping / timing: Pairs feeding the teacher-run board demo in rotation, with a showcase at the end.

Tool: School-approved child-safe AI tool run by the teacher on the board; simulated-AI cards as backup. - Teacher-mediated only; no student logins; all six test inputs are curated and contain no personal or identifying data.

Facilitation steps:

1. Confirm your solution follows the plan and the fallback branch is wired in.
2. Define what counts as a correct output for your goal.
3. Run all six curated test inputs on the board and record each result.
4. Compute accuracy (correct out of six) and handle any failure with your fallback.
5. Document the fix and complete the three-question risk check (accuracy, clarity, no personal data).

Unplugged fallback: Unplugged: use six curated output cards as the AI responses; pairs evaluate accuracy, act out the fallback

for a failing card, and complete the same risk check without any live tool.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Confirms a completed solution is evaluated across several inputs, handles a

failure, and passes a basic risk check

- Check: Test-set evaluation log (accuracy count) + documented fix + completed risk check

- Success criteria: Solution is completed, evaluated across several inputs with an accuracy count, a failure is

handled gracefully, and the risk check confirms no personal data.

- Evidence: Completed test-set log, documented fix, and risk-check sheet.

- Feedback / adjustment: Logs without an accuracy count or a completed risk check are returned with one targeted prompt.

Differentiation

Representation: Show accuracy as a simple count and as a filled bar; show the risk check as a three-tick card.

Action & expression: Students may tally accuracy in writing, with counters, or by narrating to a scribe.

Engagement: The test set uses varied UAE contexts (Eid event notices, souq stock lines, sports-day sign-ups) to keep evaluation lively and fair.

People of Determination: Provide a pre-formatted test-set log and risk-check card with enlarged text; allow extra time and a calm station.

EAL: Give frames: 'My solution got ____ out of 6 correct.', 'The fallback handled ____ by ____.', 'Risk check: accurate? ____ clear? ____ personal data used? no.'

Stretch / next tier: Add a seventh, harder edge case, predict the result, then test and explain whether your solution generalised.

Answer key / model responses

Task / item Model response Marking note

Compute accuracy
across the test set.

Five of six outputs matched the answer key, so accuracy is 5/6. Accuracy is a count across the whole set, not one case.

Handle an incorrect
output gracefully.

The two-item notice failed, so the fallback split the items and flagged the unclear one for a person. The wrong output was not presented as final.

Graceful handling uses the
fallback and keeps a person in
the loop.

Document the fix. Fix: strengthened the two-item rule and confirmed the fallback triggers on unclear notices; re-ran and accuracy rose to 6/6.

The fix is traceable and its
effect is recorded.

Run the basic risk check. Accurate? mostly (6/6 after fix). Clear? yes. Personal data used? no - all inputs curated. Biggest remaining risk: rare unseen edge cases.

The risk check covers
accuracy, clarity, and data
safety.

Assessment rubric (observable look-fors)

Criterion Emerging (expected) Approaching Not yet

Completed, plan-faithful
solution

Solution is completed, follows
the plan, and has the fallback
branch wired in.

Solution is mostly complete
but the plan or fallback is partly
missing.

Solution is not completed to
a working state.

Evaluation across inputs Evaluates outputs across the test
set and reports an accuracy
count with a failure handled
gracefully.

Evaluates some inputs but the
accuracy count or failure
handling is incomplete.

No multi-input evaluation is
recorded.

Risk check and data safety Completes the three-question
risk check and confirms no

personal or identifying data is used.

Completes part of the risk check or the data-safety confirmation is unclear.

No risk check is completed, or personal data would be used.

Resources & links

- Test-Set Evaluation Log (Lab) - Six-row log with judgement and accuracy count, plus a documented-fix strip. (None)
- Basic Risk-Check Card - Three-question card: accuracy, clarity, no personal data, with a remaining-risk line. (None)

Week 9 - Core | Signature Solution - Refining & Finishing

Strand: Integration (Signature Solution - Refining & Finishing) | Framework ref(s): AIF·CU·G6·E AIF·SD·G6·E AIF·CE·G6·E AIF·GE·G6·E | 1 x

45-minute period (Core)

Objectives & success criteria

- Finish the solution so it meets the agreed success criteria.
- Evaluate it for accuracy, usefulness and risk.
- Explain one change that improved it.

Student-facing success criteria:

- I can finish my solution so it meets the success criteria our class agreed on.
- I can judge my solution for accuracy, usefulness and risk using the rubric.
- I can explain one change I made and say why it made the solution better.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In Weeks 5-8 you designed and built a signature solution that uses rule-based and learning ideas to help a real user at your school in Abu Dhabi. You already know that a rule-based system follows fixed rules a person writes, while a learning system finds patterns in example data. You have tested an early version and collected some feedback.

Activate: Show the class the agreed success-criteria card from Week 6 on the board. Ask pairs to whisper one thing their solution already does well and one thing that is still rough.

Teach: Explain that refining means comparing your work to the criteria on purpose, not just guessing. Model one accuracy check, one usefulness check and one risk check on a curated demo solution.

Apply: Pairs mark the demo solution against three rubric rows (accuracy, usefulness, risk) and agree the single most important improvement.

Check: Cold-call two pairs to name the improvement they chose and the evidence behind it.

Critical & analytical thinking focus

Evaluate & justify - quality and risk against criteria (Evaluate | DOK 3 | SOLO: Relational to Extended abstract)

The technical concept (teacher background)

Refining an AI-supported solution means measuring it against the success criteria you agreed earlier, not against how you

feel about it. A strong evaluation looks at three different things at once. Accuracy asks how often the system gives the right

or sensible answer, and you check it by testing with examples where you already know the correct outcome. Usefulness

asks whether the solution actually helps the real user do their task faster, more easily or more fairly. Risk asks what could

go wrong, who could be harmed, and whether any private information is exposed. In a rule-based part, refining often means

fixing or adding a rule; in a learning part, it often means giving cleaner or more balanced example data. A single well-chosen

improvement, backed by evidence, is worth more than many random changes. Responsible builders always verify a change

actually helped by re-testing, because a change that fixes one case can quietly break another.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Connect

7 min

Recap the success-criteria card and the three evaluation lenses (accuracy, usefulness, risk). Share the lesson goal: finish and improve one thing with evidence.

Recall their own criteria and note where their solution stands.

- What did we agree a good solution must do?
- Which lens matters most for your user?

Model

10 min

On a curated demo solution, think aloud through one accuracy test, one usefulness check and one basic risk check. Show how to write a short refinement note.

Follow the model and annotate a copy of the demo rubric.

- How do I test accuracy fairly?
- What evidence proves this is useful?
- What could go wrong here?

Guided

evaluation

10 min

Circulate as pairs run the same three checks on their own solution and record judgements on the tiered rubric.

Self-evaluate the solution against the Emerging rubric and mark accuracy, usefulness and risk.

- Where does my evidence come from?
- Which row is not yet at Emerging?

Refine

12 min

Support each pair to choose ONE improvement, make it, and re-test to confirm it helped without breaking something else.

Make one targeted change (fix a rule or improve example data) and re-test.

- Did the change fix the problem?
- Did anything else get worse?

Consolidate

6 min

Lead 'two stars and a wish' peer feedback and collect exit notes. Reinforce the responsible-use message about never using real personal data.

Give and receive structured peer feedback; write the refinement note.

- What is one star and one wish for this solution?
- Is any private data being used?

How do we keep it out?

Anticipated misconceptions

Misconception How to address it

A change always makes the solution better. Show that a change must be re-tested. A change can fix one case and break another, so evidence after the change is required.

Refining just means making it look nicer. Refining is measured against the success criteria for accuracy,

usefulness and risk, not only appearance.

If it works on my one example, it is accurate. Accuracy needs several test cases with known correct answers,

including tricky ones, not a single lucky example.

Hands-on activity guide

Overview: Pairs run a structured self-evaluation of their signature solution against the tiered rubric, choose one

improvement, apply it, and re-test to confirm it helped. This mirrors the Lab, which goes deeper on the risk review.

Grouping / timing: Same working pairs as the build weeks, with a peer-review swap for feedback.

Tool: Board demo plus printed rubric and refinement-note sheet; school-approved child-safe tool used by the teacher only

for any live check. - Account-free for students. Teacher-mediated demo on the class board; no student logins; no personal data entered into any tool.

Facilitation steps:

1. Open the agreed success-criteria card and read the three lenses.
2. Run one accuracy test using examples with known correct answers.
3. Run one usefulness check against what the real user needs.
4. Run one basic risk check (harm, fairness, privacy).
5. Choose the single most important improvement and record why.
6. Make the change and re-test to confirm it helped and broke nothing.
7. Swap with another pair for 'two stars and a wish' feedback.

Unplugged fallback: If no tool or board is available, use the simulated-AI cards: pairs judge printed curated outputs against the

rubric and plan the improvement on paper, exactly the same thinking without any device.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Self- and peer-evaluation against the tiered rubric (Emerging = expected)

- Check: Structured self/peer review using the rubric; one improvement acted upon

- Success criteria: Meets the Emerging criteria; names an accuracy/usefulness/risk judgement and one improvement.

- Evidence: Completed self/peer rubric + refinement note.

- Feedback / adjustment: Peer feedback structured ('two stars and a wish'), teacher-checked for accuracy.

Differentiation

Representation: Provide the criteria as a visual card, spoken recap and printed checklist so every learner can access it.

Action & expression: Learners may record judgements by writing, drawing, or speaking to a scribe.

Engagement: Anchor the task in each pair's real Abu Dhabi user so the improvement feels meaningful.

People of Determination: Offer sentence stems, chunked steps, extra think-time and a partner who can scribe; allow a quieter review corner.

EAL: Pre-teach accuracy, useful, risk, improve with bilingual glossary and picture cues; accept first-language notes.

Stretch / next tier: Ask advanced pairs to justify why their chosen improvement beats a second candidate improvement, using evidence.

Answer key / model responses

Task / item Model response Marking note

Name one accuracy

check for your solution.

I tested it on five cases where I already knew the correct answer and counted how many it got right.

Look for known-answer test cases, not opinion.

State one usefulness judgement.

It helps Aisha sort library returns faster, so it is useful for her real task. Must link to the real user's task.

Describe one risk you checked.

I checked no student names or photos are used, so private data is protected.

Accept harm, fairness or privacy risks.

Explain one change that improved it.

I added a rule for empty inputs, then re-tested; accuracy went from 3/5 to 5/5 and nothing else broke.

Requires before/after evidence.

Assessment rubric (observable look-fors)

Criterion Emerging (expected) Approaching Not yet

Evaluates against criteria

(accuracy, usefulness, risk)

Judges all three lenses with

evidence from testing.

Judges one or two lenses,

evidence partly clear.

Gives opinion without

testing evidence.

Acts on one improvement Makes one targeted change and re-tests to confirm it helped.

Makes a change but does not re-test it.

No clear improvement made.

Responsible use

(privacy/fairness)

Confirms no personal data used and considers fairness.

Mentions privacy or fairness but not both.

Does not consider privacy or fairness.

Resources & links

- ADEK K-12 AI Fluency Framework - The four-strand outcomes this unit is built on.

(<https://www.adek.gov.ae>)

- Class success-criteria and rubric card - The agreed Emerging criteria used for self and peer review. (school-shared)

Week 9 - Lab | Final Iteration and Risk Review (Lab)

Strand: Integration (Signature Solution - Refining & Finishing) | Framework ref(s): AIF·CU·G6·E AIF·SD·G6·E AIF·CE·G6·E AIF·GE·G6·E | 1 x

45-minute period (Lab)

Objectives & success criteria

- Run a final evaluation across accuracy, usefulness and risk, and act on one improvement.
- Complete a risk review and confirm the solution is safe and fair to use.
- Document the final iteration with before-and-after evidence.

Student-facing success criteria:

- I can run a final evaluation across accuracy, usefulness and risk and act on one improvement.
- I can complete a risk review and confirm my solution is safe and fair.
- I can record my final iteration with clear before-and-after evidence.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Week 9 Core lesson you judged your signature solution with three lenses and made one improvement. In

this Lab you carry the same solution and the same rubric into a full final iteration, adding a deeper risk review before you

freeze the solution for showcase.

Activate: Pairs re-open their refinement log from Core and read out their last recorded judgement.

Teach: Introduce the risk matrix (how likely x how serious) and model placing one risk on it for a curated demo solution.

Apply: Pairs place one real risk from their own solution onto a printed risk matrix.

Check: Each pair states whether their placed risk needs action now or just watching.

Critical & analytical thinking focus

Evaluate & justify - quality and risk against criteria (Evaluate | DOK 3 | SOLO: Relational to Extended abstract)

The technical concept (teacher background)

A risk review is a structured check of what could go wrong with a solution before real people rely on it. Two questions shape

every risk: how likely is it to happen, and how serious would it be if it did. Plotting a risk on a matrix of likelihood against

seriousness helps you decide what needs fixing now and what only needs watching. For

AI-supported solutions, common

risks include a wrong answer that misleads a user, unfair treatment of one group because the example data was

unbalanced, and exposure of private information. The most important privacy rule for students is simple: never put personal

or identifying data into a prompt or a training set. A final iteration means you make your last planned improvement, re-test

with known-answer cases, and confirm the solution is safe and fair before you freeze it.

Freezing means you stop changing it

so the version you showcase is the version you evaluated. Good documentation of

before-and-after evidence lets others

trust your judgement.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Connect

6 min

Recap the three lenses and introduce

today's focus: a full final iteration plus a

deeper risk review.

Re-read their refinement log

and set one goal for the Lab.

- Where did we stop last lesson?

- What is left to finish?

Model the risk

matrix

8 min

Model likelihood x seriousness on a

curated demo, placing a wrong-answer

risk and a privacy risk.

Watch and annotate a copy

of the matrix.

- How likely is this risk?
- How serious would it be?

Final iteration

14 min

Support pairs to make their last planned improvement and re-test with known-answer cases.

Apply the final change and record before-and-after accuracy.

- Did the change help?
- Did anything else break?

Risk review

12 min

Guide pairs to list risks, place them on the matrix, decide actions, and confirm no

Complete the risk matrix and write a safe-and-fair

- Which risk needs action now?
- Is any private data present
personal data is used. statement. anywhere?

Freeze and log

5 min

Have pairs freeze the version and complete the final iteration log for showcase.

Freeze the solution and finish the log.

- Is this the version we evaluated?
- Is our evidence clear enough for others to trust?

Anticipated misconceptions

Misconception How to address it

A rare risk is not worth checking. A rare risk can still be very serious. Use the matrix: high seriousness

needs a plan even at low likelihood.

If data has no names, it is automatically private-safe. Other details can still identify someone.

The safe rule is to keep all

personal or identifying data out of prompts and training sets.

Once improved, we can keep changing it right up to showcase.

You freeze the evaluated version so the solution you present is the one you actually tested and judged safe.

Hands-on activity guide

Overview: Pairs complete a full final iteration on their signature solution, then run a structured risk review using a likelihood-by-seriousness matrix, confirm the solution is safe and fair, and freeze the evaluated version. This is the deeper, practical partner to the Core lesson.

Grouping / timing: Working pairs, with a cross-pair check for the risk matrix.

Tool: Printed risk matrix and final-iteration log; board demo; school-approved child-safe tool used by the teacher only for any live re-test. - Account-free and teacher-mediated. Students use paper matrices and curated outputs; no student logins and no personal data entered anywhere.

Facilitation steps:

1. Re-open the refinement log and set the Lab goal.
2. Make the final planned improvement to the solution.
3. Re-test with at least five known-answer cases and record before and after.
4. List possible risks (wrong answer, unfairness, privacy).
5. Place each risk on the likelihood-by-seriousness matrix.
6. Decide an action for any high risk and confirm no personal data is used.
7. Write a one-line safe-and-fair statement and freeze the version.
8. Swap matrices with another pair to sense-check placements.

Unplugged fallback: If no device or board is available, use simulated-AI cards and the paper matrix only: pairs judge curated outputs, plan the final change on paper, and complete the full risk review unplugged.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Self- and peer-evaluation against the tiered rubric (Emerging = expected)

- Check: Final iteration completed and a full risk review with a safe-and-fair statement
- Success criteria: Meets the Emerging criteria; completes a risk matrix, acts on one improvement, and confirms no personal data is used.
- Evidence: Final iteration log + completed risk matrix + safe-and-fair statement.
- Feedback / adjustment: Cross-pair sense-check plus teacher confirmation that risk judgements and privacy checks are sound.

Differentiation

Representation: Give the risk matrix as a labelled diagram, a spoken walkthrough and a worked example.

Action & expression: Learners may place risks with sticky notes, drawings or dictation.

Engagement: Use each pair's real user scenario so the risks feel authentic and worth managing.

People of Determination: Provide a pre-labelled matrix, chunked steps, sentence stems and extra time; allow a scribe.

EAL: Pre-teach likelihood, seriousness, fair, freeze with a bilingual glossary and icons; accept first-language notes.

Stretch / next tier: Ask pairs to propose a design change that would move a risk to a lower box and justify it.

Answer key / model responses

Task / item Model response Marking note

Place one risk on the matrix and explain.

A wrong answer is medium likelihood and medium seriousness, so I added a check rule to reduce it.

Look for both likelihood and seriousness reasoning.

State your final improvement with evidence.

I balanced my example data; accuracy rose from 4/6 to 6/6 with no new errors.

Requires before/after numbers.

Write a safe-and-fair statement.

My solution uses no personal data and treats all users the same, so it is safe and fair to use.

Must cover privacy and fairness.

Explain what freezing means.

I stopped changing it so the version I show is the one I tested and judged safe.

Accept any answer linking freeze to the evaluated version.

Assessment rubric (observable look-fors)

Criterion Emerging (expected) Approaching Not yet

Completes a structured risk review

Places risks by likelihood and seriousness and decides actions.

Lists risks but placement or action is unclear.

No structured risk review.

Final iteration with evidence Makes the final change and shows before-and-after test results.

Makes a change but evidence

is thin.

No final iteration evidence.

Privacy and fairness

confirmed

Confirms no personal data used

and solution is fair.

Confirms one of privacy or

fairness.

Neither confirmed.

Resources & links

- ADEK K-12 AI Fluency Framework - The four-strand outcomes this unit is built on.

(<https://www.adek.gov.ae>)

- Risk matrix template - Printed likelihood-by-seriousness grid for the risk review.

(school-shared)

Week 10 - Core | Showcase, Assessment & Reflection

Strand: Integration (Showcase, Assessment & Reflection) | Framework ref(s): AIF·CU·G6·E

AIF·SD·G6·E AIF·CE·G6·E AIF·GE·G6·E | 1 x 45-

minute period (Core)

Objectives & success criteria

- Present the finished solution and explain at least two design/evaluation choices.

- Demonstrate the term's key outcomes in a short assessment task.

- State one thing learned and set one goal for next term.

Student-facing success criteria:

- I can present my finished solution and explain at least two design or evaluation choices.

- I can show the term's key outcomes in a short assessment task.

- I can state one thing I learned and set one goal for next term.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: Across Term 1 you learned the difference between rule-based and learning systems,

designed a signature

solution, built it, evaluated it and, in Week 9, finished it with a risk review. You now have a

frozen, safe-and-fair solution and

a refinement log full of evidence.

Activate: Show the four-strand summary poster. Ask each learner to point to the strand they feel strongest in and say why.

Teach: Model a 60-second solution pitch that names the user, the problem, and two justified design/evaluation choices.

Apply: Learners draft two design/evaluation choices they will explain, with evidence.

Check: Cold-call three learners to state one of their two choices and its evidence.

Critical & analytical thinking focus

Evaluate & communicate - justify and reflect (Evaluate | DOK 3 | SOLO: Extended abstract (stretch))

The technical concept (teacher background)

Communicating a solution well is part of building it responsibly. A clear pitch names the real user, the problem, how the

solution works, and the evidence that it is accurate, useful and safe. Explaining a design or evaluation choice means giving the reason behind a decision, not just describing what you did; for example, choosing a rule-based check because the task had a fixed correct answer, or balancing example data because one group was under-represented. A short performance assessment lets you show each of the four strands at the Emerging level: understanding concepts, designing and building, evaluating and using responsibly, and thinking about the wider effects. Reflection turns experience into learning by naming what worked, what was hard, and what you would do next. Setting a goal for next term, aimed at the next tier of Proficient, keeps your growth deliberate. Responsible communicators are honest about limits and never claim their solution is perfect.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Connect

6 min

Share the four-strand summary and the goal: present, be assessed, reflect.

Remind students to be honest about limits.

Locate their two

design/evaluation choices to explain.

- What are the four strands?
- Which two choices will you justify?

Model the pitch

8 min

Model a 60-second pitch and a self-reflection sentence, thinking aloud about honesty and evidence.

Annotate the pitch structure and refine their own two choices.

- What makes a choice justified?
- Where is my evidence?

Assessment

task

15 min

Administer the short Term-1 performance task that samples all four strands; keep it individual and quiet.

Complete the assessment task, showing each strand at Emerging.

- Which strand does this question check?
- Have I shown evidence?

Present and justify

Run brief pitches in small groups; ensure each learner explains two choices.

Present the finished solution and justify two

- Why did you make that choice?
- How do you know it worked?

10 min design/evaluation choices.

Reflect and set a goal
6 min

Lead structured self-reflection and record each learner's next-tier goal for Term 2.

Write one thing learned and one goal for next term.

- What is your proudest outcome?
- What is your goal toward Proficient?

Anticipated misconceptions

Misconception How to address it

A good pitch means saying the solution is perfect. Responsible presenting is honest about limits. Naming a limit and how you managed it shows stronger understanding.

Explaining a choice just means describing what I did. Justifying a choice gives the reason behind it and the evidence, for example why a rule-based check fit the task.

The assessment is separate from my project. The task samples the same four strands your project used, so your project evidence supports your answers.

Hands-on activity guide

Overview: Learners complete a short Term-1 performance task sampling all four strands at Emerging, then present their finished solution and justify two design/evaluation choices, and finish with a structured reflection and a Term-2 goal.

Grouping / timing: Individual for the assessment task; small groups of three or four for the pitches.

Tool: Printed assessment task and pitch card; four-strand summary poster; board demo only where needed, teacher-mediated. - Account-free. No student logins. The assessment and reflection are on paper; no personal data is entered into any tool.

Facilitation steps:

1. Read the four-strand summary and set out your evidence.
2. Complete the short Term-1 assessment task individually.
3. Prepare a 60-second pitch naming user, problem and two justified choices.
4. Present to your small group.
5. Give one peer a specific 'why' question about their choices.
6. Write one thing learned and one Term-2 goal toward Proficient.

Unplugged fallback: If groups cannot present live, learners record their pitch on the pitch card and swap cards for silent written peer feedback.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Summative Term-1 task samples all four strands at Emerging; reflection informs

Term 2

- Check: Short Term-1 performance task marked with the analytic rubric + structured self-reflection
- Success criteria: Demonstrates each strand at Emerging; articulates two justified design/evaluation choices.
- Evidence: Assessed task + reflection; results reported per strand.
- Feedback / adjustment: Feedback flags each student's next tier target (to Proficient) for Term 2.

Differentiation

Representation: Offer the assessment in written and read-aloud forms and the pitch as a structured card.

Action & expression: Learners may present by speaking, using cue cards, or a recorded clip.

Engagement: Let learners choose which two choices to spotlight so the pitch feels personal.

People of Determination: Provide sentence stems, a pitch scaffold, extra time and a low-pressure presenting option.

EAL: Pre-teach present, justify, choice, goal with bilingual glossary; allow first-language rehearsal before the pitch.

Stretch / next tier: Ask learners to explain a trade-off between two choices, reaching Extended abstract thinking.

Answer key / model responses

Task / item Model response Marking note

Name your user and problem in one sentence.

My solution helps Omar in the canteen sort recycling faster and more accurately.

Must name a real user and a clear problem.

Justify one design choice.

I used a rule-based check because recycling categories have fixed correct answers.

Look for a reason, not just a description.

Justify one evaluation choice.

I tested with six known-answer cases because one lucky example is not enough evidence.

Should link to accuracy or fairness.

State one thing learned and one goal.

I learned to re-test after changes; my goal is to build a solution with balanced data next term.

Goal should aim toward Proficient.

Assessment rubric (observable look-fors)

Criterion Emerging (expected) Approaching Not yet Presents and justifies choices

Explains two design/evaluation choices with reasons and evidence.

Explains one choice, or reasons are partly clear.

Describes work without justifying choices.

Demonstrates four strands (assessment)

Shows each strand at Emerging in the task.

Shows most strands at Emerging.

Several strands not yet shown.

Reflection and goal States one learning and one clear next-term goal.

States a learning or a goal, not both clearly.

No clear reflection or goal.

Resources & links

- ADEK K-12 AI Fluency Framework - The four-strand outcomes and tier descriptors used for reporting.

(<https://www.adek.gov.ae>)

- Four-strand summary poster - Emerging-level summary of the term across all four strands. (school-shared)

Week 10 - Lab | Showcase Carousel and Reflection (Lab)

Strand: Integration (Showcase, Assessment & Reflection) | Framework ref(s): AIF·CU·G6·E

AIF·SD·G6·E AIF·CE·G6·E AIF·GE·G6·E | 1 x 45-

minute period (Lab)

Objectives & success criteria

- Present the solution in a showcase carousel and give structured peer feedback.
- Consolidate reflections into a Term-2 goal across the four strands.
- Confirm the responsible-use message for every solution on display.

Student-facing success criteria:

- I can present my solution in a showcase carousel and give structured peer feedback.
- I can consolidate my reflections into a Term-2 goal across the four strands.
- I can confirm my solution is responsible to use and honest about its limits.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Week 10 Core lesson you presented your solution, completed the Term-1 assessment, and drafted a

reflection and goal. This Lab makes it hands-on: a showcase carousel where you rotate, present, and give structured peer

feedback, then consolidate everything into one Term-2 goal across all four strands.

Activate: Learners set up their frozen solution and evidence at a station and read the peer-feedback stems.

Teach: Model 'two stars and a wish' feedback tied to the four strands on a curated demo station.

Apply: Learners practise one star and one wish on a partner's station before the carousel starts.

Check: Two learners share the feedback they drafted; teacher checks it is specific and kind.

Critical & analytical thinking focus

Evaluate & communicate - justify and reflect (Evaluate | DOK 3 | SOLO: Extended abstract (stretch))

The technical concept (teacher background)

A showcase carousel is a structured way for a whole class to present and give feedback in a short time: solutions stay at

stations while learners rotate, present briefly, and leave written feedback. Good peer feedback is specific, kind and tied to

the criteria, so 'two stars and a wish' works best when each comment names a strand and points to evidence. Consolidating

reflections means gathering the small notes you made across the term into one clear picture of your growth across the four

strands, then choosing a single goal for next term aimed at the Proficient tier. Every solution on display should carry a

responsible-use message: it uses no personal data, it is honest about its limits, and it treats users fairly. A four-strand

Emerging summary captures what a Grade 6 learner can now do across understanding concepts, designing and building, evaluating and using responsibly, and thinking about wider effects. Presenting and receiving feedback well is itself a fluency skill, because real builders improve through honest, respectful review.

Lesson plan

Phase | time Teacher (I do / we do) Students Key questions (targeted thinking)

Connect

5 min

Explain the carousel rules and the feedback stems; reinforce the responsible-use message.

Set up their station with the frozen solution and evidence.

- What makes feedback specific and kind?
- Is my responsible-use message on display?

Carousel round

1

12 min

Manage rotations; ensure presenters name two justified choices and visitors leave strand-tagged feedback.

Present at their station and give feedback at two others.

- Which strand does this comment address?
- Where is the evidence?

Carousel round

2

10 min

Swap roles so all learners both present and review; prompt 'why' questions.

Give a second round of structured feedback and collect their own.

- Why did they make that choice?
- What is one honest limit here?

Consolidate Guide learners to gather feedback and Complete the four-strand • What can I now do in each

reflections

12 min

term notes into the four-strand summary
and one Term-2 goal.

Emerging summary and set
one goal.

strand?

- What single goal moves me
toward Proficient?

Close the term

6 min

Celebrate growth, confirm each
responsible-use message, and collect the
summaries for Term-2 planning.

Share one goal aloud and
hand in the summary.

- What are you most proud of?
- How will you keep your solutions
responsible?

Anticipated misconceptions

Misconception How to address it

Feedback should only say nice things. Kind feedback still includes a wish. A specific, respectful
wish tied to
a strand helps a peer grow.

A goal can be vague, like 'do better'. A strong goal is specific and tied to a strand, for example
'use
balanced data to improve fairness'.

The responsible-use message is only for my project,
not the showcase.

Every solution on display carries the message: no personal data,
honest about limits, fair to users.

Hands-on activity guide

Overview: The class runs a showcase carousel: solutions stay at stations while learners rotate
to present and leave
structured, strand-tagged peer feedback. Learners then consolidate all feedback and term notes
into a four-strand

Emerging summary and one Term-2 goal, confirming each solution's responsible-use message.

Grouping / timing: Individual stations; whole-class rotation in two rounds; pairs support each
other for consolidation.

Tool: Station cards, feedback slips, four-strand summary sheet; board demo only for the
modelling step, teacher-

mediated. - Account-free and paper-based. No student logins; no personal data on any station
or in any tool.

Facilitation steps:

1. Set up your station with the frozen solution, evidence and responsible-use message.
2. Present your solution in 60 seconds, naming two justified choices.
3. Rotate to at least two other stations and leave 'two stars and a wish' feedback tagged to a strand.
4. Collect the feedback slips left at your own station.
5. Gather your term notes and feedback onto the four-strand summary.
6. Set one specific Term-2 goal aimed at Proficient.
7. Confirm your responsible-use message is complete and honest.

Unplugged fallback: If a live carousel is not possible, run a paper gallery: station cards and evidence are laid out, and learners

circulate silently leaving written feedback slips, then consolidate as above.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Summative Term-1 task samples all four strands at Emerging; reflection informs

Term 2

• Check: Structured peer feedback across the carousel plus a completed four-strand summary and Term-2 goal

• Success criteria: Gives specific strand-tagged feedback, confirms responsible use, and sets one clear Term-2

goal.

• Evidence: Feedback slips + completed four-strand Emerging summary with goal.

• Feedback / adjustment: Peer feedback structured and strand-tagged; teacher flags each learner's next-tier

target for Term 2.

Differentiation

Representation: Provide feedback stems, a modelled example and the summary as a labelled grid.

Action & expression: Learners may present live, with cue cards, or via a recorded clip at their station.

Engagement: Choice of which two choices to spotlight and a friendly rotation partner keep confidence high.

People of Determination: Offer a quieter rotation slot, pre-filled feedback stems, extra time, and a scribe for slips.

EAL: Pre-teach carousel, feedback, star, wish, consolidate with bilingual glossary; allow first-language rehearsal.

Stretch / next tier: Ask learners to synthesise feedback from several peers into one improved design idea for Term 2.

Answer key / model responses

Task / item Model response Marking note

Write one star tagged to

a strand.

Star (Evaluate & use): you tested with six known-answer cases, which is strong evidence.

Must name a strand and

evidence.

Write one wish tagged to a strand.

Wish (Wider effects): explain who might be treated unfairly and how you would prevent it.

Should be specific and kind.

State your four-strand summary in brief.

I can explain rule-based vs learning, build a solution, evaluate it for risk, and consider wider effects.

One clause per strand at

Emerging.

Set one Term-2 goal. Next term I will use balanced data so my solution is fairer to all users.

Specific and tied to a strand toward Proficient.

Assessment rubric (observable look-fors)

Criterion Emerging (expected) Approaching Not yet

Structured peer feedback Gives specific 'two stars and a wish' tagged to strands.

Gives feedback but it is general or untagged.

Little or unhelpful feedback.

Four-strand summary consolidated

Summarises all four strands at Emerging with evidence.

Summarises most strands. Summary incomplete.

Term-2 goal and responsible use

Sets a specific goal and confirms responsible use.

Sets a goal or confirms responsible use, not both.

Neither is clear.

Resources & links

- ADEK K-12 AI Fluency Framework - The four-strand outcomes and tier descriptors for the summary. (<https://www.adek.gov.ae>)

- Carousel feedback slips and four-strand summary sheet - Templates for strand-tagged peer feedback and end-of-term consolidation. (school-shared)

Appendix A - Glossary of Key Terms

Teacher reference and student-facing definitions. Introduce each term where it first appears.

Data - information a system collects and uses to make a decision.

Label - a name or tag given to an example so a system can sort or learn it.

Rule - an instruction a system follows, often IF ... THEN ...

Outcome / output - the result a system produces after processing its input.

Fair / unfair - fair treats similar people or things equally; unfair wrongly favours or leaves out some.

Bias - a slant in a system's outputs, usually from unbalanced, skewed or low-quality data.

Personal information - data about a real person (name, photo, home, age) that must be kept private.

Rule-based system - a system that follows fixed IF-THEN rules a person wrote.

Learning system / model - a system that learns patterns from many examples (training data) instead of fixed rules.

Training data - the examples a learning system learns from.

Prompt - the words we give an AI system to tell it what to do.

Data quality / noisy data - how complete and correct data is; noisy data is messy or error-filled.

Trustworthy AI - an AI system whose outputs are reliable, fair and safe to use.

Appendix B - Understanding Bias: How Data Shapes Outputs

How learning systems use training data

At Grade 6 students meet learning systems (models). Unlike a rule-based system that follows fixed IF-THEN

rules a person wrote, a learning system finds patterns in many examples called training data and uses those

patterns to decide. So the training data shapes what the system does: if the examples are unbalanced, noisy or

low quality, the model can be biased or wrong. This is why 'AI is just rules' is only half the story - modern AI also

learns from data.

A concrete, age-appropriate example

The Animal-Sorter. We train a learning helper with many cat photos but very few dogs. Shown a new dog, it

labels it 'cat'. Nobody wrote a wrong rule - the training data was unbalanced, so the model learned a biased

pattern. Balanced, good-quality training data gives fairer, more accurate models.

Misconceptions to name and correct (say them aloud, then correct)

- 'AI is just fixed rules.' -> Rule-based systems follow rules; learning systems also learn patterns from

training data. Grade 6 studies both.

- 'Data is neutral / always true.' -> Training data is chosen by people; unbalanced or noisy data biases the model.

- 'AI understands like a person.' -> A model matches learned patterns; it has no understanding.

Use as the core of the fairness/data-quality lesson and as a one-line check in every hands-on activity: 'What

data did the system learn from? Who might be left out?'

Appendix C - Rule-Based vs Learning Systems (comparison)

Rule-based system Learning system (model)

How it decides Follows fixed IF-THEN rules a person wrote. Finds patterns in training data and uses them to decide.

Where it can go

wrong

A wrong or unfair rule. Unbalanced, noisy or low-quality training data.

Example A sorting robot with IF-THEN cards. An image labeller trained on many photos.

To improve it Fix or rewrite the rule. Improve the data (more balanced, cleaner).

Appendix D - Recommended Tools & Simulations + Unplugged Alternatives (UAE-accessible)

- Primary (teacher-mediated, account-free): Teachable Machine (teacher-run, no student logins) for a rule-

vs-learning demo; Machine Learning for Kids (teacher-mediated); a prompt-based helper on the teacher

screen.

- Fully unplugged core: printed clean-vs-noisy dataset cards, prompt-refinement logs and a rule-vs-learning

comparison card - every lesson runs with no device.

- Alternatives for restricted environments: (1) paper-only clean-vs-noisy data sorts and prompt-refinement

activities; (2) a single teacher-projected simulation or a pre-recorded demo played offline.

- Device requirements: one teacher device + projector is sufficient. Offline option: all demos can be pre-

recorded; no student accounts or internet needed for the unplugged path. Confirm every tool against your

school's approved-apps list (UAE).

- UDL notes travel with every lesson (multiple means of representation, action & expression, and

engagement).

Appendix E - Criteria Checklist (accuracy, fairness, usefulness, reliability)

Use when judging any output. Print one prompt per card as 'criteria cards' for pairs.

- Accuracy: did the system give the right answer on the examples we tested?

- Coverage: did we test examples from every group?

- Fairness: does any group get the wrong output more often? Who might be left out?

- Usefulness: does the output actually help the task?

- Reliability: does it give consistent answers when we try again?

- Data check: was the data balanced, complete and clean? What was missing or noisy?

- Fix: what one change to the data, prompt or rule would improve it?

Appendix F - Privacy: Why We Protect Personal Information

Rule: we never put personal information (real names, photos, home address, age) into a system; we use

synthetic (made-up) data only.

- Why - ethical: it is each person's right to keep their information private; sharing it can hurt or embarrass

someone.

- Why - technical: once data enters a system it can be stored, copied and reused in ways we cannot take

back, and small pieces of data can be combined to identify a person. Synthetic data removes this risk.

Appendix G - Printable Resources (templates)

Prompt frames

Sentence starters students use to instruct a helper: 'Please [task] using [these examples]. Show your steps. If

unsure, say so.' Print as cards.

Clean vs noisy dataset cards

Example Clean or noisy? Problem to fix

Prompt-refinement log

Version Prompt we tried What the output missed Improved prompt

1

2

Improvement / iteration log

Version What we tested What went wrong Change we made Better?

1

2

Appendix H - Real-World & Cross-Curricular Connections

- Real AI: data-and-model decisions power photo tagging, recommendation and image/voice recognition;

helper prompting is how people use assistants and chat tools responsibly.

- UAE example: a school library or an app store that sorts and recommends by labels and past choices - a

real 'uses data to decide' system students can see.

- Cross-curricular anchors: Maths (data, proportion, sets); Science (classification, evidence);

Humanities /

Moral education (fairness, privacy, responsible use); Literacy (clear prompts and success criteria).

- Deployment norms: real teams test a system for accuracy and fairness before people rely on it - exactly

what the criteria checklist rehearses.

Appendix I - Looking Ahead

This unit prepares you for Grade 7 and beyond, where you build, test and evaluate learning systems more

Deeply.

STUDENT LESSON CONTENT

Grade 6 (Year 7) | Phase 3 / Cycle 2 | AI Solution Builder

Main Course, Term 1 - Rule-Based vs Learning Systems

(ML Foundations)

Emerging tier | 10 weeks (Core + Lab)

Field Detail

Program Grade 6 Main Course, Term 1 (Emerging tier)

Unit theme Rule-Based vs Learning Systems - ML Foundations

Signature activity A structured, prompt-driven AI solution to a simple task, evaluated for accuracy and risk

Strands AI Conceptual Understanding | AI Solution Design & Development | Critical Evaluation & Informed Interaction | Ethics & Governance

Delivery model 2 x 45-minute periods per week - a Core lesson + a paired Lab lesson each week

Lessons in this pack 20 lessons (10 Core + 10 Lab)

Week 1 - Core | Unit Launch - Rule-Based vs Learning Systems

Some machines are told every single step. Others are shown lots of examples and work out the pattern

themselves. This term you become a system detective - can you tell which kind you are looking at?

I can...

- I can ask at least two genuine questions about rule-based and learning systems.
- I can sort three systems into 'follows fixed rules' or 'learns from data' and give a reason for each.
- I can help write our class responsible-use rules for AI tools.

Words to know

Word What it means

rule-based system A system that follows fixed steps a person wrote for it.

learning system A system that finds patterns from many examples instead of fixed rules.

data The examples or information a learning system is shown.

model The pattern a learning system builds from its data.

example One piece of data, like one photo or one email.

Two kinds of machine

A rule-based system is like a recipe. A person writes every step, and the machine follows them exactly. A Salik toll

gate charges you when your car passes - a person set that rule.

A learning system is different. Nobody writes a rule for every case. Instead it is shown many examples - this is data -

and it finds the pattern by itself. The pattern it builds is called a model.

Why it matters

If you know which kind a system is, you can guess when it might fail. A rule-based system fails when something

happens that has no rule. A learning system can fail if it was shown poor or unfair examples.

Neither kind is magic, and neither understands like a person. That is why we always check what a machine tells us.

Let's try it - hands-on

Time to be a detective. Sort three systems, then post the questions you most want answered this term.

1. Watch the two clips your teacher shows and spot which machine was told every step.
2. Sort the three cards into 'fixed rules' or 'learns from data'.
3. Write a reason under each card.
4. Post two questions on the class question wall.
5. Help agree one class rule for safe AI use.

Think about it

- Which system was hardest to sort, and why?
- What is one clue that a system learned from examples?

My reflection

- Write one thing you now understand about how machines make decisions.
- Write one question you still have.

Week 1 - Lab | Rule vs Learning - Hands-On Exploration (Lab)

Today you get your hands on two machines - one that never changes and one that learns from examples. Watch

closely: their differences will jump out.

I can...

- I can run the rule-card machine and the learn-from-examples game and record what each does.
- I can log at least three differences I observed between them.
- I can design a fair test that would reveal whether a system follows rules or learns.

Words to know

Word What it means

rule-card machine A fixed stack of if-then cards that always behaves the same way.

example card A labelled card the group learns a pattern from.

pattern The rule-of-thumb your group builds from the example cards.

test input A new input you give a system to see how it behaves.

predictable Behaves the same way every time for the same input.

Two stations, two behaviours

At Station A the rule cards never change. Feed the same input twice and you get the same output twice. That is a rule-based system.

At Station B nobody gives you rules. You study example cards, build a pattern, then classify new cards. That is a learning system in miniature.

Design a fair test

To tell the two apart, give each a brand-new, unusual input. A rule-based machine reacts the same as always or

breaks. A learning system tries to match the pattern it built.

The best detectives write down exactly what they will try and what each result would mean.

Let's try it - hands-on

Run both stations, log the differences, then design a test that would reveal which kind an unknown machine is.

1. Take a role: feeder, recorder or tester.
2. Run Station A with five inputs, including one strange one.

3. Run Station B: study examples, then sort five new cards.
4. Log at least three differences.
5. Write one test that would separate rules from learning.

Think about it

- Which station was more predictable, and how did you know?
- What made the learning game guess wrongly?

My reflection

- Write the clearest difference you saw today.
- Write what you would test first on a mystery machine.

Week 2 - Core | Rules vs Learning - How Each System Works

Two machines can do the same job in totally different ways. Today you learn the two features that reveal which

is which - and you will never be fooled by a shiny screen again.

I can...

- I can name two features that separate rule-based and learning systems.
- I can give one real example of each and explain why it fits.
- I can use the words rule-based, learning, data and model correctly.

Words to know

Word What it means

rule-based system Logic written by a person as fixed steps.

learning system A system that builds a model from data instead of fixed rules.

data The labelled examples a learning system is shown.

model The pattern a learning system builds from its data.

feature A true characteristic you use to classify a system, not a surface look.

Two features that decide it

Feature one - where does the logic come from? In a rule-based system a person writes it. In a learning system it comes from data.

Feature two - what happens with something new? A rule-based system only handles cases its rules cover. A learning system can handle new cases that look like its examples.

Data is not the model

Be careful with words. The data is the examples. The model is the pattern built from them.

Change the data and you

can change the model.

A learning system does not understand meaning - it matches patterns. Always check its answers.

Let's try it - hands-on

Build the feature chart, then use it to classify everyday systems you meet in the UAE.

1. Copy the two-column feature chart.
2. Add an example to each row.
3. Classify two everyday systems.
4. Underline the feature that decided each answer.
5. Write two sentences using data and model correctly.

Think about it

- Which feature is the strongest proof of a learning system?
- Why is 'it looks modern' not a good reason?

My reflection

- Write the two features you now use to classify systems.
- Write one system you are still unsure about and why.

Week 2 - Lab | Train a Simple Model (Lab)

Today you become a model trainer. You choose the examples, and the examples decide what your model learns

- so choose wisely and keep them safe.

I can...

- I can sort examples into training and testing sets and train a simple classifier.
- I can test the model on new examples and record how it did.
- I can explain how my training examples changed what the model learned, and name one data-safety rule.

Words to know

Word What it means

training data The examples the model learns from.

testing data New examples, held back, used to check the model.

model The pattern the system builds from the training data.

classifier A model that sorts examples into categories.

data-safety rule Use only non-personal examples - never faces, voices or names.

Train, then test

First you show the model many labelled training examples. It finds the pattern and builds a model.

Then you test it on new examples it has never seen. If it still gets them right, it really learned. If not, it only memorised.

Your examples shape the model

If all your apple pictures are red, the model learns 'red', not 'apple', and fails on a green one.

Variety and fairness

matter, not just how many.

Golden rule: only non-personal examples. No faces, no voices, no names ever go into a training set.

Let's try it - hands-on

Build a training set of safe, non-personal examples, train a model, then test it on new examples and check your data

is safe.

1. Pick two categories of non-personal items.
2. Collect and label 8-10 training examples each.
3. Hide 3 testing examples per category.
4. Train using training examples only.
5. Test the held-back examples and log the results.
6. Complete the data-safety and risk check.

Think about it

- Which testing example fooled the model, and why?
- How did your choice of examples change what it learned?

My reflection

- Write how your training examples shaped the model.
- Write one data-safety rule you followed today.

Week 3 - Core | Structured Prompts to Complete a Task

Ask an AI tool a vague question and you get a vague answer. Learn to structure your prompt and you can

actually get a task done. Today you learn the trick: Goal, Context, Format.

I can...

- I can write a structured prompt with a clear goal, context and format.
- I can say whether the task was completed and why.
- I can improve my prompt once to complete the task more fully.

Words to know

Word What it means

prompt The instruction you give an AI tool.

structured prompt A prompt with a clear goal, context and format.

goal What you want the tool to do.

context The details the tool needs to do it.

format How you want the answer to look.

Goal, Context, Format

A vague prompt like 'tell me about the desert' leaves the tool guessing. A structured prompt tells it exactly what to do.

Goal: what you want. Context: the details it needs. Format: how the answer should look.

Together they make the task

far more likely to be completed.

Check, then improve

Once you get an output, compare it to the task. Did it do everything you asked? If not, change one thing in your prompt and try again.

Two safety rules: never put personal data in a prompt, and always verify the answer - a confident output can still be wrong.

Let's try it - hands-on

Write a structured prompt for a real task, judge whether it completed the task, and improve it once.

1. Underline what 'complete' means for your task.
2. Write your prompt with Goal, Context and Format.
3. Watch the teacher run it on the approved tool.
4. Judge whether the task was completed.
5. Adjust your prompt once and verify the output.

Think about it

- Which part of the structure helped most?
- How did you know the task was or was not complete?

My reflection

- Write the one adjustment that improved your result.
- Write one safety rule you used today.

Week 3 - Lab | Prompt-to-Task Workshop (Lab)

You can structure one prompt - now build something reusable. Today you compare prompt structures on a

richer task and design a template you and your class can use again and again.

I can...

- I can complete a richer task with a structured prompt and compare two structures.
- I can design a reusable prompt template for similar tasks.
- I can check an output for accuracy and complete a risk check.

Words to know

Word What it means

richer task A task with several requirements to complete.

prompt structure The order and parts of a prompt, such as Goal-Context-Format.

reusable template A prompt skeleton with slots that works for many similar tasks.

verification Checking an output against a reliable source.

risk check A quick check for personal data and possible errors.

Compare, then reuse

A richer task has several requirements, so a weak prompt structure leaves parts unfinished. Run two structures on

the same task and compare which completes it and which is more accurate.

Then turn the winner into a reusable template - a prompt with labelled slots you can fill in for any similar task.

Build in the checks

Every good template forces two checks: verify the output against a reliable source, and run a risk check for personal data and errors.

This is how real AI solutions are designed - and it is the foundation for your signature activity in the coming weeks.

Let's try it - hands-on

Complete a richer task with two structures, compare them, then design a reusable template with checks built in.

1. List the three requirements of the task.
2. Draft two structured prompts with different orders.
3. Compare which output completed the task and check accuracy.
4. Turn the stronger structure into a reusable template.
5. Add a verification step and a risk check.

Think about it

- Which structure completed more of the task, and how do you know?
- Where should the verify step sit in your template?

My reflection

- Write the biggest difference between your two structures.
- Write how your template forces an accuracy check.

Week 4 - Core | Judging Outputs - Correct, Incorrect, Unclear

An answer can sound perfectly sure of itself and still be wrong. Today you become the judge who decides

whether an output is correct, incorrect, or simply unclear - and you learn to back up every judgement with a reason.

I can...

- I can sort outputs into correct, incorrect, and unclear.
- I can give a reason for each judgement I make.
- I can name one action to take when an output cannot be trusted.

Words to know

Word What it means

Output The text or answer an AI tool produces after you give it a prompt.

Verdict The judgement you give an output: correct, incorrect, or unclear.

Source A trusted place, like a textbook or official website, you check an answer against.

Hallucination When an AI tool states something made up as if it were a fact.

Verify To check whether something is true using a reliable source.

Sounds right is not the same as is right

A language model writes by predicting which words usually come next. That makes its answers smooth and

confident, but confidence is a writing style, not proof.

Your job is to separate 'sounds right' from 'is right' by checking the output against the task you set and a source you trust.

Three verdicts

Correct means the output matches the task and agrees with a trusted source.

Incorrect means it clashes with something you can check now - a fact, a number, or your instruction.

Unclear means it is vague, incomplete, or you cannot verify it yet, so it needs more checking before use.

Let's try it - hands-on

You will triage three short outputs. For each, give a verdict and a six-word reason. Remember: a reason must point to a check, not a feeling.

1. Read the task the output was meant to answer.
2. Read the output carefully.
3. Check it against the task and any source given.
4. Write your verdict: correct, incorrect, or unclear.
5. Add a short reason and, if needed, one action.

Think about it

- Which signal most often warned you an output was not trustworthy?
- When an output is unclear, is it safer to pause or to proceed?

My reflection

- Write one thing you will now check before trusting an AI answer.

- Describe a time a confident answer turned out to be wrong.

Week 4 - Lab | Output Triage Clinic (Lab)

Welcome to the triage clinic. Each output is a patient waiting for a diagnosis - correct, incorrect, or unclear - and

a treatment: the action you would take to verify it. By the end you will have ranked them all on a trust ladder.

I can...

- I can run the triage routine across a full set of outputs.
- I can rank outputs by trustworthiness and defend my order.
- I can choose a verification action for outputs I cannot trust.

Words to know

Word What it means

Triage Quickly sorting items by how they need to be handled.

Verification action The concrete step you take to check whether an output is true.

Trust ladder An ordering of outputs from most to least trustworthy.

Risk check Asking what could go wrong if an output were used as-is.

Match the check to the output

A fact needs a trusted source. A summary needs comparing back to the original text. A vague instruction needs a

clearer prompt before you can even judge it.

Choosing the right check is what turns a guess into a verified judgement.

Rank and defend

After triaging, place your outputs on the trust ladder from most to least trustworthy.

Be ready to defend your order using signals, not preferences: what evidence puts one output above another?

Let's try it - hands-on

Rotate through four stations. Triage each output, choose a verification action, then rank them all on the trust ladder

and prepare to defend your order.

1. Read the task and output at each station.
2. Assign a verdict and write a checkable reason.
3. Choose a verification action that fits the output type.
4. Rank all outputs on the trust ladder.
5. Agree your group's defence and one risk of the least trustworthy output.

Think about it

- Which verification action was hardest to choose, and why?
- Did any output move up or down the ladder after you discussed it?

My reflection

- Write one verification action you will reuse in a future task.
- Describe how you would explain your ranking to another group.

Week 5 - Core | Risks and Responsible Use

An AI answer can look helpful and still put someone at risk - through a mistake you did not check, or through

personal data you should never have shared. Today you learn to spot those risks and to name the rule that keeps people safe.

I can...

- I can name two genuine risks of using an AI tool.
- I can explain who each risk could affect.
- I can state a responsible-use rule that reduces a risk.

Words to know

Word What it means

Risk Something that could go wrong when using a tool.

Accuracy risk The danger of acting on an incorrect output.

Privacy Keeping personal or identifying information safe and unshared.

Personal data Information that identifies a person, like a name, address, or photo.

Responsible-use rule An agreed habit that reduces a risk.

Two risk families

Accuracy risk is the danger of using a confident but incorrect output without checking it. The mistake can spread into your work and anyone who trusts it.

Safety and privacy risk is the danger of entering personal or identifying data, which may be stored or used to train future systems and could expose you or someone else.

Every risk has a who

A real risk always names who could be harmed - you, a classmate, or your family.

Responsible-use rules are simple habits, agreed together, that reduce a risk: verify before you use, never enter personal data, and tell an adult if you are unsure.

Let's try it - hands-on

Sort each scenario into a risk family, name who is affected, and choose a responsible-use rule that would reduce the risk.

1. Read the scenario card.
2. Decide: accuracy risk or safety/privacy risk.
3. Name who could be affected.
4. Attach a responsible-use rule that reduces the risk.

Think about it

- Which risk do you think is easiest to forget about, and why?
- Is a rule useful if it is too hard to follow?

My reflection

- Write one responsible-use rule you will follow from now on.
- Describe a situation where you would refuse to enter data into a tool.

Week 5 - Lab | Class AI Risk Audit (Lab)

Today your group becomes a class safety board. You will audit how AI has actually been used, rank the risks you find, and decide which one to fix first - then write the rule that fixes it.

I can...

- I can audit our AI use against a rule set and find a gap.
- I can place risks on a matrix by likelihood and impact.
- I can propose one rule to reduce the highest-priority risk.

Words to know

Word What it means

Audit Checking real practice against a set of rules to find gaps.

Likelihood How often a risk could happen.

Impact How much harm a risk would cause.

Mitigate To reduce a risk, often by adding a rule.

Gap A place where practice does not meet the rule.

Audit against the rules

An audit compares what actually happened with what should happen. You read the anonymised class use-log and

flag any entry that breaks or stretches a responsible-use rule.

The log is anonymised on purpose: you fix systems and habits, never blame a person.

Rank, then rule

Place each flagged risk on the matrix by its likelihood and impact, then pick the highest-priority one.

Finally, write one realistic responsible-use rule that would close that gap - and be ready to pitch it for the class charter.

Let's try it - hands-on

Audit the anonymised use-log, flag the gaps, plot the risks on the matrix, and propose one rule to fix the top-priority risk.

1. Read each log entry against the rule set.
2. Flag gaps and name the risk and who it affects.
3. Plot risks on the likelihood-impact matrix.
4. Circle the highest-priority risk.
5. Write and pitch one mitigation rule.

Think about it

- Which risk surprised you as higher priority than expected?
- How do you make a rule strong but still realistic?

My reflection

- Write one change you will make to your own AI use after this audit.
- Explain why anonymising the log matters.

Week 6 - Core | Bringing the Strands Together

This term you learned four skills one at a time. Today you discover they were never separate - they form one

chain, from understanding the system to writing a prompt, checking the output, and using it responsibly. Put all

four hats on at once.

I can...

- I can complete a guided challenge that uses all four strands.
- I can explain how the system type and prompt shaped the output.
- I can name one responsible-use choice I made.

Words to know

Word What it means

Integration Connecting separate skills into one process.

System type Whether a system is rule-based or learning-based.

Prompt The instruction you give an AI tool to guide its output.

Triage Judging an output as correct, incorrect, or unclear.

Risk check Asking what could go wrong and choosing a responsible action.

One connected chain

The system type shapes what you can expect. The prompt shapes the output. The output needs a triage. And the

whole process needs a risk check.

Each strand feeds the next - miss one link and the chain weakens.

Explain the links

It is not enough to complete each step; you should be able to explain how one shaped the next.

For example: because the prompt was specific, the output stayed on task, which made triage easier and the risk

lower.

Let's try it - hands-on

Run one guided challenge through all four strands, then explain how the system and prompt shaped the output and

name one responsible-use choice you made.

1. Name the system type behind the task.
2. Write or choose a clear prompt.
3. Triage the curated output.
4. Run a basic risk check and note a responsible-use choice.
5. Explain how the system and prompt shaped the output.

Think about it

- Which strand was strongest in your work, and which needs practice?
- How would a weaker prompt have changed the whole chain?

My reflection

- Write which strand you will focus on before the signature solution.
- Explain one link in the chain in your own words.

Week 6 - Lab | Integrated Build-and-Check (Lab)

This is your dress rehearsal. You will run a real task from start to finish - build it with a system and a prompt,

then check it for accuracy and risk before deciding whether to use it. By the end you will know exactly which

strand to sharpen before the signature solution.

I can...

- I can run a build-and-check task through all four strands end-to-end.
- I can record the accuracy and risk checks I made.

- I can name which strand I most need to improve.

Words to know

Word What it means

Build-and-check Producing an AI solution and then testing it before use.

Accuracy verdict Whether the output is correct, incorrect, or unclear.

Risk verdict How risky the output is and what would reduce it.

Mitigation An action that reduces a risk.

Weakest link The strand most likely to fail under pressure.

Build, then check

Building means choosing a suitable system and designing a clear prompt to reach a useful output.

Checking means triaging that output for accuracy and running a risk assessment before you decide to use it. Building

without checking just produces confident mistakes.

Decide and diagnose

Your final call is use, do not use, or use-with-changes - and you must justify it with your two verdicts.

Then diagnose your weakest link: the strand where your evidence was thinnest. That is your focus before the

signature solution.

Let's try it - hands-on

Run the four stations end-to-end. Build with a system and prompt, check for accuracy and risk, decide whether to use

the output, and diagnose your weakest strand.

1. Name the system type and why it suits the task.
2. Design a clear, specific prompt.
3. Triage the curated output for accuracy.
4. Run a risk assessment and choose a mitigation.
5. Decide use / do not use / use-with-changes and diagnose your weakest strand.

Think about it

- Did your accuracy verdict or your risk verdict decide your use decision?
- Which strand was your weakest link, and how will you strengthen it?

My reflection

- Write your focus strand and one action to improve it before the signature solution.
- Explain why building without checking is risky.

Week 7 - Core | Signature Solution - Planning

Today you become a designer, not just a prompter. Before anyone types a single word into an AI tool, you will

plan a small solution with five parts that fit together: a goal, a prompt structure, inputs, a check, and a risk plan.

A good plan is like a bridge design drawn before the first bolt is placed - it lets you find weak spots on paper,

where fixing them is cheap.

I can...

- I can state the goal of my solution in one clear sentence.
- I can lay out a prompt structure with a role, a task, and one rule.
- I can name the inputs my solution needs and how I will check the output.
- I can describe one risk and one way my plan reduces it.

Words to know

Word What it means

Signature solution A small, reusable AI design with a clear goal and connected parts that solves one simple task.

Prompt structure The shape of a prompt: a role for the AI, a clear task, and at least one rule or limit.

Input The information a prompt works on, chosen to contain no personal or identifying data.

Evaluation method The agreed way you will judge whether an output is accurate.

Risk mitigation A design choice that lowers the chance or impact of something going wrong.

From one prompt to a whole solution

In Grade 5 you improved a single prompt. This term you design a solution: a set of parts that work together to do one simple job well.

The five parts are the goal, the prompt structure, the inputs, the evaluation method, and the risk plan. If any part is missing, the solution is harder to trust.

Writing a prompt structure

A prompt structure names a role ('You are a timetable helper'), a task ('list the three earliest buses'), and one rule ('if a time is missing, say so').

The role focuses the AI, the task says exactly what to do, and the rule keeps the output honest and easy to check.

Checks and risks before you build

An evaluation method says how you will judge accuracy, such as comparing the output to a known-correct answer key.

A risk plan names one thing that could go wrong, like a confident wrong answer, and one design choice that reduces

it, like a 'say if unsure' rule plus a human check. Never put personal or identifying data into a prompt.

Let's try it - hands-on

You will now plan your own signature solution on the plan sheet. You will not log in to any tool; the teacher runs any AI

on the board. Your job is to make the plan complete and safe.

1. Read your task card and write the goal in one clear sentence.
2. Fill the role-task-rule slots for your prompt structure.
3. List the inputs your prompt needs, checking there is no personal data.
4. Write your evaluation method and one risk with its mitigation.
5. Self-check against the five-part list, then submit for the approve-to-build gate.

Think about it

- Which part of a plan is easiest to forget, and why does that matter?
- How does a 'say if unsure' rule make an output safer to trust?

My reflection

- Write the one part of your plan you are most confident about.
- Write the one part you will strengthen before building next week.

Week 7 - Lab | Prototype and Pressure-Test the Plan (Lab)

A plan on paper looks tidy. Today you find out how it behaves under pressure. You will build a quick prototype,

then throw your hardest inputs at it on purpose - the missing item, the messy note, the two-in-one line. When it

stumbles, you will not hide the failure; you will design a fallback branch that catches it safely.

Engineers stress-

test bridges before people cross them. You will stress-test your solution before it is trusted.

I can...

- I can run my prompt structure on a normal input and read the output.
- I can pressure-test it with a tricky input and spot where it fails.
- I can add a fallback branch that says what to do when a check fails.
- I can record the change I made and the reason for it.

Words to know

Word What it means

Prototype A first working version built quickly to see how a design really behaves.

Pressure test Deliberately trying a tricky input to reveal a weak spot.

Edge case An unusual input at the edge of what a solution expects, like a missing or double value.

Fallback branch A planned 'if a check fails, then...' response that handles a failure safely.

Error type A label for what went wrong: wrong, unclear, or missing.

Prototype first, judge second

A prototype is a fast first version. Running it shows how your design behaves, not just how you hoped it would.

The teacher runs your prompt on the board with a normal curated input. You read the output and decide: did it meet

the goal?

Design the fallback branch

When a check fails, a good solution follows a fallback branch instead of giving a confident wrong answer.

Write it as 'If the check fails, then ____ and ask a person.' Keeping a human in the loop is part of responsible design.

Remember: only curated, made-up inputs - never real names or personal data.

Let's try it - hands-on

You will prototype your approved plan, pressure-test it, and add a fallback branch. The teacher runs the AI on the

board; you do not log in. Log every run.

1. Copy your prompt structure into the prototype slot of your log.

2. Watch the normal-input run on the board and record whether it met the goal.
3. Pressure-test with a tricky curated input and record the error type.
4. Write your fallback branch as an if-then sentence that asks a person.
5. Log the one change you made and the reason it will help.

Think about it

- Which tricky input broke the most prototypes, and why?
- Why is a handled failure safer than a confident wrong answer?

My reflection

- Write the weak spot your pressure test exposed.
- Write how your fallback branch will handle it next week.

Week 8 - Core | Signature Solution - Building

Today you build for real. You will follow your approved plan, run your inputs, and look hard at every output - not

just the ones that look right. When you find an answer that is wrong or unclear, you flag it, change one thing, and

watch what happens. This is how real systems get better: one careful change, one honest result. Keep every

prompt free of personal data, and keep a person in the loop for anything you flag.

I can...

- I can build my solution by following my approved plan.
- I can evaluate outputs and flag ones that are wrong or unclear.
- I can make one improvement and explain what changed and why.
- I can keep personal data out of every prompt.

Words to know

Word What it means

Build Turning your approved plan into a working solution by following it step by step.

Evaluate Judging an output against your check to decide if it is correct and clear.

Flag Marking an output that is wrong or unclear so it gets attention and a fix.

Improvement One deliberate change made in response to an evaluation.

Iteration loop The evaluate -> change -> result cycle that makes a solution better over time.

Follow the plan, do not improvise

Building means running your approved prompt structure on defined inputs, not inventing new prompts on the spot.

Record each output before you change anything. That way you can see what your solution really does.

Evaluate, flag, improve

Evaluate each output against your check: correct, unclear, or wrong. Flag anything that fails.

Then make one improvement and re-run. Record it as evaluate -> change -> result so the cause of the improvement is

clear.

Let's try it - hands-on

You will build your solution from your approved plan, evaluate the outputs, flag errors, and log one improvement. The

teacher runs the AI on the board; you do not log in.

1. Restate your goal and check on your log.
2. Run your prompt structure on defined curated inputs and record each output.
3. Evaluate each output and flag the wrong or unclear ones.
4. Make one improvement in response to a flag and re-run it.
5. Log the loop as evaluate -> change -> result.

Think about it

- How did you tell a subtly wrong output from a correct one?
- Why is one change at a time better than many at once?

My reflection

- Write the one improvement you made and why.
- Write what you would evaluate more carefully next time.

Week 8 - Lab | Extended Build and Evaluation (Lab)

This is the finish line for your signature solution - but finishing is not just getting one right answer. Today you

prove it works by running it across several test inputs and counting how many it gets right.

When one fails, your

fallback catches it gracefully and you document the fix. Finally you run a basic risk check: is it accurate, is it

clear, and is any prompt free of personal data? A solution you can measure and defend is one people can trust.

I can...

- I can complete a working solution that follows my plan.
- I can evaluate its outputs across several test inputs and measure accuracy.
- I can handle an incorrect output gracefully using my fallback.
- I can run a basic risk check and document the fix I made.

Words to know

Word What it means

Test set A small, curated group of inputs, including edge cases, used to measure how a solution performs.

Accuracy count The number of correct outputs out of the total tested, such as 5 out of 6.

Graceful handling Using the fallback so an incorrect output is caught safely, not presented as final.

Documented fix A written record of the change made to correct a failure and its effect.

Risk check A quick review of accuracy, clarity, and data safety before a solution is trusted.

Prove it across several inputs

One correct answer is not proof. Run your finished solution across a test set of several curated inputs and count how many outputs are correct.

Report accuracy as a simple count, like 5 out of 6, so your claim is honest and checkable.

Handle failures and check the risks

When an output is wrong, let your fallback catch it and keep a person in the loop, then document the fix and its effect.

Finish with a basic risk check: accurate enough? clear enough? any personal data used? The data answer must

always be no, because we use only curated inputs.

Let's try it - hands-on

You will complete your solution, evaluate it across six curated inputs, handle any failure with your fallback, document

the fix, and run a basic risk check. The teacher runs the AI on the board; you do not log in.

1. Confirm your build follows the plan and the fallback is wired in.
2. Define what counts as a correct output.
3. Run all six curated inputs and record each result.
4. Compute accuracy and handle any failure with your fallback, documenting the fix.
5. Complete the three-question risk check and note the biggest remaining risk.

Think about it

- Why is an accuracy count across several inputs fairer than judging one output?
- What is the biggest remaining risk in your solution, and who should check it?

My reflection

- Write your accuracy result and the fix you documented.
- Write one thing your risk check made you notice.

Week 9 - Core | Signature Solution - Refining & Finishing

You are almost there. This week you become the judge of your own work, checking it for accuracy, usefulness

and risk, then making one smart change that you can prove made it better.

I can...

- I can finish my solution so it meets the success criteria our class agreed on.
- I can judge my solution for accuracy, usefulness and risk using the rubric.
- I can explain one change I made and say why it made the solution better.

Words to know

Word What it means

Refine To improve something on purpose by comparing it to your goals.

Accuracy How often the system gives the right or sensible answer.

Usefulness How much the solution really helps the user do their task.

Risk What could go wrong or who could be harmed, including privacy.

Evidence Proof from testing, not just an opinion.

Judging your solution three ways

A good builder does not just ask 'do I like it?'. You measure your work against the criteria the class agreed.

Accuracy: test with examples where you know the right answer. Usefulness: check it helps your real user. Risk: ask

what could go wrong and whether any private data is exposed.

One change, proven

You do not need many changes. Choose the one improvement that matters most to your user.

Make the change, then re-test. If accuracy went up and nothing else broke, you have evidence it worked. Write it in

your refinement note.

Let's try it - hands-on

Work with your partner. Judge your solution against the rubric, choose one improvement, make it, and prove it helped.

1. Read the three lenses on the criteria card.
2. Run one accuracy test with known answers and record the score.
3. Check usefulness against your real user's task.
4. Do a basic risk check for harm, fairness and privacy.
5. Pick one improvement, make it, and re-test.
6. Swap for 'two stars and a wish' feedback.

Think about it

- Which lens was hardest to judge, and why?
- How did re-testing change your confidence in your solution?

My reflection

- Write the one change you made and the evidence it helped.
- Name one risk you are now sure your solution avoids.

Week 9 - Lab | Final Iteration and Risk Review (Lab)

Time for the final polish. In this Lab you make your last improvement, then act like a safety checker, running a

full risk review so you can promise your solution is safe and fair before showcase.

I can...

- I can run a final evaluation across accuracy, usefulness and risk and act on one improvement.
- I can complete a risk review and confirm my solution is safe and fair.
- I can record my final iteration with clear before-and-after evidence.

Words to know

Word What it means

Iteration One more round of improving and re-testing.

Likelihood How likely a risk is to actually happen.

Seriousness How much harm a risk would cause if it happened.

Risk matrix A grid that sorts risks by likelihood and seriousness.

Freeze To stop changing the version you have evaluated.

Sorting risks the smart way

Not every risk is equal. A risk that is both likely and serious needs action now. A rare but very serious risk still needs a plan.

Place each risk on the matrix, then decide: fix it now, or watch it. Always confirm no personal data is anywhere in your solution.

Finish, then freeze

Make your final planned change and re-test with known answers. Record before and after so others can trust your evidence.

Then freeze the version. The solution you show at the carousel must be the exact one you tested and judged safe.

Let's try it - hands-on

With your partner, make the final change, re-test, run the full risk review, and freeze your solution ready for showcase.

1. Set your Lab goal from the refinement log.
2. Make the final planned improvement.
3. Re-test with at least five known-answer cases and record before and after.
4. List risks and place each on the matrix.
5. Decide actions and confirm no personal data is used.
6. Write a safe-and-fair statement and freeze the version.

Think about it

- Which risk surprised you most when you placed it on the matrix?
- How did freezing change how carefully you tested?

My reflection

- Write your safe-and-fair statement in one sentence.
- Name the risk you are most confident you have managed.

Week 10 - Core | Showcase, Assessment & Reflection

This is your moment to show what you built and what you learned. You will present your finished solution, explain the choices behind it, complete a short assessment, and set a goal to grow even more next term.

I can...

- I can present my finished solution and explain at least two design or evaluation choices.
- I can show the term's key outcomes in a short assessment task.
- I can state one thing I learned and set one goal for next term.

Words to know

Word What it means

Pitch A short, clear explanation of your solution and its value.

Justify To give the reason and evidence behind a choice.

Assessment A task that shows what you have learned across the strands.

Reflection Thinking about what worked and what you would do next.

Goal A clear target for your next step of learning.

A pitch that tells the truth

A strong pitch names your user, the problem, how your solution works, and the evidence that it is accurate, useful and safe.

Explain two choices by giving reasons, not just descriptions. Be honest about one limit. Honesty makes your work more trustworthy, not weaker.

Show the whole term, then look ahead

The short assessment lets you show all four strands: understanding, building, evaluating and using responsibly, and wider effects.

Finish by naming one thing you learned and setting a goal for next term, aiming toward the Proficient tier.

Let's try it - hands-on

Complete the short assessment, then present your solution and justify two choices before setting your next-term goal.

1. Set out your evidence from the term.
2. Complete the short assessment task on your own.
3. Prepare a 60-second pitch with two justified choices.
4. Present to your group and answer one 'why' question.
5. Write one thing learned and one Term-2 goal.

Think about it

- Which of your two choices are you proudest of, and why?
- What limit of your solution is important to be honest about?

My reflection

- Write one thing you learned this term.
- Set one goal for Term 2 that moves you toward Proficient.

Week 10 - Lab | Showcase Carousel and Reflection (Lab)

Welcome to the showcase. Today your class becomes a gallery of AI-supported solutions. You will present, learn from each other with honest feedback, and pull your whole term together into one clear goal for next time.

I can...

- I can present my solution in a showcase carousel and give structured peer feedback.
- I can consolidate my reflections into a Term-2 goal across the four strands.
- I can confirm my solution is responsible to use and honest about its limits.

Words to know

Word What it means

Carousel A rotation where you present at your station and visit others.

Two stars and a wish Two things done well and one thing to improve.

Consolidate To gather many notes into one clear summary.

Responsible use No personal data, honest about limits, fair to users.

Proficient The next tier of skill you aim for in Term 2.

Feedback that helps

In the carousel you present in 60 seconds and then visit other stations. Leave 'two stars and a wish', and tag each comment to a strand.

Good feedback is specific, kind and tied to evidence. A helpful wish is a gift, not a criticism.

Your term in one picture

Gather your feedback and notes into the four-strand summary: what you can now do in concepts, design and build, evaluate and use, and wider effects.

Then set one specific Term-2 goal toward Proficient, and confirm every solution is responsible to use.

Let's try it - hands-on

Set up your station, present in the carousel, give and collect feedback, then consolidate your term into a four-strand

summary and one goal.

1. Set up your station with your frozen solution, evidence and responsible-use message.
2. Present in 60 seconds with two justified choices.
3. Rotate to at least two stations and leave strand-tagged feedback.
4. Collect the feedback left at your station.
5. Complete the four-strand Emerging summary.
6. Set one Term-2 goal and confirm responsible use.

Think about it

- What is the most useful wish someone gave you, and why?
- How will you keep your future solutions responsible and fair?

My reflection

- Write your four-strand summary, one line per strand.
- Set one specific Term-2 goal that moves you toward Proficient.

Figure list (for the graphic designer)

Every visual to be produced, in order. Full briefs appear at the point of use in the lessons above.

Figure ID Lesson One-line description

G6-W1-01 Week 1 - Core Two kinds of system: one follows written rules, one learns from examples.

G6-W1L-01 Week 1 - Lab How a learning game moves from examples to sorting new cards.

G6-W2-01 Week 2 - Core The two features that separate rule-based from learning systems.

G6-W2L-01 Week 2 - Lab We train on one set and test on a separate, unseen set.

G6-W2L-02 Week 2 - Lab A teachable classifier learns from safe, non-personal shapes.

G6-W3-01 Week 3 - Core A structured prompt: goal, context and format lead to a checkable result.

G6-W3L-01 Week 3 - Lab Compare two structures, then build the winner into a reusable template.

G6-W4-01 Week 4 - Core Use this routine every time before you trust an output.

G6-W4-02 Week 4 - Lab Different outputs need different checks.

G6-W4-03 Week 4 - Lab Trust is a scale, not a switch.

G6-W5-01 Week 5 - Core Some risks are more likely or more harmful than others.

G6-W5-02 Week 5 - Lab Fix the top-right risks before the rest.

G6-W6-01 Week 6 - Core Four strands, one connected process.

G6-W6-02 Week 6 - Lab Every solution is built and then checked before use.

G6-W6-03 Week 6 - Lab Accuracy and risk together decide if an output is ready.

G6-W7-01 Week 7 - Core A signature solution is parts that connect, not a single question.

G6-W7-02 Week 7 - Core Role plus task plus one rule is a structure you can check.

G6-W7-04 Week 7 - Lab Log every run so the pattern of failures is visible.

G6-W7-05 Week 7 - Lab A fallback turns a silent failure into a safe, handled outcome.

G6-W8-01 Week 8 - Core Build by following the plan, then evaluate every output.

G6-W8-02 Week 8 - Core One change, one result - that is trustworthy iteration.

G6-W8-04 Week 8 - Lab Accuracy is measured across the set, not from one case.

G6-W8-05 Week 8 - Lab Finish with a risk check that always confirms no personal data.

G6-W9-01 Week 9 - Core Judge your solution three ways before you improve it.

G6-W9-02 Week 9 - Core Prove your change helped by re-testing.

G6-W9-03 Week 9 - Lab Use the matrix to decide which risks need action now.

G6-W9-04 Week 9 - Lab Prove the improvement, then freeze the version.

G6-W10-01 Week 10 - Core A clear pitch names the user, problem, choices and evidence.

G6-W10-02 Week 10 - Core Your term at a glance across all four strands.

G6-W10-03 Week 10 - Lab Rotate, present, and leave strand-tagged feedback.

G6-W10-04 Week 10 - Lab Consolidate all four strands and set one goal.

Word bank

Rule-based system - follows fixed IF-THEN rules a person wrote.

Learning system (model) - learns patterns from many examples (training data) instead of fixed rules.

Training data - the examples a learning system learns from.

Prompt - the words we give an AI system to tell it what to do.

Data quality / noisy data - how complete and correct data is; noisy data is messy or has errors.

Bias - when a model is unfair because its training data was unbalanced or noisy.

Trustworthy AI - an AI whose answers are reliable, fair and safe to use.

Personal information - facts about a real person that we keep private.

Two kinds of AI

A rule-based system follows fixed rules a person wrote (IF ... THEN ...). A learning system (model) is

different: it learns patterns from lots of examples called training data. If the training data is unbalanced or

messy, the model can be biased - even though no one wrote a wrong rule.

Fair or unfair? A quick story

We train a helper with many cat photos but very few dogs. Shown a new dog, it says 'cat'. No one wrote a

wrong rule - the training data was unbalanced, so the model learned a biased pattern.

Balanced, clean

training data gives fairer, more accurate answers.

My check

- Is the training data balanced?

- Is the data clean or noisy?

- Did I test different kinds of examples?

- Who might be left out or treated unfairly?

- What one change to the data would make it more trustworthy?

Keep it private

We never type a real name, photo, home or age into a tool. We use made-up (synthetic) data only. Once

information goes into a system it is hard to take back.

Sample of a bridging lesson:

Data Quality and AI Outputs

What you will learn

- ✓ I can explain, using a clear example, how the quality of data changes what an AI produces.
- ✓ I can predict whether an output will get better or worse when the data is made cleaner or noisier, and say why.
- ✓ I can tell the difference between the amount of data and the quality of data.

Lesson route

Work through each part in order.

1

The dataset teaches

A model learns from its examples

2

Problems and effects

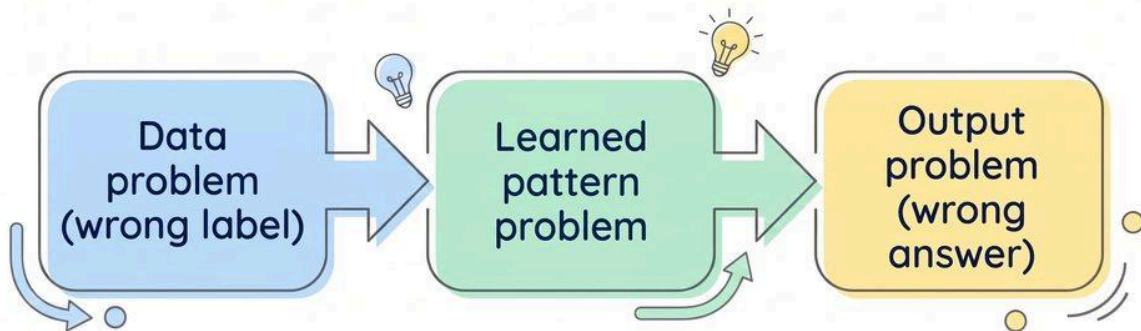
Errors, gaps and noise change the output

3

Quality beats quantity

Clean and small can beat big and messy

An AI learns the way you would teach a younger cousin - from the examples it is shown. Today you become a data detective who can predict an AI's mistakes just by studying its data. | About 45 minutes



Trace the chain: a data problem becomes an output problem.

B1C · Data Quality and AI Outputs

Overview

Part 1

Part 2

Part 3

Summary

Checks passed:

Part 1 of 3

The dataset teaches

Where does a model get its lessons?

1Figure 1. A model studies a dataset and adjusts itself to match the patterns it finds, so the dataset is its teacher.

Figure 1. A model studies a dataset and adjusts itself to match the patterns it finds, so the dataset is its teacher.

The model learns from its examples

An AI model is not filled with rules a person typed. Instead it studies a dataset - a set of labelled examples - and adjusts itself to match the patterns it finds.

This means the dataset is really the model's teacher. If the teacher shows wrong examples, the model learns the wrong lesson - and it does so confidently.



Good-quality data or poor-quality data?

Objective 1 · link data quality to what a model learns

Do it

Tap a card, then drop it in a bin. Ask: are the labels correct, complete and consistent?



Every date photo clearly labelled 'date'



A photo of a stone labelled 'date'



Labels spelled the same way every time



Examples that cover many kinds of dates



Half the photos too blurry to see



The same fruit labelled 'date', 'dat', and 'DATE'



Good-quality data



Poor-quality data



Verify Crew · Case 1

The Animal-Sorter

“

A helper is trained on many cat photos but very few dogs. Shown a brand-new dog, it labels it 'cat' - and sounds completely sure. Verify Crew, why did this happen?”



many cats



few dogs



says 'cat'

Why did the model call the dog a 'cat'?

A

Someone secretly typed a rule saying every animal in the whole world must always be called a cat.

B

The model understands animals like a person does.

C

It learned from data with too few dogs.

D

It ran out of battery.

Part 2 of 3

Problems and effects

How does a data problem become an output problem?

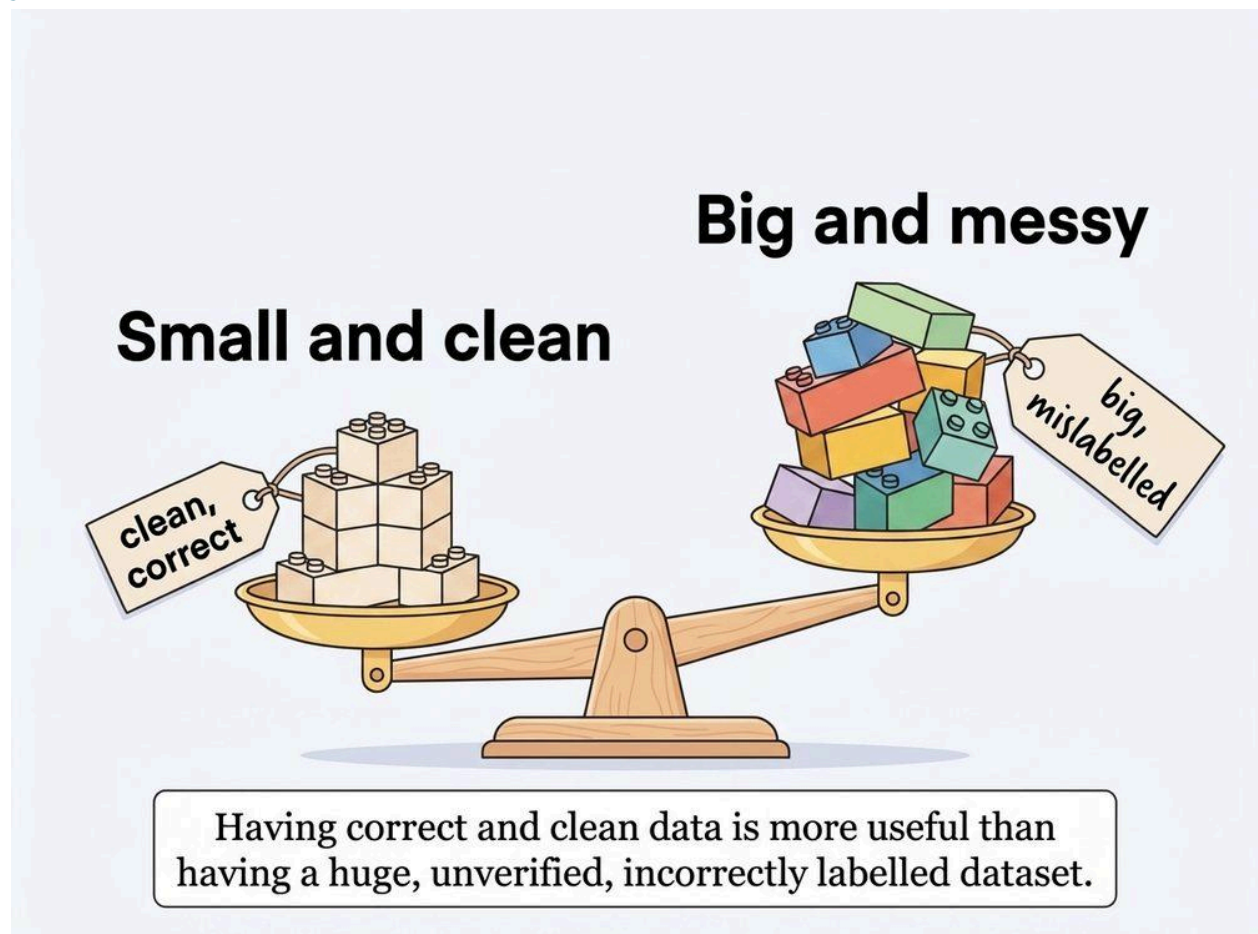
2Figure 2. Each kind of data problem - error, gap, or noise - leads to a matching problem in the output.

Figure 2. Each kind of data problem - error, gap, or noise - leads to a matching problem in the output.

Trace the chain

Three common data problems are errors (labels are wrong), gaps (some cases are missing), and noise (messy or inconsistent detail).

Each one shapes the output in its own way. If you know the input has a gap, you can predict the output will be blind to that gap. Data problem leads to learned-pattern problem leads to output problem - and you can trace each link.



Match each data problem to its effect

Objective 2 · predict the effect of a data problem

Do it

Match each data problem on the left to the output effect it causes on the right.

Data problem



Wrong labels (error)



Missing cases (gap)



Messy, inconsistent detail (noise)



Unbalanced examples

Effect on the output



is blind to what it never saw



treats one group unfairly



learns the wrong name for things



gives unsure, wobbly answers



Verify Crew · Case 2

The Missing Night Shots

“

A date-sorting dataset has clear daytime photos but zero night-time ones. The model will meet night photos in real life. Verify Crew, what will its output most likely get wrong?”



day only



no night



predict

What will the model most likely get wrong?

A

Only photos it was actually trained on.

B

Nothing at all, because a model with any data is always completely correct on every new photo.

C

Dates photographed at night.

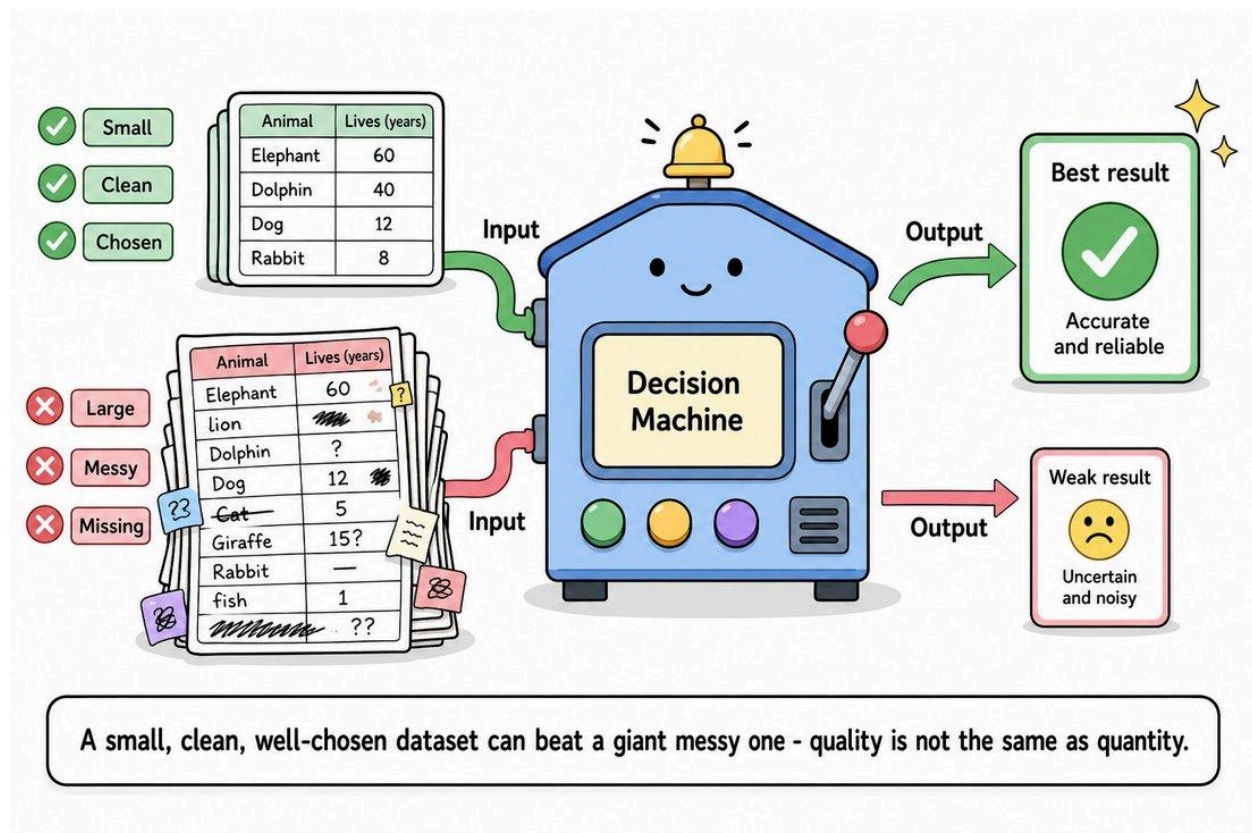
D

The colour of the sky, not the fruit.

Part 3 of 3

Quality beats quantity

Is more data always better?



3Figure 3. A small, clean, well-chosen dataset can beat a giant messy one - quality is not the same as quantity.

Figure 3. A small, clean, well-chosen dataset can beat a giant messy one - quality is not the same as quantity.

Quality is not quantity

It is tempting to think more data always helps. But a huge dataset full of mistakes just teaches the wrong pattern more strongly.

A small, clean, well-chosen dataset can beat a giant messy one. Before trusting any output, a careful builder checks the quality of the data behind it - and never puts personal or identifying information into a tool.



Which show a builder who values quality?

Objective 3 · tell quality apart from quantity

Do it

Tap the THREE statements a careful, responsible builder would agree with.



Skip checking the data if there is a lot of it.



A small, clean dataset can beat a giant messy one.



More data always makes an AI better, no matter what.



Use made-up data, never a real person's details.



Keep every mislabelled example - it still counts.



Check the quality of the data before trusting an output.



Verify Crew · Case 3

Two Piles of Photos

“

A builder must choose training data. Pile A has 10,000 date photos, but many are mislabelled. Pile B has just 200 clean, well-chosen photos. Verify Crew, which trains a more trustworthy model?”



10,000 messy



200 clean



choose

Which dataset would train a more trustworthy model?

A

The 200 clean, well-chosen photos.

B

The 10,000 photos, because a bigger pile of examples is always better no matter how many are mislabelled.

C

Both are exactly equal.

D

Neither - only the number of photos matters.



Away from the screen · Hands-on mission

Data detective: predict the mistake

Study one dataset card, predict the exact mistake the AI will make, then check your detective work against the revealed outputs.

What you need:

Dataset quality cards

Clean vs degraded output sets

A sentence-stem strip

A pencil

Read your dataset card and note how each item is labelled.

Name the quality problem: error, gap, or noise.

Predict one exact output mistake and write it down before the reveal.

Compare your prediction with the revealed clean and degraded outputs.

Write one sentence linking the data cause to the output effect.

 No screen? Compare pre-printed clean-data and noisy-data outputs on paper - underline where the noisy output goes wrong and label the matching data problem.

Lesson summary

You learned to predict an AI's mistakes just by studying its data.

Key ideas to remember

A model learns from its dataset, so poor examples teach the wrong pattern - confidently.

Three common data problems are errors (wrong labels), gaps (missing cases) and noise (messy detail).

Quality is not quantity: a small clean dataset can beat a giant messy one.

Never put a real person's details into a tool - use made-up (synthetic) data.

What to do next

Next: the Data Quality Investigation Lab, where you compare four datasets and rank them by evidence.

Worksheet

Practise what you learned, then print your work to show your teacher.

Name:

Class:

Date:

Task 1

Trace the chain

For your dataset card, fill each box: the data problem, the learned pattern problem, and the predicted output problem. Use an arrow between each.

The data problem is

.

So the model learns

.

So the output will

.

Task 2

Quality or quantity?

Read four short scenarios and decide whether the dataset's problem is about quality or quantity. Justify each choice in one sentence.

This is a
problem because

.

Task 3

Prediction vs reveal log

Record your prediction before the reveal, then what the outputs actually showed, then whether you were right.

Dataset version and why	My predicted output error	Actual revealed output	Match? (Y/N)
----------------------------	---------------------------	------------------------	--------------

Exit ticket

Show what you know

In one or two sentences, explain how making a dataset noisier would change an AI's output, and why.

Challenge ★

Try the challenge

Design your own dataset card with a hidden quality problem, swap with a partner, and see if they can predict the exact output mistake it would cause.

LAB Week 1:

Data Quality Investigation

Four versions of the same dataset, four sets of outputs, one big question: which data-quality problem does the most damage? Today you investigate, rank them, and defend your order with evidence - just like a real AI auditor. | About 45 minutes

What you will learn

- ✓ I can run a fair comparison of clean and noisy data and record what changes in the output.
- ✓ I can rank several datasets from best to worst expected output and justify my ranking.
- ✓ I can use evidence from the comparison to explain the data-to-output link.

Lesson route

Work through each part in order.

- 1 Make it a fair test**
Change only the data
- 2 Read the evidence**
Record how each output changed
- 3 Rank with evidence**
Order best to worst and justify

Grade 6 · Bridging · Term 1

Week 1 · Lab

Data Quality Investigation

Four versions of the same dataset, four sets of outputs, one big question: which data-quality problem does the most damage? Today you investigate, rank them, and defend your order with evidence - just like a real AI auditor. | About 45 minutes

B1L · Data Quality Investigation

Overview

Part 1

Part 2

Part 3

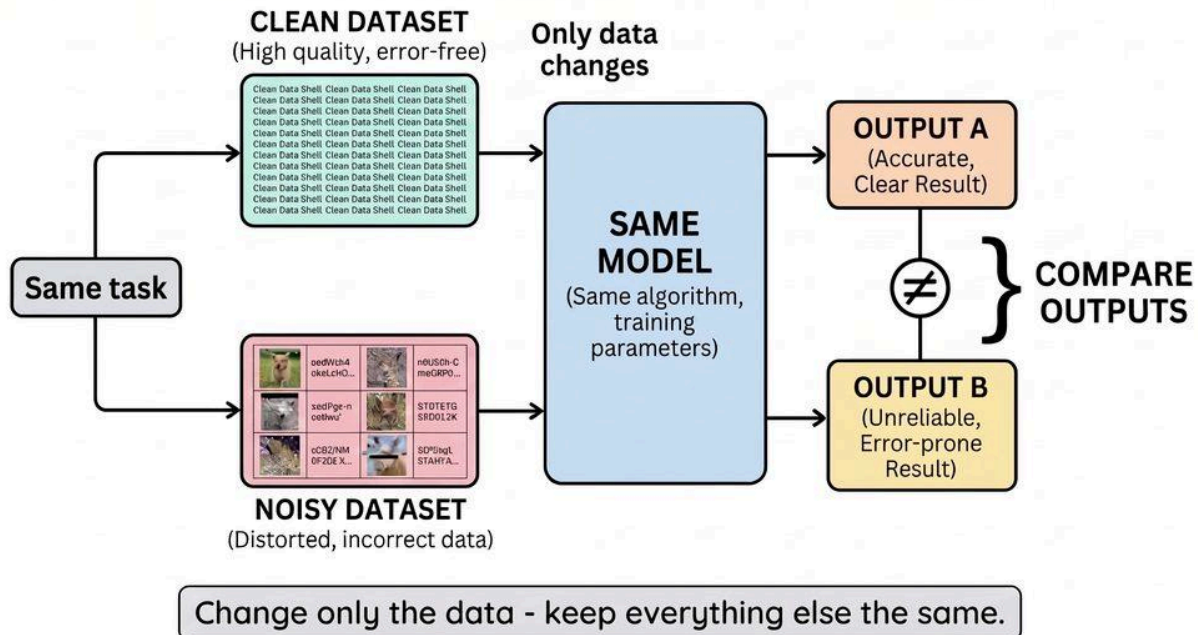
Summary

Checks passed:

Part 1 of 3

Make it a fair test

How do you compare data quality honestly?



1Figure 1. In a fair test you keep the task and model the same and change only the dataset.

Figure 1. In a fair test you keep the task and model the same and change only the dataset.

Make the comparison fair

To find out how data quality changes an output, you must be fair: keep the task and the model the same, and change only the dataset.

Then any difference you see in the output can be traced back to the data - and that is what makes your evidence trustworthy.



Put the investigation steps in order

Objective 1 · run a fair comparison and record it

Do it

Drag the steps into the order a careful auditor would follow.

Tap the steps in the right order, from first to last.

?



For each degraded version, record exactly how the output changed

?



Place all four on the ranking line and justify the order

?



Match each change to its data flaw

?



Study the clean baseline so you know what correct looks like

?



Score each dataset on accuracy, completeness and consistency



Verify Crew · Case 1

The Sloppy Experiment

“

One team swapped the dataset AND changed the task at the same time, then blamed the data for the different output. Verify Crew, why can't they trust their result?”



changed both



one output



no proof

Why can't the team trust their result?

A

The model was switched off the whole time.

B

They changed the task too, not just the data.

C

They should trust it fully, because changing lots of things at once always makes an experiment more fair.

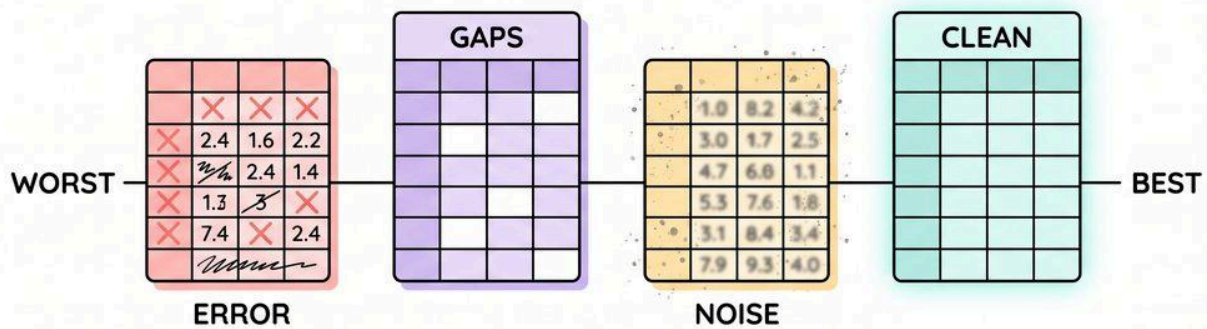
D

They used too few colours in their chart.

Part 2 of 3

Read the evidence

How healthy is each dataset?



Order the datasets and defend the hardest placements.

2Figure 2. Judge each dataset from the evidence: fine to use, needs care, or broken.

Figure 2. Judge each dataset from the evidence: fine to use, needs care, or broken.

Judge from the output

You work with several versions of the same dataset that differ in known ways: one clean, one with label errors, one with gaps, one with noise.

By watching how the output shifts across versions, you build evidence rather than opinion. Some datasets are fine, some need care, and some are broken - and the output shows you which.



Judge each dataset's health

Objective 2 · judge datasets by expected output quality

Do it

Tap a dataset, then drop it under Fine to use, Needs care, or Broken.



Clean baseline: labels correct, complete and consistent



Balanced examples across many kinds of dates



A few labels misspelled but still readable



Missing every night-time example



The same item labelled three different ways



Half the items mislabelled as the wrong fruit



So noisy the real pattern is buried



Fine to use



Needs care



Broken



Verify Crew · Case 2

The Missing December

“

Same task, same model - only the dataset changed. The clean baseline lists all twelve months, but the degraded output never mentions December. Verify Crew, which data flaw explains this?”



same task



no December



name flaw

Which data flaw best explains the missing December?

A

A gap - December was missing from the data.

B

A label error on every single month.

C

The task was made harder.

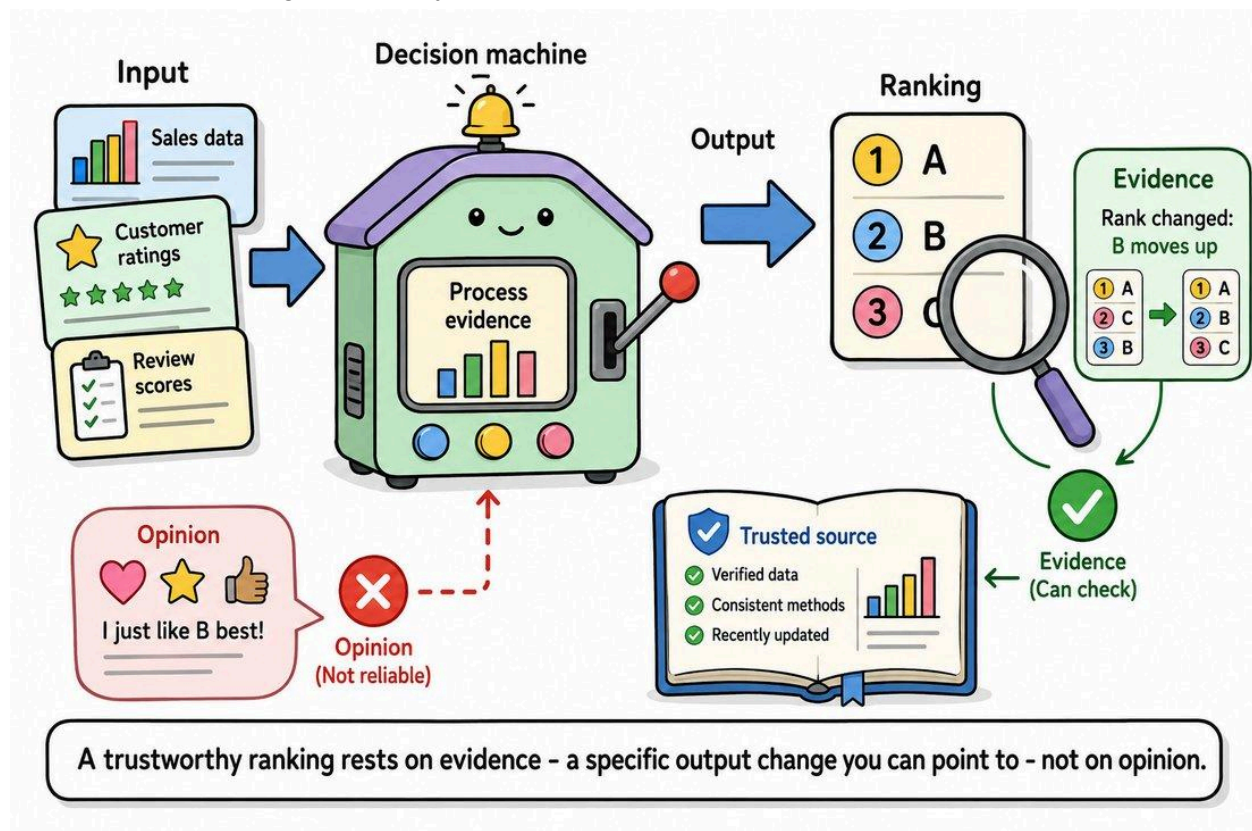
D

Nothing is wrong; the model simply decided on its own that December should not exist anymore.

Part 3 of 3

Rank with evidence

What makes a ranking trustworthy?



3Figure 3. A trustworthy ranking rests on evidence - a specific output change you can point to - not on opinion.

Figure 3. A trustworthy ranking rests on evidence - a specific output change you can point to - not on opinion.

Rank with evidence

After you see how each dataset changed the output, you rank them from best to worst expected quality.

The hard part is the middle: when two datasets are both weak, you decide which flaw hurt the output more, and you point to the exact output change that proves it.



Evidence or just an opinion?

Objective 2 · justify a ranking with evidence

Do it

Tap a reason, then decide: is it evidence you can point to, or just an opinion?

The gap output missed December that the baseline got right



The error output swapped 'date' for 'stone' on three cards



The noisy output changed its answer each time we checked



This dataset just feels the worst



I like the clean one best, no real reason



I ranked them in a random order



Evidence you can point to



Just an opinion



Verify Crew · Case 3

The Tricky Middle

“

Two datasets are both weak. Dataset E has wrong labels on half its cards; Dataset G is only missing one rare case. Verify Crew, which one should rank worse?”



E: half wrong



G: one gap



rank

Which dataset should rank worse?

A

Dataset G, because missing one rare case always damages an output more than getting half the labels wrong.

B

Dataset E - wrong labels harm more answers.

C

Whichever one has fewer cards.

D

They must always tie exactly.



Away from the screen · Hands-on mission

Audit four datasets and rank them

Compare four output sets against a clean baseline, record what changed, then rank the datasets from best to worst with evidence.

What you need:

Four-version output packs

A ranking line

A three-dimension scoring grid

A pencil

Study the clean baseline so you know what correct looks like.

For each degraded version, write exactly how the output changed.

Match each change to its data flaw: error, gap, or noise.

Score each dataset on accuracy, completeness and consistency.

Place all four on the ranking line and justify your order.

Focus your justification on the two datasets hardest to separate.

 No output packs? Sort pre-printed output cards along a best-to-worst line and annotate each with its data flaw and output effect.

Lesson summary

You ran a fair investigation and ranked datasets with real evidence.

Key ideas to remember

A fair test changes only the data, so any output change traces back to the data.

Evidence is a specific output change you can point to, not an opinion.

Ranking forces a judgement: when two datasets are weak, decide which flaw hurt more.

This is what real AI builders do - audit training data before trusting a system.

What to do next

Next: turning trustworthy data into clear prompts for a reliable AI helper.

Worksheet

Practise what you learned, then print your work to show your teacher.

Name:

Class:

Date:

Task 1

Change from baseline

For each degraded version, describe exactly how the output changed from the clean baseline, and name the data flaw responsible.

Compared to the baseline, this output

.

The flaw responsible is

.

Task 2

Defend the ranking

Write your full best-to-worst ranking and justify the two placements you found hardest to separate, using output evidence.

My order is

.

The hardest pair was
because the output

.

Task 3

Investigation log

Record each dataset version, its output change, its data flaw, and its dimension scores.

Version	Output change vs baseline	Data flaw	Accuracy/Complete/Consistent score
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Exit ticket

Show what you know

Which single data flaw would you fix first to improve the output the most, and why?

Challenge ★

Try the challenge

Invent a fifth dataset that combines two flaws. Predict where it lands on the ranking line and justify your prediction with the evidence you gathered.

 Print my worksheet

ASSESSMENT DOCUMENT:

FINAL ASSESSMENT

Grade 6 (Year 7)

Unit: Rule-Based vs Learning Systems (ML Foundations)

ASSESSMENT IDENTIFICATION

Assessment End-of-module final assessment

Module 8 of 14

Phase / Cycle Phase 3 - AI Solution Builder | Cycle 2

Term / Tier Term 1 | Emerging tier

Competency strands AI Conceptual Understanding; AI Solution Design and Development; Critical Evaluation and Informed

Interaction; Ethics & Governance

Framework reference ADEK K-12 AI Fluency Framework - AIF - Grade 6 -

Emerging

Format Performance-based (formative checkpoints + summative task)

Administration 55-70 minutes | individual work with teacher-mediated tool demonstration

Version / Date Version 1.0 | 2026

Prepared by AI Curriculum Unit

A curriculum-aligned, performance-based assessment of the four AI Fluency competency strands, with formative checkpoints, an authentic solution-builder task, an analytic rubric, moderation guidance, annotated exemplars, and a full teacher guide.

For teacher use in ADEK schools. Aligned to the delivered Grade 6 Main Course (Term 1, ten weeks of Core and

Lab). Not for distribution to students.

ADEK K-12 AI Fluency | Final Assessment | Grade 6 - Module 8: Rule-Based vs Learning Systems

1. Assessment overview and design rationale

This final assessment measures how well Grade 6 students have met the four AI Fluency competency outcomes taught across the ten weeks of the Term 1 Main Course, "Rule-Based vs Learning Systems (ML Foundations)". Grade 6 is the first module of Phase 3, AI Solution Builder. The key shift is that students no longer only use AI: they reason about how a system learns, plan and build a small structured AI-assisted solution, evaluate its outputs, and reason about bias, privacy and responsible use. The assessment is designed to be valid (it measures what was taught), reliable (two teachers reach the same judgement), and fair (every learner has an equitable route to show what they know).

It is performance-based. Students demonstrate learning by running a defined input-to-process-to-output process with a teacher-mediated AI tool and by explaining their reasoning, rather than by recalling isolated facts. As in the delivered course, all AI and ML use is teacher-mediated and account-free, and no personal data is ever entered into a tool.

What this assessment contains, at a glance

Component Purpose Evidence type Time

Formative checkpoints

(2)

Assessment FOR learning - check readiness mid-unit and adjust teaching before the summative task.

Teacher observation

and short written

responses

10-15 min

each, in-

lesson

Summative

performance task (4

parts)

Assessment OF learning - one authentic solution-builder task covering all four strands.

Worksheets, teacher-

mediated tool run, brief

conversation

55-70 min

(may be

split)

Analytic rubric Judge each strand at Emerging / Proficient /

Advanced against observable descriptors.

Teacher judgement

record

During or

right after

the task

Evidence and

moderation tools

Record evidence consistently and agree

judgements across teachers.

Observation record,

moderation protocol

As needed

Annotated exemplars Anchor judgements to worked examples at each level.

Reference material Reference

Research-based design principles

The design draws on well-established assessment research so that the results can be

trusted and used to move learning forward:

Backward design (Wiggins and McTighe). The task and rubric were built from the framework outcomes first, then the evidence, then the activity, so every part earns its place.

Constructive alignment (Biggs). Each task part is aligned to one primary strand, its success criteria, and a cognitive-demand level, shown in the alignment map in Section 2.

Page 2 of 22

ADEK K-12 AI Fluency | Final Assessment | Grade 6 - Module 8: Rule-Based vs Learning Systems

Formative assessment (Black and Wiliam). Two checkpoints make thinking visible early and carry specific feedback moves, so teaching can adapt before the summative task.

Triangulation of evidence. Judgements draw on three sources - what students produce, what they do, and what they say - rather than a single worksheet.

Cognitive demand (Webb's Depth of Knowledge and the SOLO taxonomy).

Parts progress from recognising system types (DOK 1-2) to planning a process, evaluating outputs, and reasoning about how data affects fairness and quality (DOK 3).

Universal Design for Learning (CAST). Multiple means of engagement, representation, and expression are built in, so access arrangements are part of the design rather than an afterthought.

Academic integrity and safe use. The plan (Part 2) and the evaluation (Part 3) are recorded individually so each student's reasoning is their own. Students may talk during exploration, but recorded answers must be independent. All tool use is teacher-operated on the board; students never sign in. No personal or identifying data is entered into any tool. Teachers may scribe or accept alternative response modes without changing the content of a student's answer.

Page 3 of 22

ADEK K-12 AI Fluency | Final Assessment | Grade 6 - Module 8: Rule-Based vs Learning Systems

2. Curriculum alignment map

This section makes the link between the delivered curriculum and every part of the assessment explicit and auditable. A teacher or reviewer can trace each strand from the framework outcome, to where it was taught, to where it is assessed.

Module identification

Field Detail

Programme Grade 6 Main Course, Term 1 (Emerging tier)

Module 8 of 14

Phase / Cycle Phase 3 - AI Solution Builder | Cycle 2

Unit theme Rule-Based vs Learning Systems (ML Foundations)

Signature activity

assessed

A structured, prompt-driven AI-assisted solution (input -> process -> output) that sorts labelled items for a real school user, built, evaluated and refined
Lessons covered Weeks 1-10 (Core + Lab, twenty 45-minute periods)

Framework reference AIF - Grade 6 - Emerging (CU, SD, CE, GE)

Framework outcomes assessed (the four strands)

Competency strand Emerging outcome
(target)

Proficient outcome Advanced outcome
(stretch)

AI Conceptual

Understanding

Recognizes the
difference between rule-
based and learning
systems.

Explains how AI systems
learn from data to
generate outputs

Explains how training
data affects output
quality and performance

AI Solution Design and
Development

Uses AI tools with
structured prompts to
complete tasks

Applies simple structured
processes (input to
output) to generate
results

Builds structured AI-
assisted solutions using
guided inputs and data

Critical Evaluation and
Informed Interaction

Identifies when outputs
are incorrect or unclear

Evaluates outputs for
accuracy and relevance

Explains how changes to
inputs or data improve
outputs

Ethics & Governance Recognizes basic risks
(e.g. incorrect outputs,
unsafe use)

Explains bias, privacy,
and the need for

responsible use

Identifies biased outputs

and suggests simple

ways to improve fairness

and safety

Alignment matrix: taught, then assessed

Each strand is mapped to the lessons that teach it and the assessment component that provides evidence, with the cognitive-demand level for the summative part.

Strand Where taught Success criteria assessed Assessed by Demand

AI Conceptual

Understanding

W1-W2 (Core +

Lab)

Name features that separate

rule-based and learning systems;

Checkpoint 1;

Summative Part

DOK 1-2 /

SOLO Multi

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explain how labelled examples

train a model; explain how the

training data shapes output

quality

1

AI Solution Design

and Development

W3 (Core +

Lab); W7-W8

Write a structured prompt (goal,

context, format); lay out an

input-process-output plan; build

a solution from the plan

Checkpoint 1;

Summative Part

2

DOK 2-3 /

SOLO

Relational

Critical Evaluation

and Informed

Interaction

W4 (Core +
Lab); W9

Triage outputs as correct /
incorrect / unclear with
checkable reasons; explain how
changing an input or the data
improves the output

Checkpoint 2;
Summative Part

3

DOK 2-3 /

SOLO

Relational

Ethics & Governance W5 (Core +
Lab); W9-W10

Name genuine risks and who
they affect; explain bias and
privacy; identify a biased output
and suggest a fairness or safety
fix

Checkpoint 2;
Summative Part

4

DOK 2-3 /

SOLO

Relational-

Extended

Vertical coherence (progression)

From Grade 5: students began direct AI interaction - they composed structured prompts,
evaluated outputs for accuracy and fairness, and started to see how training data affects
what AI generates. Grade 6 marks the entry into Phase 3 - AI Solution Builder.

Students now explain the difference between rule-based and learning-based systems,
apply a structured input-to-output process to generate and evaluate results, assess
outputs for bias and privacy risks, and reason about responsible AI governance. The shift is
from using AI as consumers to evaluating and building with it as informed, critical builders.

Towards Grade 7: students will chain these skills into multi-step AI workflows and justify
design decisions across a whole pipeline, so a strong Advanced performance here -
especially reasoning about how training data affects quality and fairness - is early
evidence of readiness for the Grade 7 outcomes.

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Systems

3. Learning outcomes and student-facing success
criteria

Teachers share the student-facing "I can" statements before and during the unit so learners know what success looks like. They are written in the same language used in the lessons.

Strand Student-facing success criteria ("I can...")

AI Conceptual

Understanding

I can name two features that separate a rule-based system from a learning system.

I can explain how a learning system builds a model from labelled example data.

I can explain how the training examples change what the model learns and how good its outputs are.

I can use the words rule-based, learning, data and model correctly.

AI Solution Design and

Development

I can write a structured prompt with a clear goal, context and format.

I can lay out a solution as input, then process, then output.

I can list the inputs my solution needs and keep personal data out of them.

I can build the solution by following my plan.

Critical Evaluation and

Informed Interaction

I can sort outputs into correct, incorrect and unclear and give a checkable reason for each.

I can measure how accurate a solution is across a set of test inputs.

I can explain how changing an input or the training data would improve the output.

Ethics & Governance I can name genuine risks of using an AI tool and say who they affect.

I can explain what bias and a privacy risk are in my own example.

I can spot an output that is unfair because of the data and suggest a way to make it fairer or safer.

I can state a responsible-use rule that reduces a risk.

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4. Assessment blueprint

The blueprint shows how the summative task distributes emphasis and cognitive demand across the four strands. Each strand carries roughly equal weight so no single strand dominates the overall judgement. As set out in the module design note, students are assessed on the quality of their process, evaluation and reasoning about improvements - not on what the AI happens to produce.

Summative part Primary strand Cognitive demand

Evidence Weight

Part 1 - How the sorter

learns

AI Conceptual

Understanding

DOK 1-2 Worksheet + talk 25%

Part 2 - Plan and build

the sorter

AI Solution Design and

Development

DOK 2-3 Plan sheet

(individual) + tool

run

25%

Part 3 - Evaluate the

outputs

Critical Evaluation and

Informed Interaction

DOK 2-3 Evaluation sheet

(individual)

25%

Part 4 - Fair, safe and

responsible

Ethics & Governance DOK 2-3 Worksheet +

short

conversation

25%

Cognitive-demand spread. The task rises from recognising the system type (DOK 1-2) to planning a structured process, evaluating outputs against a known answer key, and reasoning about how training data drives quality and fairness (DOK 3). Advanced probes in each part give every student a route to show higher-order reasoning without penalising those still consolidating the Emerging outcome.

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5. Formative checkpoints (Assessment FOR learning)

Run these two short checkpoints during the unit, not on the summative day. Each names when to use it, what to look for at each level, and a specific feedback move that closes the gap before the summative task.

Checkpoint 1 - "Rules, learning, and a structured prompt" (use after Week 3)

Strands: AI Conceptual Understanding and AI Solution Design and Development.

Activity. Show three system cards on the board: a metro ticket gate, a spam-email filter, and a photo app that groups pictures. Ask students to sort each as "follows fixed rules" or "learns from data" and give one feature-based reason. Then give a defined task ("write a two-line welcome

notice for new students") and ask each student to write a structured prompt for it using Goal, Context and Format. Allow three minutes of quiet work.

Look-for Emerging Proficient Advanced

Rule vs learning Sorts one card with a

surface reason (for

example, "it is on a

phone")

Sorts all three correctly

using a real feature

(where the logic comes

from)

Explains that the learning

system builds a model

from examples and would

handle a new case rule-

based logic could not

Structured prompt Writes a vague or one-

part prompt

Writes a prompt with a

clear goal, context and

format

Writes a structured

prompt and predicts how a

weak prompt would

change the output

Feedback move. If students sort by surface traits, re-teach the two-feature test from Week 2

(where the logic comes from, and how each system handles a new case). If a prompt is only a goal,

model adding context and a format slot using the Goal-Context-Format frame and have them redraft

one line.

Checkpoint 2 - "Triage and risk" (use after Week 5)

Strands: Critical Evaluation and Informed Interaction, and Ethics & Governance.

Activity. Show two curated outputs for the same task: a correct one, and a fluent summary that adds a fact not in the source. Ask students to (a) give each a verdict - correct, incorrect, or unclear - with a checkable reason, and (b) name one risk of using the untrustworthy output as-is and who it could affect.

Look-for Emerging Proficient Advanced

Triage with reason Gives a verdict but the

reason is a feeling ("it

Gives a correct verdict

tied to the task or a source

Names the exact added

detail and the verification

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looks wrong") action that would confirm it

Risk and who Says the output could be a problem without naming who

Names a genuine risk and a specific person or group it affects

Links the risk to a responsible-use rule that would reduce it

Feedback move. If a verdict rests on a feeling, model one check - compare the output to the source - and ask for a six-word checkable reason. If risk stays at "it is bad", prompt with "who is affected, and what rule would stop it?" to move reasoning from the tool in general to the specific situation.

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6. Summative performance task - "The Heritage-Day Souq Sorter"

An authentic task built directly on the unit's signature solution: a small, structured AI-assisted helper that sorts short item cards for the school Heritage-Day souq into category labels. The four parts move through all four strands and follow the input-to-process-to-output method used across Weeks 7-10. Read the scene-setting script, then run the parts in order. All tool use is teacher-operated on the board; students plan, evaluate and reason on their own sheets.

Scene-setting script (read aloud). "Our school is holding a Heritage-Day souq. Dozens of stalls have handed in short cards describing what they will sell, and a helper is needed to sort each card into a category so visitors can find things. Today you will design a small AI helper called

the Souq Sorter. First you will decide what kind of system it should be and explain how a learning

system would learn to sort. Then you will plan it as input, process and output and we will run it together on the board. Then you will check its answers against the correct labels and improve it. Finally you will decide whether it is fair, safe and responsible to use. Remember: I will run the tool

- you never type personal data, and there are no wrong answers when you explain your thinking.

Show it by writing, marking, or telling me."

Materials to prepare (see Appendix)

Souq Sorter Example Set - a labelled table of item cards used to "train" the sorter, with

a deliberate imbalance (Dates and Food examples far outnumber the other categories).

Souq Sorter Output Sheet - five curated outputs the sorter produced for five new item cards, with an answer key of the correct labels, including two seeded problems (one incorrect label and one unclear label).

Solution Plan Sheet (input -> process -> output) - a one-page frame with a goal line, a structured-prompt box (role, task, one rule), an inputs list, an output-format line, and an evaluation-and-risk strip.

Two "tricky" item cards for the fairness and privacy part: one that fits a rarely-seen category, and one that wrongly includes a stallholder's full name and phone number.

A pencil and coloured pens or highlighters for marking the outputs.

Part 1 - How the Souq Sorter learns

Strand: AI Conceptual Understanding | DOK 1-2

Show the Souq Sorter Example Set and read the category labels aloud. Ask:

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"Should the Souq Sorter follow fixed rules a person writes, or learn from these labelled example cards? Give one feature-based reason."

"If it learns, explain in your own words how it uses the labelled examples to build a model that can sort a new card."

Advanced probe: "Look at the whole Example Set. Most cards are labelled Dates and Food. How will that shape what the sorter learns, and how good will its labels be for a craft or textile card?"

Part 2 - Plan and build the sorter

Strand: AI Solution Design and Development | DOK 2-3 | Plan recorded individually

Give each student the Solution Plan Sheet. Say: "Plan the Souq Sorter as input, then process, then output." Ask students to complete, on their own sheet:

the goal in one clear sentence (what it sorts and who it helps);

a structured prompt in the box: a role, the task, and one rule (for example, a rule that no personal data is used);

the inputs the sorter needs, with personal or identifying data left out;

the output format (for example, "one category label per card").

Then run two or three students' prompt structures on the board with a defined item card and show the output. Ask: "Did the process turn the input into the output you planned?"

What one change would make it complete the task more fully?"

Worked frame on the sheet. Input: a short item card. Process: "You are a souq stall sorter.

Read the card and give one category label from Dates and Food, Crafts and Pottery, Textiles and

Clothing, or Other. Use no personal data." Output: one category label. Students complete their own version and keep personal data out of every input.

Part 3 - Evaluate the outputs

Strand: Critical Evaluation and Informed Interaction | DOK 2-3 | Completed individually

Give each student the Souq Sorter Output Sheet (five outputs plus the answer key). Say:

"Check each output against the correct label. Mark it correct, incorrect, or unclear, and write a checkable reason." Then ask students to work out the accuracy count (how many of five are correct) and answer:

"Which output is incorrect, and which is unclear? How do you know?"

"Choose one wrong or unclear output. What one change to the input or to the example data would make it correct?"

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Advanced probe: "How would adding more varied example cards for the rare categories change the sorter's accuracy?"

Seeded problems (for the teacher). 1. An incorrect label - a pottery water-jug card is labelled Dates and Food; it clashes with the answer key.

2. An unclear label - a card that lists two items ("dates and a woven basket") is given one vague label, so it cannot be judged until the input is clarified.

3. The imbalance link - the rare categories are labelled less accurately because the Example Set has few of them.

Part 4 - Fair, safe and responsible

Strand: Ethics & Governance | DOK 2-3

Give each student the two tricky item cards and say: "Before we let the Souq Sorter be used at the fair, we need to check it is fair and safe." Then ask:

"The sorter often mislabels craft and textile cards. Is that fair to those stalls? What in the example data causes it?"

"One card includes a stallholder's full name and phone number. What is the privacy risk, and who could it affect?"

"State one responsible-use rule the Souq Sorter should follow."

Advanced probe: "Suggest one simple way to make the sorter fairer or safer - for the bias, for the privacy risk, or for both."

How the task maps to the strands

Part Strand What strong evidence looks like Demand

1 AI Conceptual Understanding Chooses a system type with a feature reason, explains how labelled examples build a model, and links data imbalance to output quality

DOK 1-2

2 AI Solution Design and Development

Lays out a clear input-process-output plan with a structured prompt and safe inputs, and builds it by following the plan

DOK 2-3

3 Critical Evaluation and Informed Interaction

Triages all five outputs with checkable reasons,

reports an accuracy count, and explains a data or input change that improves an output

DOK 2-3

4 Ethics & Governance Explains the bias comes from the imbalanced data and the privacy risk from the personal details, and proposes a fairness or safety fix

DOK 2-3

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7. Analytic rubric

Judge each strand independently against the descriptors below. Circle the observed level. The rubric describes observable performance, so two teachers using it on the same evidence should reach the same level. Emerging is the expected Grade 6 standard.

Competency

strand

Emerging Proficient Advanced

AI Conceptual

Understanding

(Part 1)

Recognises that the sorter could follow rules or learn, but cannot yet say how a learning system uses the example data.

Explains that the sorter learns a model from the labelled examples and uses it to label a new card.

Explains how the imbalanced training data lowers output quality for rare categories - training data affects performance.

AI Solution Design and Development

(Part 2)

Uses a structured prompt to complete the sorting task but the input-process-output plan is partial.

Lays out a clear input-process-output plan with a structured prompt and safe inputs, and builds it by following the plan.

Builds a structured solution with guided inputs, adds a rule or output format that hardens it, and keeps all inputs data-safe.

Critical Evaluation
and Informed
Interaction

(Part 3)

Identifies that at least one output is incorrect or unclear but the reasons rely on feeling.

Triages the outputs for accuracy with checkable reasons and reports a correct accuracy count.

Explains how a specific change to the input or the example data would improve the outputs.

Ethics &
Governance

(Part 4)

Recognises a basic risk (a wrong label or the personal details) but cannot yet explain bias or privacy.

Explains the bias and the privacy risk and states why responsible use matters and who is affected.

Identifies the biased outputs, traces them to the imbalanced data, and suggests a simple

fairness or safety fix.

Overall level judgement

Overall level Decision rule

Emerging

(expected)

At least 3 of 4 strands at Emerging or above. The student recognises the system type, uses a structured prompt, spots an incorrect or unclear output, and recognises a basic risk. This is the expected Grade 6 standard at the end of the first Phase 3 module.

Proficient At least 3 of 4 strands at Proficient or above, with none below Emerging.

The student explains how the model learns from data, lays out and builds an input-process-output solution, evaluates outputs for accuracy, and explains bias, privacy and responsible use.

Advanced At least 3 strands at Advanced, with no strand below Proficient. The student reasons about how training data affects quality, hardens the solution, explains data-driven improvements to outputs, and proposes fairness and safety fixes.

Working towards Emerging. If a student cannot yet meet the Emerging descriptor in a strand even with support, record "working towards" and use the linked bridging content (Section 11)

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before re-checking. This is diagnostic information, not a failure.

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8. Evidence recording and triangulation

Base each judgement on three sources of evidence, not one worksheet. This raises reliability and gives students more than one way to show what they know.

Evidence source What to collect Strands it best evidences

Products (what students make)

The completed Solution Plan Sheet (Part 2) and the marked Souq Sorter Output Sheet (Part 3)

AI Solution Design and

Development; Critical

Evaluation

Observations (what students do)

How students sort the system cards, build the plan, and mark outputs against the answer key

AI Conceptual

Understanding; Critical

Evaluation

Conversations (what students say)

Answers to the Part 1 and Part 4 questions and the "why" and "what one change" probes

AI Conceptual

Understanding; Ethics &

Governance

Teacher observation record (one per student)

Field Record

Student name / Class _____ Date: _____

AI Conceptual Understanding E P A Note: _____

AI Solution Design and

Development

E P A Note: _____

Critical Evaluation and

Informed Interaction

E P A Note: _____

Ethics & Governance E P A Note: _____

Overall judgement EMERGING PROFICIENT ADVANCED

Access arrangements used _____

Next step / bridging _____

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9. Standardisation and moderation guide

Moderation makes judgements consistent across teachers and classes. Run a short moderation meeting after the first set of students has been assessed.

1. Each teacher independently levels the same 2-3 anonymised student pieces using the rubric in Section 7.

2. Compare levels. Where they match, note the rubric language that decided it. Where they differ, return to the observable evidence and the annotated exemplars in Section 10 to agree a level.

3. Agree borderline decisions: a student who is Proficient on three strands but Emerging on one is Proficient overall; a student Advanced on two strands and Proficient on two is Proficient (Advanced needs three).

4. Record any task-specific decisions (for example, how much teacher support during the tool run still counts as an independent plan) so the whole team applies them the same way.

Common pitfalls to avoid

Crediting the AI's output, not the student's process. The module design note is explicit: assess the plan, the evaluation and the reasoning, never the polish of what the AI produced.

Rewarding fluency over reasoning. A confident talker who cannot explain how the

model learns from data is still Emerging on that strand.

Over-crediting a lucky verdict. In Part 3, credit Proficient only when the verdict has a checkable reason and the accuracy count is right, not only that a guess matched.

Halo effect. Level each strand on its own evidence; a strong plan in Part 2 does not lift a weak fairness argument in Part 4.

Counting scaffolding as independence. If a plan or an evaluation needed heavy teacher prompting, it is Emerging, not Proficient.

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10. Annotated exemplars

Use these worked responses to anchor judgements. Each shows a plausible Grade 6 performance and the annotations that place it at a level.

Exemplar - EMERGING

Student response (observed) Annotation (why this level)

Part 1: "It could learn from the cards." When asked how, says "it just knows." Part 2: writes "Sort the souq cards" as the prompt; the plan has a goal but no clear process or output. Part 3: marks the pottery card "wrong" but the reason is "it looks off"; does not give an accuracy count. Part 4: "Putting a phone number is bad," but cannot say what the privacy risk is or who it affects.

Recognises the system could learn, uses a rough prompt, spots one problem output, and recognises a basic risk - all at the Emerging descriptor. No explanation of how the model learns, an incomplete plan, feeling-based verdicts, and no bias or privacy reasoning. At least 3 strands at Emerging - Emerging overall, the expected Grade 6 standard. Next step: the how-a-model-learns routine (Week 2) and the triage reason frame from Checkpoint 2.

Exemplar - PROFICIENT

Student response (observed) Annotation (why this level)

Part 1: "It learns a model from the labelled example cards, then uses the pattern to label a new card."

Part 2: a full input-process-output plan with a role, task and a no-personal-data rule, and one label per card as the output; builds it by following the plan.

Part 3: triages all five outputs with checkable reasons, reports "3 out of 5 correct," and flags the two-item card as unclear. Part 4: "It is biased against crafts because most examples were dates,

and the phone number is a privacy risk for the stallholder; we should never enter personal data." Explains how the model learns from data, lays out and builds an input-process-output solution, evaluates outputs for accuracy, and explains bias, privacy and responsible use. At least 3 strands at Proficient with none below Emerging - Proficient overall.

Exemplar - ADVANCED

Student response (observed) Annotation (why this level)

As Proficient, and: Part 1 explains that because the Example Set is mostly Dates and Food, the sorter learns that pattern strongly and labels rare categories poorly - the training data lowers its performance there. Part 2 adds an output-format rule and a fallback for two-item cards. Part 3 explains that adding varied craft and textile examples would raise accuracy on those cards. Part 4 traces the biased labels to the imbalanced data and proposes collecting a balanced example set and removing personal details before any card is used.

Reasons about how training data affects quality and performance, hardens the solution with guided inputs, explains data-driven improvements to outputs, and proposes fairness and safety fixes.

At least 3 strands at Advanced with none below Proficient - Advanced overall.

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11. Reporting, feedback and next steps

Report each student against the four strands and the overall level, and turn the result into a next step. Feedback is most useful when it is specific and forward-looking.

If the student is... Feed-forward for the student Teacher next step / bridging link

Working towards /

Emerging

"You can spot the system type and a basic risk. Next, practise saying how a learning system uses example data, and give a checkable reason for each verdict."

Re-run the Week 2 train-a-simple-model routine and the Week 4 triage routine; use the Goal-Context-Format frame from Week 3.

Emerging
(expected)

"You use a structured prompt and spot problems. Next, complete a full input-process-output plan and report an accuracy count."

Extend with the Week 7-8 plan-and-build cycle and the Week 8 test-set evaluation; introduce data-balance thinking.

Proficient "You explain how models learn and evaluate outputs well. Next, reason about how the training data changes quality and fairness across the whole set."

Offer the Week 9 refine-and-justify stretch and the Week 5 risk-matrix task; begin the multi-step workflow thinking that opens Grade 7.

Advanced "You reason about data quality, bias and safety. Next, design a balanced example set and defend your fairness and safety fixes."

Provide early Grade 7 workflow-chaining tasks and a data-collection redesign challenge.

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12. Accessibility and Students of Determination

Access is designed in, following Universal Design for Learning and the school's Best Endeavours and Reasonable Adjustments (BERA). None of these changes the construct being assessed - they change how a student accesses the task or shows what they know.

Universal Design for Learning - built-in options

UDL principle Options offered in this assessment

Multiple means of
engagement

Familiar UAE context (a Heritage-Day souq); choice of which item cards to reason about; a low-stakes "no wrong answers when you explain" framing; teacher-run tool so no login pressure.

Multiple means of
representation

Category labels and prompts read aloud; the input-process-output flow shown as a labelled arrow; colour-coded example cards; a word card (rule-based, learning, data, model, bias, privacy) with icons.

Multiple means of action
and expression

Students may write, mark, highlight, place cards, or explain aloud; a teacher may scribe; the plan can be completed with cut-and-place labels.

Reasonable adjustments by need (BERA)

Need Adjustment

Visual impairment Large-print, high-contrast cards and sheets; read all card text and labels aloud;

allow a magnifier or screen reader; tactile category tokens for sorting.

Hearing impairment Provide the scene-setting script and prompts in writing; face the student when

speaking; check understanding visually; captions for any board clip.

Physical / fine-motor Allow pointing or verbal answers scribed by an adult; use cut-and-place labels

instead of writing on the plan and output sheets.

Attention / cognitive

load

Chunk into two sessions (Parts 1-2, then 3-4); one part at a time; extra processing time; a checklist of the four parts.

Language / EAL Sentence stems ("It learns because ____", "This output is ____ because ____", "The

privacy risk is ____"); bilingual key-word card; accept answers in the student's stronger language and scribe.

Sensory needs Quiet space; movement break between parts; reduce visual clutter on printed cards and sheets.

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13. Administration guide

Field Guidance

Time 55-70 minutes total, about 12-16 minutes per part. May be split into two sessions (Parts 1-2, then 3-4).

Grouping Part 1 may be discussed in pairs; the Part 2 plan and the Part 3 evaluation are recorded individually; Part 4 is answered individually with a short conversation.

Conditions Calm, low-stakes. All AI/ML tool use is teacher-operated on the board; students never sign in and never enter personal data. Reassure students that exploration and talk are expected; recorded answers are their own.

When to run At the end of the unit (Week 10), once the signature solution has been planned, built and refined, so students have run the full input-process-output cycle before the task.

Preparation Print the Example Set, Output Sheet with answer key, Solution Plan Sheet, and the two tricky cards (Appendix); prepare coloured pens; display the input-process-output arrow and the word card; ready the approved teacher-operated tool or the simulated-AI cards.

Integrity Keep the seeded Output Sheet consistent across the cohort; do not pre-mark the incorrect or unclear outputs for students; assess process and reasoning, not the tool's polish.

Before / during / after checklist

Before: share the success criteria; print materials; plan access arrangements; run both checkpoints earlier in the unit; confirm the tool is teacher-operated and account-free.

During: read the script; run parts in order; operate the tool on the board; observe and note; ask the "why" and "what one change" probes; scribe where needed.

After: complete the rubric and observation record; moderate a sample; record next steps and bridging links.

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Appendix. Printable stimulus materials

Print one copy of each per student. The Output Sheet contains two deliberate problems (one incorrect label and one unclear label) and the Example Set contains a deliberate imbalance; do not mark these for students.

A. Souq Sorter Example Set (the labelled training cards)

These labelled cards are what the sorter "learns" from. Notice how many are Dates and Food.

Card Item description (no personal data) Category label

1 Fresh khalas dates by the box Dates and Food

2 Date-filled maamoul biscuits Dates and Food

3 Honey and date syrup jars Dates and Food

4 Bags of mixed dried dates Dates and Food

5 Fresh laban and cheese Dates and Food

6 Hand-painted clay water jug Crafts and Pottery

7 Woven palm-leaf basket Textiles and Clothing

8 Embroidered shawl Textiles and Clothing

Teacher note: six of the eight cards are Dates and Food; Crafts and Pottery and Textiles have very few

examples. This imbalance is the source of the bias explored in Parts 1, 3 and 4.

B. Souq Sorter Output Sheet (five new cards) with answer key

The sorter labelled five new cards. Mark each output against the correct label: correct, incorrect, or

unclear, with a checkable reason.

New card Item description Sorter's output Correct label (key)

1 Sealed tubs of fresh dates Dates and Food Dates and Food

2 Painted clay water jug Dates and Food Crafts and Pottery

3 Cotton kandura for boys Textiles and Clothing Textiles and Clothing

4 Dates and a woven basket Dates and Food (two items - needs splitting)

5 Carved wooden coffee pot Other Crafts and Pottery

Teacher note: Card 2 is incorrect (a jug labelled Dates and Food). Card 4 is unclear (two items, one vague

label) and cannot be judged until the input is split. Card 5 shows the imbalance effect - a craft item

pushed into Other. Accuracy on this set is 2 out of 5.

C. Solution Plan Sheet (input -> process -> output)

Complete each row for your Souq Sorter. Keep all inputs free of personal or identifying data.

Plan part Your plan

Goal (one sentence) _____

Input The item card(s) the sorter reads: _____

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Process - structured

prompt

Role: _____

Task: _____

One rule: _____

Output format _____ (e.g. one category

label per

card)

Evaluation and risk I will check the output by _____. One risk is

_____;

I will reduce it by _____.

D. Tricky item cards for Part 4 (fairness and privacy)

Card Item description What to notice

Tricky 1 Carved wooden coffee pot and incense

holder

A craft item the sorter rarely sees - likely

mislabelled because of the imbalanced

example data (bias).

Tricky 2 Handmade soap - contact Mr Ahmed Al

Blooshi, 050 123 4567

Includes a stallholder's full name and phone

number - a privacy risk; this personal data

must be removed before any use.

After Part 4, be ready to say: what in the data makes the sorter unfair to some stalls, what the privacy risk is and who it affects, and one simple way to make the sorter fairer or safer.

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